

雅思阅读段落信息匹配题真题

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Test 1

READING PASSAGE 1

You should spend about 20 minutes on Questions 1-13 which are based on Reading Passage 1 below.

Bamboo, A Wonder Plant

The wonder plant with an uncertain future: more than a billion people rely on bamboo for either their shelter or income, while many endangered species depend on it for their survival. Despite its apparent abundance, a new report says that species of bamboo may be under serious threat.

Section A

Every year, during the rainy season, the mountain gorillas of Central Africa migrate to the foothills and lower slopes of the Virunga Mountains to graze on bamboo. For the 650 or so that remain in the wild, it's a vital food source. Although they eat almost 150 types of plant, as well as various insects and other invertebrates, at this time of year bamboo accounts for up to 90 per cent of their diet. Without it, says Ian Redmond, chairman of the Ape Alliance, their chances of survival would be reduced significantly. Gorillas aren't the only locals keen on bamboo. For the people who live close to the Virungas, it's a valuable and versatile raw material used for building houses and making household items such as mats and baskets. But in the past 100 years or so, resources have come under increasing pressure as populations have exploded and large areas of bamboo forest have been cleared to make way for farms and commercial plantations.

Section B

Sadly, this isn't an isolated story. All over the world, the ranges of many bamboo species appear to be shrinking, endangering the people and animals (that depend upon them). But despite bamboo's importance, we know surprisingly little about it. A recent report published by the UN Environment Programme (UNEP) and the International Network for Bamboo and Rattan (INBAR) has revealed just how profound is our ignorance of global bamboo resources, particularly in relation to conservation. There are almost 1,600 recognised species of bamboo, but the report concentrated on the 1,200 or so woody varieties distinguished by the strong stems, or culms, that most people associate with this versatile plant. Of these, only 38 'priority species' identified for their commercial value have been the subject of any real scientific research, and this has focused mostly on matters relating to their viability as a commodity. This problem isn't confined to bamboo. Compared to the work carried out on animals, the science of assessing the conservation status of plants is still in its infancy. "People have only started looking hard at this during the past 10-15 years, and only now are

they getting a handle on how to go about it systematically,” says Dr Valerie Kapos, one of the report’s authors and a senior adviser in forest ecology and conservation to the UNEP.

Section C

Bamboo is a type of grass. It comes in a wide variety of forms, ranging in height from 30 centimetres to more than 40 metres. It is also the world’s fastest-growing woody plant: some species can grow more than a metre in a day. Bamboo’s ecological role extends beyond providing food and habitat for animals. Bamboo tends to grow in stands made up of groups of individual plants that grow from root systems known as rhizomes. Its extensive rhizome systems, which tie in the top layers of the soil, are crucial in preventing soil erosion. And there is growing evidence that bamboo plays an important part in determining forest structure and dynamics. “Bamboo’s pattern of mass flowering and mass death leaves behind large areas of dry biomass that attract wildfire,” says Kapos. “When these burn, they create patches of open ground within the forest far bigger than would be left by a fallen tree.” Patchiness helps to preserve diversity because certain plant species do better during the early stages of regeneration when there are gaps in the canopy.

Section D

However, bamboo’s most immediate significance lies in its economic value. Modern processing techniques mean that it can be used in a variety of ways, for example, as flooring and laminates. One of the fastest-growing bamboo products is paper—25 per cent of paper produced in India is made from bamboo fiber, and in Brazil, 100,000 hectares of bamboo is grown for its production. Of course, bamboo’s main function has always been in domestic applications, and as a locally traded commodity, it’s worth about US\$4.5 billion annually. Because of its versatility, flexibility and strength (its tensile strength compares to that of some steel), it has traditionally been used in construction. Today, more than one billion people worldwide live in bamboo houses. Bamboo is often the only readily available raw material for people in many developing countries, says Chris Stapleton, a research associate at the Royal Botanic Gardens. “Bamboo can be harvested from forest areas or grown quickly elsewhere, and then converted simply without expensive machinery or facilities,” he says. “In this way, it contributes substantially to poverty alleviation and wealth creation.”

Section E

Given bamboo’s value in economic and ecological terms, the picture painted by the UNEP report is all the more worrying. But keen horticulturists will spot an apparent contradiction here. Those who’ve followed the recent vogue for cultivating exotic species in their gardens will point out that if it isn’t kept in check, bamboo can cause real problems. “In a lot of places, the people who live with bamboo don’t perceive it as being endangered in any way,” says Kapos. “In fact, a lot of bamboo species are actually very invasive if they’ve been introduced.” So why are so many species endangered? There are two separate issues here, says Ray Townsend, vice president of the British Bamboo Society and arboretum manager at the Royal Botanic Gardens. “Some plants are threatened because they can’t survive in the habitat – they aren’t strong enough or there aren’t enough of them, perhaps. But bamboo can take care of

itself – it is strong enough to survive if left alone. What is under threat is its habitat.” It is the physical disturbance that is the threat to bamboo, says Kapos. “When forest goes, it is converted into something else: there isn’t any-where for forest plants such as bamboo to grow if you create a cattle pasture.”

Section F

Around the world, bamboo species are routinely protected as part of forest eco-systems in national parks and reserves, but there is next to nothing that protects bamboo in the wild for its own sake. However, some small steps are being taken to address this situation. The UNEP-INBAR report will help conservationists to establish effective measures aimed at protecting valuable wild bamboo species. Towns end, too, see the UNEP report as an important step forward in promoting the cause of bamboo conservation. “Until now, bamboo has been perceived as a second-class plant. When you talk about places such as the Amazon, everyone always thinks about the hardwoods. Of course, these are significant, but there is a tendency to overlook the plants they are associated with, which are often bamboo species. In many ways, it is the most important plant known to man. I can’t think of another plant that is used so much and is so commercially important in so many countries.” He believes that the most important first step is to get scientists into the field. “We need to go out there, look at these plants and see how they survive and then use that information to conserve them for the future.”

Questions 1-7

Reading Passage 1 has six section A-F.

Which section contains the following information?

Write the correct letter A-F in boxes 1-7 on your answer sheet

NB You may use any letter more than once

- 1 The limited extent of existing research
- 2 Comparison of bamboo with other plant species
- 3 Commercial application of bamboo
- 4 Example of an animal which relies on bamboos for survival
- 5 The human activity that damaged large areas of bamboo
- 6 The approaches used to study bamboo
- 7 Bamboo helps the survival of a range of plants

Questions 8-11

Use the information in the passage to match the people (listed A-D) with opinions or deeds below.

Write the appropriate letters A-D in boxes 8-11 on your answer sheet.

NB You may use any letter more than once

- | | |
|---|-----------------|
| A | Ian Redmond |
| B | Valerie Kapos |
| C | Ray Townsend |
| D | Chris Stapleton |

答案

1. B
2. E
3. D
4. D
5. A
6. B
7. C
8. A
9. B
10. B
11. D

READING PASSAGE 2

You should spend about 20 minutes on Questions 14-26 which are based on Reading Passage 2 below.

Activities for Children

A

Twenty-five years ago, children in London walked to school and played in parks and playing fields after school and at the weekend. Today they are usually driven to school by parents anxious about safety and spend hours glued to television screens or computer games. Meanwhile, community playing fields are being sold off to property developers at an alarming rate. ‘This change in lifestyle has, sadly, meant greater restrictions on children,’ says Neil Armstrong, Professor of Health and Exercise Sciences at the University of Exeter. ‘If children continue to be this inactive, they’ll be storing up big problems for the future.’

B

In 1985, Professor Armstrong headed a five-year research project into children’s fitness. The results, published in 1990, were alarming. The survey, which monitored 700 11-16-year-olds, found that 48 per cent of girls and 41 per cent of boys already exceeded safe cholesterol levels set for children by the American Heart Foundation. Armstrong adds, ‘heart is a muscle and need exercise, or it loses its strength.’ It also found that 13 per cent of boys and 10 per cent of girls were overweight. More disturbingly, the survey found that over a four-day period, half the girls and one-third of the boys did less exercise than the equivalent of a brisk 10-minute walk. High levels of cholesterol, excess body fat and inactivity are believed to increase the risk of coronary heart disease.

C

Physical education is under pressure in the UK – most schools devote little more than 100 minutes a week to it in curriculum time, which is less than many other European countries. Three European countries are giving children a head start in PE, France, Austria and Switzerland – offer at least two hours in primary and secondary schools. These findings, from the European Union of Physical Education Associations, prompted specialists in children’s physiology to call on European governments to give youngsters a daily PE programme. The survey shows that the UK ranks 13th out of the 25 countries, with Ireland’s bottom, averaging under an hour a week for PE. From age six to 18, British children received, on average, 106 minutes of PE a week. Professor Armstrong, who presented the findings at the meeting, noted that since the introduction of the national curriculum there had been a marked fall in the time devoted to PE in UK schools, with only a minority of pupils getting two hours a week.

D

As a former junior football international, Professor Armstrong is a passionate advocate for the sport. Although the Government has poured millions into beefing up the sport in the community, there is less commitment to it as part of the crammed school curriculum. This means that many children never acquire the necessary skills

to thrive in team games. If they are no good at them, they lose interest and establish an inactive pattern of behaviour. When this is coupled with a poor diet, it will lead inevitably to weight gain. Seventy per cent of British children gives up all sport when they leave school, compared with only 20 per cent of French teenagers. Professor Armstrong believes that there is far too great an emphasis on team games at school. “We need to look at the time devoted to PE and balance it between individual and pair activities, such as aerobics and badminton, as well as team sports. “He added that children need to have the opportunity to take part in a wide variety of individual, partner and team sports.

E

The good news, however, is that a few small companies and children’s activity groups have reacted positively and creatively to the problem. ‘Take That, shouts Gloria Thomas, striking a disco pose astride her mini-space hopper. ‘Take That, echo a flock of toddlers, adopting outrageous postures astride their space hoppers. ‘Michael Jackson, she shouts, and they all do a spoof fan-crazed shriek. During the wild and chaotic hopper race across the studio floor, commands like this are issued and responded to with untrammelled glee. The sight of 15 bouncing seven-year-olds who seem about to launch into orbit at every bounce brings tears to the eyes. Uncoordinated, loud, excited and emotional, children provide raw comedy.

F

Any cardiovascular exercise is a good option, and it doesn’t necessarily have to be high intensity. It can be anything that gets your heart rate up: such as walking the dog, swimming, running, skipping, hiking. “Even walking through the grocery store can be exercise,” Samis-Smith said. What they don’t know is that they’re at a Fit Kids class and that the fun is a disguise for the serious exercise plan they’re covertly being taken through. Fit Kids trains parents to run fitness classes for children. ‘Ninety per cent of children don’t like team sports,’ says company director, Gillian Gale.

G

A Prevention survey found that children whose parents keep in shape are much more likely to have healthy body weights themselves. “There’s nothing worse than telling a child what he needs to do and not doing it yourself,” says Elizabeth Ward, R.D., a Boston nutritional consultant and author of Healthy Foods, Healthy Kids. “Set a good example and get your nutritional house in order first.” In the 1930s and ‘40s, kids expended 800 calories a day just walking, carrying water, and doing other chores, notes Fima Lifshitz, M.D., a pediatric endocrinologist in Santa Barbara. “Now, kids in obese families are expending only 200 calories a day in physical activity,” says Lifshitz, “incorporate more movement in your family’s life – park farther away from the stores at the mall, take stairs instead of the elevator, and walk to nearby friends’ houses instead of driving.”

Questions 14-17

The Reading Passage has seven paragraphs A-G

Which paragraph contains the following information?

Write the correct letter A-G, in boxes 14-17 on your answer sheet.

-
- 14 Health and living condition of children
 - 15 Health organization monitored physical activity
 - 16 Comparison of exercise time between the UK and other countries
 - 17 Wrong approach for school activity

答案

14. A
15. B
16. C
17. D

READING PASSAGE 3

You should spend about 20 minutes on Questions 27-40 which are based on Reading Passage 3 below.

Save Endangered Language

“Obviously we must do some serious rethinking of our priorities, lest linguistics go down in history as the only science that presided obviously over the disappearance of 90 percent of the very field to which it is dedicated.” – Michael Krauss, “The World’s Languages in Crisis”.

A

Ten years ago Michael Krauss sent a shudder through the discipline of linguistics with his prediction that half the 6,000 or so languages spoken in the world would cease to be uttered within a century. Unless scientists and community leaders directed a worldwide effort to stabilize the decline of local languages, he warned, nine-tenths of the linguistic diversity of humankind would probably be doomed to extinction. Krauss’s prediction was little more than an educated guess, but other respected linguists had been clanging out similar alarms. Kenneth L. Hale of the Massachusetts Institute of Technology noted in the same journal issue that eight languages on which he had done fieldwork had since passed into extinction. A 1990 survey in Australia found that 70 of the 90 surviving Aboriginal languages were no longer used regularly by all age groups. The same was true for all but 20 of the 175 Native American languages spoken or remembered in the US., Krauss told a congressional panel in 1992.

B

Many experts in the field mourn the loss of rare languages, for several reasons. To start, there is scientific self-interest: some of the most basic questions in linguistics have to do with the limits of human speech, which are far from fully explored. Many researchers would like to know which structural elements of grammar and vocabulary – if any – are truly universal and probably, therefore, hardwired into the human brain. Other scientists try to reconstruct ancient migration patterns by comparing borrowed words that appear in otherwise unrelated languages. In each of these cases, the wider the portfolio of languages you study, the more likely you are to get the right answers.

C

Despite the near-constant buzz in linguistics about endangered languages over the past 10 years, the field has accomplished depressingly little. “You would think that there would be some organized response to this dire situation,” some attempt to determine which language can be saved and which should be documented before they disappear, says Sarah G. Thomason, a linguist at the University of Michigan at Ann Arbor. “But there isn’t any such effort organized in the profession. It is only recently that it has become fashionable enough to work on endangered languages.” Six years ago, recalls Douglas H. Whalen of Yale University, “when I asked linguists who were raising money to deal with these problems, I mostly got blank stares.” So Whalen and a few other linguists founded the Endangered Languages Fund. In the five years to 2001,

they were able to collect only \$80,000 for research grants. A similar foundation in England, directed by Nicholas Ostler, has raised just \$8,000 since 1995.

D

But there are encouraging signs that the field has turned a corner. The Volkswagen Foundation, a German charity, just issued its second round of grants totaling more than \$2 million. It has created a multimedia archive at the Max Planck Institute for Psycholinguistics in the Netherlands that can house recordings, grammars, dictionaries and other data on endangered languages. To fill the archive, the foundation has dispatched field linguists to document Aweti (100 or so speakers in Brazil), Ega (about 300 speakers in Ivory Coast), Waima'a (a few hundred speakers in East Timor), and a dozen or so other languages unlikely to survive the century. The Ford Foundation has also edged into the arena. Its contributions helped to reinvigorate a master-apprentice program created in 1992 by Leanne Hinton of Berkeley and Native Americans worried about the imminent demise of about 50 indigenous languages in California. Fluent speakers receive \$3,000 to teach a younger relative (who is also paid) their native tongue through 360 hours of shared activities, spread over six months. So far about 5 teams have completed the program, Hinton says, transmitting at least some knowledge of 25 languages. "It's too early to call this language revitalization," Hinton admits. "In California, the death rate of elderly speakers will always be greater than the recruitment rate of young speakers. But at least we prolong the survival of the language." That will give linguists more time to record these tongues before they vanish.

E

But the master-apprentice approach hasn't caught on outside the U.S., and Hinton's effort is a drop in the sea. At least 440 languages have been reduced to a mere handful of elders, according to the Ethnologue, a catalogue of languages produced by the Dallas-based group SIL International that comes closest to global coverage. For the vast majority of these languages, there is little or no record of their grammar, vocabulary, pronunciation or use in daily life. Even if a language has been fully documented, all that remains once it vanishes from active use is a fossil skeleton, a scattering of features that the scientist was lucky and astute enough to capture. Linguists may be able to sketch an outline of the forgotten language and fix its place on the evolutionary tree, but little more. "How did people start conversations and talk to babies? How do husbands and wives converse?" Hinton asks. "Those are the first things you want to learn when you want to revitalize the language."

F

But there is as yet no discipline of "conservation linguistics," as there is for biology. Almost every strategy tried so far has succeeded in some places but failed in others, and there seems to be no way to predict with certainty what will work where. Twenty years ago in New Zealand, Maori speakers set up "language nests," in which preschoolers were immersed in the native language. Additional Maori-only classes were added as the children progressed through elementary and secondary school. A similar approach was tried in Hawaii, with some success – the number of native speakers has stabilized at 1,000 or so, reports Joseph E. Grimes of SIL International,

who is working on Oahu. Students can now get instruction in Hawaiian all the way through university.

G

One factor that always seems to occur in the demise of a language is that the speakers begin to have collective doubts about the usefulness of language loyalty. Once they start regarding their own language as inferior to the majority language, people stop using it in all situations. Kids pick up on the attitude and prefer the dominant language. In many cases, people don't notice until they suddenly realize that their kids never speak the language, even at home. This is how Cornish and some dialects of Scottish Gaelic is still only rarely used for daily home life in Ireland, 80 years after the republic was founded with Irish as its first official language.

H

Linguists agree that ultimately, the answer to the problem of language extinction is multilingualism. Even uneducated people can learn several languages, as long as they start as children. Indeed, most people in the world speak more than one tongue, and in places such as Cameroon (279 languages), Papua New Guinea (823) and India (387) it is common to speak three of four distinct languages and a dialect or two as well. Most Americans and Canadians, to the west of Quebec, have a gut reaction that anyone speaking another language in front of them is committing an immoral act. You get the same reaction in Australia and Russia. It is no coincidence that these are the areas where languages are disappearing the fastest. The first step in saving dying languages is to persuade the world's majorities to allow the minorities among them to speak with their own voices.

Questions 27-33

The reading passage has eight paragraphs, A-H

Choose the correct heading for paragraphs A-H from the list below.

Write the correct number, i-xi, in boxes 27-33 on your answer sheet.

List of headings

- i data consistency needed for language
 - ii consensuses on an initial recommendation for saving dying out languages
 - iii positive gains for protection
 - iv minimum requirement for saving a language
 - v Potential threat to minority language
 - vi a period when there was absent of real effort made.
 - vii native language programs launched
 - viii Lack of confidence in young speakers as a negative factor
 - ix Practise in several developing countries
 - x Value of minority language to linguists
 - xi government participation in the language field
- 27 Paragraph A
- 28 Paragraph B
- Example: Paragraph C vi
- 29 Paragraph D

- 30 Paragraph E
- 31 Paragraph F
- 32 Paragraph G
- 33 Paragraph H

Questions 34-38

Use the information in the passage to match the people (listed A-F) with opinions or deeds below.

Write the appropriate letters A-F in boxes 34-38 on your answer sheet.

- | | |
|---|-------------------|
| A | Nicholas Ostler |
| B | Michael Krauss |
| C | Joseph E. Grimes |
| D | Sarah G. Thomason |
| E | Keneth L. Hale |
| F | Douglas H. Whalen |
- 34 Reported language conservation practice in Hawaii
 - 35 Predicted that many languages would disappear soon
 - 36 The experienced process that languages die out personally
 - 37 Raised language fund in England
 - 38 Not enough effort on saving until recent work

答案

- 27. v
- 28. x
- 29. iii
- 30. i
- 31. vii
- 32. viii
- 33. ii
- 34. C
- 35. B
- 36. E
- 37. A
- 38. D

Test 2

READING PASSAGE 1

You should spend about 20 minutes on Questions 1-13 which are based on Reading Passage 1 below.

LEISURE TIME IN AMERICA

A

As most Americans will tell you if you can stop them long enough to ask, working people in the United States are as busy as ever. Sure, technology and competition are boosting the economy; but nearly everyone thinks they have increased the demands on people at home and in the workplace. But is the overworked American a creature of myth?

B

A pair of economists have looked closely at how Americans actually spend their time. Mark Aguiar, at the Federal Reserve Bank of Boston and Erik Hurst, at the University of Chicago's Graduate School of Business constructed four different measures of leisure. The narrowest includes only activities that nearly everyone considers relaxing or fun; the broadest counts anything that is not related to a paying job, housework or errands as "leisure". No matter how the two economists slice the data, Americans seem to have much more free time than before.

C

Over the past four decades, depending on which of their measures one uses, the amount of time that working-age Americans are devoting to leisure activities has risen by 4-8 hours a week. For somebody working 40 hours a week, that is equivalent to 5-10 weeks of extra holiday a year. Nearly every category of American has more spare time: single or married, with or without children, both men and women. Americans may put in longer hours at the office than other countries, but that is because average hours in the workplace in other rich countries have dropped sharply.

D

How then have Messrs Aguiar and Hurst uncovered a more relaxed America, where leisure has actually increased? It is partly to do with the definition of work, and partly to do with the data they base their research upon. Most American labour studies rely on well-known official surveys, such as those collected by the Bureau of Labour Statistics (BLS) and the Census Bureau, that concentrate on paid work. These are good at gleaning trends in factories and offices, but they give only a murky impression of how Americans use the rest of their time. Messrs Aguiar and Hurst think that the hours spent at your employer's are too narrow a definition of work. Americans also spend lots of time shopping, cooking, running errands and keeping house. These chores are among the main reasons why people say they are so overstretched, especially working women with children.

E

However, Messrs Aguiar and Hurst show that Americans actually spend much less time doing them than they did 40 years ago. There has been a revolution in the household economy. Appliances, home delivery, the internet, 24-hour shopping, and more varied and affordable domestic services have increased flexibility and freed up people's time.

F

The data for Messrs Aguiar and Hurst's study comes from time-use diaries that American social scientists have been collecting methodically, once a decade, since 1965. These diaries ask people to give detailed information on everything they did the day before, and for how long they did it. The beauty of such surveys, which are also collected in Australia and many European countries, is that they cover the whole day, not just the time at work, and they also have a built-in accuracy check, since they must always account for every hour of the day.

G

Do the numbers add up? One thing missing in Messrs Aguiar's and Hurst's work is that they have deliberately ignored the biggest leisure-gainers in the population, the growing number of retired folk. The two economists excluded anyone who has reached 65 years old, as well as anyone under that age who retired early. So America's true leisure boom is even bigger than their estimate.

H

The biggest theoretical problem with time diaries is "multi-tasking". Do you measure the time you spend cleaning your house while listening to portable music as "leisure" or "work"? This problem may be exaggerated: usually people seem to combine two work activities, using a laptop computer on a plane, or two leisure ones, watching television and doing something else. The two economists counted many combinations of work and leisure, such as reading a novel while commuting or goofing off on the internet at the office, as time spent working.

I

Is all this leisure a good thing? Some part-time workers might well wish they had less leisure and more income. For most Americans, however, the leisure dividend appears to be a bonus. Using average hourly wages after tax, Steven Davis, a colleague of Mr Hurst's, reckons that the national value of five extra hours of leisure per week is \$570 billion, or \$3,300 per worker, every year.

Questions 1-9

Match each heading to the most suitable paragraph.

- i One possible source of inaccuracies
- ii Less time doing chores
- iii A difference between perception and reality
- iv The value of extra leisure time
- v Americans are working harder
- vi Significantly more free time
- vii The effect of including retirees

-
- viii The need for a wider description of work
ix An effective system for measuring time spent
x How Americans think about their time
1 Paragraph A
2 Paragraph B
3 Paragraph C
4 Paragraph D
5 Paragraph E
6 Paragraph F
7 Paragraph G
8 Paragraph H
9 Paragraph I

答案

1. x
2. iii
3. vi
4. viii
5. ii
6. ix
7. vii
8. i
9. iv

READING PASSAGE 2

You should spend about 20 minutes on Questions 14-26 which are based on Reading Passage 2 below.

RECLAIMING THE NIGHT

A

On a summer's day, apart from the intermittent drizzle and lowering sky, South Street in Romford looks as close to an Englishman's dream of a continental-style piazza as it is possible to get. Leafy trees line the extended pavements crowded with seats and tables as young families, pensioners, teenagers and businessmen tuck into a variety of faux-European dishes for lunch. Local cafes serve the full range of meaningless variations on the theme of coffee, from cappuccino through mochaccino to doppos, all at top prices. Round the corner, in the Market Place, it is French week. There are several stalls, complete with real Frenchmen, selling claret and cheeses.

B

The cafes are open during the day, and the clubs stay open until two or three in the morning most nights. In this respect, Romford is typical of contemporary Britain. In the late 1980s, the centres of many towns and cities went into decline as retailers, and particularly supermarkets, moved to new big, out-of-town shopping centres. So in the early 1990s, many local councils, in league with local businesses, re-developed their increasingly desolate town centres into "leisure zones". They looked to continental Europe for the inspiration to create modern 24-hour environments, mixing cafes, bars and clubs to keep people in the centres spending money for as long as possible.

C

By night however, South Street turns into a very different place. The street becomes a mass of 18-26-year-olds, drinking as much as they can. For anyone else, the place becomes almost a no-go area. Gillian Balfe, the council's town-centre manager and a strong supporter of the "leisureing" of South Street, concedes that the crowds become uncontrollable, and the atmosphere quickly turns "hostile and threatening". Buses are now barred from going down South Street after 9.30pm: there are too many drunken people milling about.

D

In a survey for the local council done last year, 49% of the residents of the surrounding areas of South Street confessed that they did not want to come to the city centre any more for fear of crime. The local police concede that they are virtually overwhelmed. Violence is commonplace. There has only been one consequent fatality in the area in the past couple of years, but the police say that this is mainly thanks to the merciful proximity of the local hospital. Romford's dilemma is typical of what has happened in the other "leisure zones" in towns and cities throughout the country. What were meant to be civilised places for entertainment and shopping have too often turned into alcoholic ghettos for the young.

E

For all the problems, however, Romford's local authority thinks that the idea of a

24-hour city is already too profitable to be stopped. Local authorities think that new repressive legislation, or even a decision not to reform the licensing laws, would be unworkable. So instead of trying to pack everyone back off to bed, Romford is trying to reclaim the town centre for a broader mix of people, and so to fulfil the original ambitions of the 24-hour-city dreamers.

F

The first part of the strategy involves security. The police accept that, with their current resources, they will never be able to make South Street safe on their own. So they now work closely with the 528 “door-staff”, previously known as bouncers, to target drug-dealers in the bars and clubs. In the year since that scheme came into effect, there have been more than 300 arrests for drugs. In the six months before that, there had been only one. All the premises now have a radio link to the police station for an instant response to trouble.

G

The second part of the strategy involves trying to encourage more, and different kinds of people to use the town centre at night. New attractions are opening next year to rival the pubs. On the site of the old Romford brewery there will be a 16-screen cinema and a 24-hour supermarket. A new health and leisure centre, open on some nights until 9pm, starts up soon. The hope is that these facilities will draw in a different, more sober and ethnically diverse crowd. The police have bravely encouraged one club to start a gay night on Wednesdays.

H

Together with other measures such as better street lighting, Romford hopes that it can show that the phrase “24-hour city” can be more than a euphemism for an all-night drinkathon. As the new licensing laws delegate the job of granting alcohol licences to local councils, cities across England will be trying to reclaim the night.

Questions 14-18

Which paragraph contains the following information?

Write the correct letter, A-F, in boxes 14-18 on your answer sheet.

NB You may use any letter more than once.

- 14 why some local people stay out of the centre at night
- 15 how communication with the police has been made faster
- 16 reasons behind the growth in inner-city leisure venues
- 17 examples of Romford's similarity to mainland Europe
- 18 how illegal substances are being controlled

答案

- 14. D
- 15. F
- 16. B
- 17. A
- 18. F

READING PASSAGE 3

You should spend about 20 minutes on Questions 27-40 which are based on Reading Passage 3 below.

THE DRINKING OF WINE

THE birth of the cult of fine wine can be dated precisely. On April 10th 1663, Samuel Pepys, diarist and man-about-London, noted that he had enjoyed “a sort of French wine called Ho Bryan that hath a good and most particular taste that I never met with”.

The owners of Ho Bryan were the Pontacs. They were the top winemaking family of their day, and founded a fashionable restaurant, called Pontack’s Head, in London, in 1663. John Locke, the philosopher whose theory of the social contract inspired America’s revolutionaries, but who had worldlier interests too, spotted the reasons for the superiority of Ho Bryan on a visit to the vineyard in 1667. He found “a little rise of ground...white sand mixed with a little gravel; scarce fit to bear anything.” He added that “they say the wine in the next vineyard to it, tho’ seeming equal to me, is not so good.” Today that vineyard is still rated just below its neighbour.

Locke had seized on the essential concept of terroir, the combination of soil, subsoil, drainage and microclimate which provide the conditions for the production of fine wine. Another connoisseur, the 18th-century economist Adam Smith, noted that “the vine is more affected by the difference of soils than any other fruit tree. From some it derives a flavour which no culture or management can equal.”

By the early 18th century claret was getting more popular partly because it was getting better. The craft of claret-making had developed. The wine was designed to be kept for years not months, notably by being carefully stored in oak casks. Better corks allowed wine to be stored longer and more safely. Bottles were produced that could be “binned”—laid down on their sides to mature.

In the latter part of the 18th century drinking claret helped the rich to distinguish themselves from England’s port-sodden squirearchy. Port was not only the more traditional drink, but also—because it attracted much lower duties—far cheaper. John Hervey, the first Earl of Bristol, spent four times as much on claret as on port, whereas the lusty trenchermen who gathered in the Barbers Hall in the City of London spent a mere £2 on claret as against £850 on port.

When Britain made peace with France in 1713, claret became more accessible and the wine trade flourished. Claret was pricey but rich Londoners, who were also by then big spenders on theatres, spas and music produced by fashionable immigrants, such as Handel, consumed conspicuous quantities. Sir Robert Walpole, Britain’s first prime minister, used navy ships to smuggle his favourite wines from France. The most

expensive one he bought was old burgundy, but that—as now—was available only in tiny quantities. So he relied largely on claret, buying four hogsheads of 24 dozen bottles of Margaux and one hogshead of Lafite every three months. In a single year his wine bill amounted to over £1,200 (£100,000 today). British consumers bought the best stuff and paid top prices. By the time of the French revolution, the British were paying five times as much for their claret as the wine's other main customers, the notoriously parsimonious Dutch, who preferred the cheaper, lower-grade stuff.

By the late 19th century claret was beginning to flow down the social hierarchy. A free-trade treaty between Britain and France in 1860 drastically reduced the duty on French wines, thus encouraging the British middle classes to ape their social superiors; and in that year the chancellor of the exchequer, William Gladstone, keen to stiffen the nation's moral spine, cut the duty on table wines to 40% of that on more intoxicating fortified wines such as port and sherry.

The following year came the Single Bottle Act, allowing grocers to sell wine by the bottle. A much-despised, enormously popular drink called “grocers' claret” was born, with the result that, between 1859 and 1878, sales of French wines, largely from Bordeaux, rose sixfold to 36m bottles. The Gilbey family, one of the most remarkable commercial dynasties of Victorian England, franchised 2,000 grocers licensed to sell wine, largely claret. Their business grew so fast that by 1875 they were able to buy Château Loudenne in the Médoc to hold their gigantic stocks of claret. As the middle classes turned to claret, so the upper classes abandoned this increasingly common tittle in favour of hock and champagne.

Then the fortunes of the claret business turned. In the late 1870s and 1880s an attack of mildew tainted the wines: the reputation of Lafite, for instance, was ruined when the 1884 vintage turned mouldy after only a couple of years in bottle. At the same time, the phylloxera bug began to devastate Bordeaux's vineyards.

Claret came back into its own in 1960 when the splendid 1959 vintage coincided with the arrival of big American buyers. Its popularity has risen steadily since. London remains at the centre of the fine-wine business—home of organisations such as the Institute of Masters of Wine, of Decanter and World of Fine Wine magazines, and of most of the world's biggest wine auctions. Liv-Ex, the world's first stockmarket for fine wine, is based in London; and its figures show that nine-tenths of the wine trade is still in “classed growth” (leading) clarets. Newcomers from vineyards in a dozen countries trying to launch their finest wines on the world market come to London first for validation. Yet though London may still have much of the knowledge and the market, as consumers the British may be past their best. This year, 57% of the fine wine that Sotheby's sold globally, by value, was bought by Asians; four-fifths of those buyers were from China and Hong Kong.

Questions 27-32

Match each name to the sentences below.

- A John Hervey
 - B Adam Smith
 - C John Locke
 - D William Gladstone
 - E Robert Walpole
- 27 was perhaps the first person to notice why Ho Bryan tasted so good
 - 28 imported wine illegally
 - 29 wanted to discourage people from drinking strong wines
 - 30 drank more claret than port
 - 31 was a specialist in wine and economics
 - 32 bought more claret than any other kind of wine

答案

- 27. C
- 28. E
- 29. D
- 30. A
- 31. B
- 32. E

Test 3

READING PASSAGE 1

You should spend about 20 minutes on Questions 1-13 which are based on Reading Passage 1 below.

LONGAEVA: Ancient Bristlecone Pine

A

To understand more about the earth's history, humans have often looked to the natural environment for insight into the past. The bristlecone pine (*Pinus longaeva*), of the White Mountains in California, has served this purpose greater than any other species of tree on the planet. Conditions here are brutal: scant precipitation and low average temperatures mean a short growing season, only intensified by ferocious wind and mal-nutritious rocky. Nevertheless, bristlecone pines have claimed these barren slopes as their permanent home. Evolving here in this harsh environment, super-adapted and without much competition, bristlecones have earned their seat on the longevity throne by becoming the oldest living trees on the planet. Results of extensive studies on bristlecone pine stands have shown that in fact such, environmental limitations are positively associated with the attainment of great age. This intriguing phenomenon will be discussed further on.

B

But exactly how old is old? Sprouted before the invention of Egyptian hieroglyphs and long before the teachings of Jesus of Nazareth, Methuselah is the oldest bristlecone alive at roughly 4,700 years. Although specimens of this age do not represent the species' average, there are 200 trees more than 3,000 years old, and two dozen more than 4,000. Considering that these high ages are obtained in the face of such remarkable environmental adversity, the bristlecone pines have become the focus of much scientific examination over the past half-century.

C

Perhaps most interested in the bristlecone pine are dendrochronologists or tree-ring daters. With every strenuous year that passes in the White Mountains, each bristlecone grows and forms a new outer layer of cambium that reflects a season's particular ease or hardship. So while growing seasons may expand or shrink, the trees carry on, their growth rings faithfully recording the bad years alongside the goods. Through examining the annual growth rings of both living and dead specimens, taking thousands of core samples, and by processes of cross-dating between trees and other qualitative records, scientists have compiled a continuous tree-ring record that dates back to the last Ice Age between eight and ten thousand years ago. Among other linked accomplishments, this record has enhanced the dating process, helping to double-check and correct the radiocarbon-14 method to more accurately estimate the

age of organic material.

D

Now more than ever the importance of monitoring the bristlecone is being realized. As our global climate continues to undergo its most recent and abrupt atmospheric change, these ancient scribes continue to respond. Since, the rings of wood formed each year reveal the trees' response to climatic conditions during a particular growing season, in their persistence they have left us natural recordings of the past, markers of the present, and clues to the future.

E

The species' name originates from the appearance of its unusual cones and needles. The bristlecone's short, pale needles are also trademarks, bunching together to form foxtail-like bundles. As is the case of most conifer needles, these specialized leaves cluster together to shelter the stomata so very little moisture is lost through them. This adaptation helps the bristlecone photosynthesize during particularly brutal months. Saving the energy of constant needle replacement and providing a stable supply of chlorophyll. For a plant trying to store so much energy, bristlecone seeds are relatively large in size. They are first reproduced when trees reach ages between thirty and seventy-five years old. Germination rates are generally high, in part because seeds require little to no initial stratification. Perhaps the most intriguing physical characteristic of a mature bristlecone, however, is its ratio of living to deadwood on harsh sites and how this relates to old age. In older trees, however, especially in individuals over 1,500 years, a strip-bark trait is adaptive. This condition occurs as a result of cambium dieback, which erodes and thereby exposes certain areas of the bole, leaving only narrow bands of bark intact.

F

The technique of cambial edge retreat has helped promote old age in bristlecone pine, but that certainly is not the only reason. Most crucial to these trees' longevity is their compact size and slow rates of growth. By remaining in most cases under ten meters tall, bristlecones stay close to the limited water supply and can hence support more branches and photosynthesizing. Combined with the dry, windy, and often freezing mountain air, slow growth guarantees the bristlecones tight, fibrous rings with a high resin content and structural strength. The absence of natural disaster has also safeguarded the bristlecone's lengthy lifespan. Due to a lack of ground cover vegetation and an evenly spaced layout, bristlecone stands on the White Mountain peaks have been practically unaffected by the fire. This lack of vegetation also means a lack of competition for the bristlecones.

G

Bristlecone pines restricted to numerous, rather isolated stands at higher altitudes in the southwestern United States. Stands occur from the Rocky Mountains, through the Colorado Plateau, to the western margin of the Great Basin. Within this natural range, the oldest and most widely researched stands of bristlecones occur in California's the White Mountains. Even just 200 miles away from the Pacific Ocean, the White Mountains are home to one of this country's few high-elevation deserts. Located in the extreme eastern rain shadow of the Sierra Nevada, this region receives only 12.54

inches of precipitation per year and experiences temperatures between -20F and +50F. The peaks south of the Owens Valley, are higher up than they might appear from a distance. Although most summits exist somewhere around 11,000 feet, snow-capped White Mountain Peak, for which the range is named, stands at 14,246 feet above sea level. That said, to reach areas of a pure bristlecone is an intense journey all to itself.

H

With seemingly endless areas of wonder and interest, the bristlecone pines have become subject to much research over the past half-century. Since the annual growth of these ancient organisms directly reflects the climatic conditions of a particular time period, bristlecones are of greatest significance to dendrochronologists or tree-ring specialists. Dating any tree is simple and can be done within reasonable accuracy just by counting out the rings made each year by the plant's natural means of growth. By carefully compiling a nearly 10,000-year-old bristlecone pine record, these patient scientists have accurately corrected the carbon-14 dating method and estimated ages of past periods of global climate change. What makes this record so special to dendrochronologists, too, is that, nowhere, throughout time, is precisely the same long-term sequence of wide and narrow rings repeated, because year-to-year variations in climate are never exactly the same.

I

Historically the bristlecone's remote location and gnarled wood have deterred commercial extraction, but nothing on earth will go unaffected by global warming. If temperatures rise by only 6 degrees F, which many experts say is likely this century, about two-thirds of the bristlecones' ideal habitat in the White Mountains effectively will be gone. Almost 30,000 acres of National Forest now preserves the ancient bristlecone, but paved roads, campsites, and self-guided trails have led only to more human impact. In 1966, the U.S.F.S reported over 20,000 visitors to the Ancient Bristlecone Pine Forest, a figure which could exceed 40,000 today. Over the past hundreds of thousands of years, this species has endured in one of the earth's most trying environments; they deserve our respect and reverence. As global climate change slowly alters their environment, we as humans must do our part to raise awareness and lower our impact.

Questions 1-4

The Reading Passage has nine paragraphs A-I

Which paragraph contains the following information?

Write the correct letter A-I, in boxes 1-4 on your answer sheet.

- 1 Human activity threatens bristlecone pines habitat
- 2 Explanations for a ring of bristlecone pines
- 3 An accountable recording provided from the past until now
- 4 Survived in a hostile environment

答案

1. I
2. C
3. D
4. A

READING PASSAGE 2

You should spend about 20 minutes on Questions 14-26 which are based on Reading Passage 2 below.

Corporate Social Responsibility

Broadly speaking, proponents of CSR have used four arguments to make their case: moral obligation, sustainability, license to operate, and reputation. The moral appeal – arguing that companies have a duty to be good citizens and to “do the right thing” – is prominent in the goal of Business for Social Responsibility, the leading nonprofit CSR business association in the United States. It asks that its members “achieve commercial success in ways that honor ethical values and respect people, communities, and the natural environment.” Sustainability emphasizes the environment and community stewardship.

A

An excellent definition was developed in the 1980s by Norwegian Prime Minister Gro Harlem Brundtland and used by the World Business Council for Sustainable Development: “Meeting the needs of the present without compromising the ability of future generations to meet their own needs.” The notion of a license to operate derives from the fact that every company needs tacit or explicit permission from governments, communities, and numerous other stakeholders to do business. Finally, reputation is used by many companies to justify CSR initiatives on the grounds that they will improve a company’s image, strengthen its brand, enliven morale, and even raise the value of its stock.

B

To advance CSR, we must root it in a broad understanding of the interrelationship between a corporation and society while at the same time anchoring it in the strategies and activities of specific companies. To say broadly that business and society need each other might seem like a cliché, but it is also the basic truth that will pull companies out of the muddle that their current corporate-responsibility thinking has created. Successful corporations need a healthy society. Education, health care, and equal opportunity are essential to a productive workforce. Safe products and working conditions not only attract customers but lower the internal costs of accidents. Efficient utilization of land, water, energy, and other natural resources makes business more productive. Good government, the rule of law, and property rights are essential for efficiency and innovation. Strong regulatory standards protect both consumers and competitive companies from exploitation. Ultimately, a healthy society creates expanding demand for business, as more human needs are met and aspirations grow. Any business that pursues its ends at the expense of the society in which it operates will find its success to be illusory and ultimately temporary. At the same time, a healthy society needs successful companies. No social program can rival the business sector when it comes to creating the jobs, wealth, and innovation that improve standards of living and social conditions over time.

C

A company's impact on society also changes over time, as social standards evolve and science progresses. Asbestos, now understood as a serious health risk, was thought to be safe in the early 1900s, given the scientific knowledge then available. Evidence of its risks gradually mounted for more than 50 years before any company was held liable for the harms it can cause. Many firms that failed to anticipate the consequences of this evolving body of research have been bankrupted by the results. No longer can companies be content to monitor only the obvious social impacts of today. Without a careful process for identifying evolving social effects of tomorrow, firms may risk their very survival.

D

No business can solve all of society's problems or bear the cost of doing so. Instead, each company must select issues that intersect with its particular business. Other social agendas are best left to those companies in other industries, NGOs, or government institutions that are better positioned to address them. The essential test that should guide CSR is not whether a cause is worthy but whether it presents an opportunity to create shared value – that is, a meaningful benefit for society that is also valuable to the business. However, Corporations are not responsible for all the world's problems, nor do they have the resources to solve them all. Each company can identify the particular set of societal problems that it is best equipped to help resolve and from which it can gain the greatest competitive benefit. Addressing social issues by creating shared value will lead to self-sustaining solutions that do not depend on private or government subsidies. When a well-run business applies its vast resources, expertise, and management talent to problems that it understands and in which it has a stake, it can have a greater impact on social good than any other institution or philanthropic organization.

E

The best corporate citizenship initiatives involve far more than writing a check: They specify clear, measurable goals and track results over time. A good example is GE's program to adopt underperforming public high schools near several of its major U.S. facilities. The company contributes between \$250,000 and \$1 million over a five-year period to each school and makes in-kind donations as well. GE managers and employees take an active role by working with school administrators to assess needs and mentor or tutor students. In an independent study of ten schools in the program between 1989 and 1999, nearly all showed significant improvement, while the graduation rate in four of the five worst-performing schools doubled from an average of 30% to 60%. Effective corporate citizenship initiatives such as this one create goodwill and improve relations with local governments and other important constituencies. What's more, GE's employees feel great pride in their participation. Their effect is inherently limited, however. No matter how beneficial the program is, it remains incidental to the company's business, and the direct effect on GE's recruiting and retention is modest.

F

Microsoft's Working Connections partnership with the American Association of Community Colleges (AACC) is a good example of a shared-value opportunity

arising from investments in context. The shortage of information technology workers is a significant constraint on Microsoft's growth; currently, there are more than 450,000 unfilled IT positions in the United States alone. Community colleges, with an enrollment of 11.6 million students, representing 45% of all U.S. undergraduates, could be a major solution. Microsoft recognizes, however, that community colleges face special challenges: IT curricula are not standardized, the technology used in classrooms is often outdated, and there are no systematic professional development programs to keep faculty up to date. Microsoft's \$50 million five-year initiative was aimed at all three problems. In addition to contributing money and products, Microsoft sent employee volunteers to colleges to assess needs, contribute to curriculum development, and create faculty development institutes. Note that in this case, volunteers and assigned staff were able to use their core professional skills to address a social need, a far cry from typical volunteer programs. Microsoft has achieved results that have benefited many communities while having a direct – and potentially significant – impact on the company.

G

At the heart of any strategy is a unique value proposition: a set of needs a company can meet for its chosen customers that others cannot. The most strategic CSR occurs when a company adds a social dimension to its value proposition, making social impact integral to the overall strategy. Consider Whole Foods Market, whose value proposition is to sell organic, natural, and healthy food products to customers who are passionate about food and the environment. The company's sourcing emphasizes purchases from local farmers through each store's procurement process. Buyers screen out foods containing any of nearly 100 common ingredients that the company considers unhealthy or environmentally damaging. The same standards apply to products made internally. Whole Foods' commitment to natural and environmentally friendly operating practices extends well beyond sourcing. Stores are constructed using a minimum of virgin raw materials. Recently, the company purchased renewable wind energy credits equal to 100% of its electricity use in all of its stores and facilities, the only Fortune 500 company to offset its electricity consumption entirely. Spoiled produce and biodegradable waste are trucked to regional centers for composting. Whole Foods' vehicles are being converted to run on biofuels. Even the cleaning products used in its stores are environmentally friendly. And through its philanthropy, the company has created the Animal Compassion Foundation to develop more natural and humane ways of raising farm animals. In short, nearly every aspect of the company's value chain reinforces the social dimensions of its value proposition, distinguishing Whole Foods from its competitors.

From Harvard business review 2007

Questions 14-20

The Reading Passage has seven paragraphs, A-G

Choose the correct heading for paragraphs A-G from the list below.

Write the correct number, i-xi, in boxes 14-20 on your answer sheet.

List of Headings

- i How CSR may help one business to expand
- ii CSR in many aspects of a company's business
- iii A CSR initiative without a financial gain
- iv Lack of action by the state of social issues
- v Drives or pressures motivate companies to address CSR
- vi the past illustrates business are responsible for future outcomes
- vii Companies applying CSR should be selective
- viii Reasons that business and society benefit each other
- 14 Paragraph A
- 15 Paragraph B
- 16 Paragraph C
- 17 Paragraph D
- 18 Paragraph E
- 19 Paragraph F
- 20 Paragraph G

Questions 23-26

Use the information in the passage to match the companies (listed A-C) with opinions or deeds below.

Write the appropriate letters A, B or C in boxes 23-26 on your answer sheet.

List of companies

- A General Electronics
 - B Microsoft
 - C Whole foods market
- NB You may use any letter more than once
- 23 The disposable waste
 - 24 The way company purchases as goods
 - 25 Helping the undeveloped
 - 26 ensuring the people have the latest information

答案

- 14. v
- 15. viii
- 16. vi
- 17. vii
- 18. iii
- 19. i
- 20. ii
- 23. C
- 24. C
- 25. A
- 26. B

READING PASSAGE 3

You should spend about 20 minutes on Questions 27-40 which are based on Reading Passage 3 below.

The Exploration of Mars

A

In 1877, Giovanni Schiaparelli, an Italian astronomer, made drawings and maps of the Martian surface that suggested strange features. The images from telescopes at this time were not as sharp as today. Schiaparelli said he could see a network of lines or canali. In 1894, an American astronomer, Percival Lowell, made a series of observations of Mars from his own observatory at Flagstaff, Arizona, USA. Lowell was convinced a great network of canals had been dug to irrigate crops for the Martian race! He suggested that each canal had fertile vegetation on either side, making them noticeable from Earth. Drawings and globes he made show a network of canals and oases all over the planet.

B

The idea that there was intelligent life on Mars gained strength in the late 19th century. In 1898, H.G. Wells wrote a science fiction classic, *The War of the Worlds* about an invading force of Martians who try to conquer Earth. They use highly advanced technology (advanced for 1898) to crush human resistance in their path. In 1917, Edgar Rice Burroughs wrote the first in a series of 11 novels about Mars. Strange beings and rampaging Martian monsters gripped the public's imagination. A radio broadcast by Orson Welles on Halloween night in 1938 of *The War of the Worlds* caused widespread panic across America. People ran into the streets in their pyjamas-millions believed the dramatic reports of a Martian invasion.

C

Probes are very important to our understanding of other planets. Much of our recent knowledge comes from these robotic missions into space. The first images sent back from Mars came from Mariner 4 in July 1965. They showed a cratered and barren landscape, more like the surface of our moon than Earth. In 1969, Mariners 6 and 7 were launched and took 200 photographs of Mars's southern hemisphere and pole on fly-by missions. But these showed little more information. In 1971, Mariner 9's mission was to orbit the planet every 12 hours. In 1975, The USA sent two Viking probes to the planet, each with a lander and an orbiter. The Landers had sampler arms to scoop up Martian rocks and did experiments to try and find signs of life. Although no life was found, they sent back the first colour pictures of the planet's surface and atmosphere from pivoting cameras.

D

The Martian meteorite found in Earth aroused doubts to the above analysis. ALH84001 meteorite was discovered in December 1984 in Antarctica, by members of the ANSMET project; The sample was ejected from Mars about 17 million years ago and spent 11,000 years in or on the Antarctic ice sheets. Composition analysis by NASA revealed a kind of magnetite that on Earth, is only found in association with

certain microorganisms. Some structures resembling the mineralized casts of terrestrial bacteria and their appendages fibrils of by-products occur in the rims of carbonate globules and pre-terrestrial aqueous alteration regions. The size and shape of the objects are consistent with Earthly fossilized nanobacteria, but the existence of nanobacteria itself is still controversial.

E

In 1965, the Mariner 4 probe discovered that Mars had no global magnetic field that would protect the planet from potentially life-threatening cosmic radiation and solar radiation; observations made in the late 1990s by the Mars Global Surveyor confirmed this discovery. Scientists speculate that the lack of magnetic shielding helped the solar wind blow away much of Mars's atmosphere over the course of several billion years. After mapping cosmic radiation levels at various depths on Mars, researchers have concluded that any life within the first several meters of the planet's surface would be killed by lethal doses of cosmic radiation. In 2007, it was calculated that DNA and RNA damage by cosmic radiation would limit life on Mars to depths greater than 7.5 metres below the planet's surface. Therefore, the best potential locations for discovering life on Mars may be at subsurface environments that have not been studied yet. The disappearance of the magnetic field may be played a significant role in the process of Martian climate change. According to the valuation of the scientists, the climate of Mars gradually transits from warm and wet to cold and dry after magnetic field vanished.

F

NASA's recent missions have focused on another question: whether Mars held lakes or oceans of liquid water on its surface in the ancient past. Scientists have found hematite, a mineral that forms in the presence of water. Thus, the mission of the Mars Exploration Rovers of 2004 was not to look for present or past life, but for evidence of liquid water on the surface of Mars in the planet's ancient past. Liquid water, necessary for Earth life and for metabolism as generally conducted by species on Earth, cannot exist on the surface of Mars under its present low atmospheric pressure and temperature, except at the lowest shaded elevations for short periods and liquid water does not appear at the surface itself. In March 2004, NASA announced that its rover Opportunity had discovered evidence that Mars was, in the ancient past, a wet planet. This had raised hopes that evidence of past life might be found on the planet today. ESA confirmed that the Mars Express orbiter had directly detected huge reserves of water ice at Mars' south pole in January 2004.

G

Researchers from the Center of Astrobiology (Spain) and the Catholic University of the North in Chile have found an 'oasis' of microorganisms two meters below the surface of the Atacama Desert, SOLID, a detector for signs of life which could be used in environments similar to subsoil on Mars. "We have named it a 'microbial oasis' because we found microorganisms developing in a habitat that was rich in rock salt and other highly hygroscopic compounds that absorb water" explained Victor Parro, a researcher from the Center of Astrobiology in Spain. "If there are similar microbes on Mars or remains in similar conditions to the ones we have found in the

Atacama, we could detect them with instruments like SOLID” Parro highlighted.

H

Even more intriguing, however, is the alternative scenario by Spanish scientists: If those samples could be found to that use DNA, as Earthly life does, as their genetic code. It is extremely unlikely that such a highly specialised, complex molecule like DNA could have evolved separately on the two planets, indicating that there must be a common origin for Martian and Earthly life. Life-based on DNA first appeared on Mars and then spread to Earth, where it then evolved into the myriad forms of plants and creatures that exist today. If this was found to be the case, we would have to face the logical conclusion: we are all Martian. If not, we would continue to search the life of signs.

Questions 27-32

The Reading Passage has seven paragraphs A-H.

Which paragraph contains the following information?

Write the correct letter A-H, in boxes 27-32 on your answer sheet.

- 27 Martian evidence on Earth
- 28 Mars and Earth may share the same life origin
- 29 certain agricultural construction was depicted specifically
- 30 the project which aims to identify life under similar condition of Mars
- 31 Mars had experienced terrifying climate transformation
- 32 Attempts in scientific investigation to find liquid water

答案

- 27. D
- 28. H
- 29. A
- 30. G
- 31. E
- 32. F

Test 4

READING PASSAGE 2

You should spend about 20 minutes on Questions 14-26 which are based on Reading Passage 2 below.

Ancient Storytelling

A

It was told, we suppose, to people crouched around a fire: a tale of adventure, most likely-relating some close encounter with death; a remarkable hunt, an escape from mortal danger; a vision, or something else out of the ordinary. Whatever its thread, the weaving of this story was done with a prime purpose. The listeners must be kept listening. They must not fall asleep. So, as the story went on, its audience should be sustained by one question above all. What happens next?

B

The first fireside stories in human history can never be known. They were kept in the heads of those who told them. This method of storage is not necessarily inefficient. From documented oral traditions in Australia, the Balkans and other parts of the world we know that specialised storytellers and poets can recite from memory literally thousands of lines, in verse or prose, verbatim-word for word. But while memory is rightly considered an art in itself, it is clear that a primary purpose of making symbols is to have a system of reminders or mnemonic cues – signs that assist us to recall certain information in the mind’s eye.

C

In some Polynesian communities, a notched memory stick may help to guide a storyteller through successive stages of recitation. But in other parts of the world, the activity of storytelling historically resulted in the development or even the invention of writing systems. One theory about the arrival of literacy in ancient Greece, for example, argues that the epic tales about the Trojan War and the wanderings of Odysseus – traditionally attributed to Homer – were just so enchanting to hear that they had to be preserved. So the Greeks, c.750-700BC, borrowed an alphabet from their neighbors in the eastern Mediterranean, the Phoenicians.

D

The custom of recording stories on parchment and other materials can be traced in many manifestations around the world, from the priestly papyrus archives of ancient Egypt to the birch-bark scrolls on which the North American Ojibway Indians set down their creation-myth. It is a well-tried and universal practice: so much so that to this day storytime is probably most often associated with words on paper. The formal practice of narrating a story aloud would seem-so we assume to have given way to newspapers, novels and comic strips. This, however, is not the case. Statistically, it is doubtful that the majority of humans currently rely upon the written word to get

access to stories. So what is the alternative source?

E

Each year, over 7 billion people will go to watch the latest offering from Hollywood, Bollywood and beyond. The supreme storyteller of today is cinema. The movies, as distinct from still photography, seem to be an essential modern phenomenon. This is an illusion, for there are, as we shall see, certain ways in which the medium of film is indebted to very old precedents of arranging ‘sequences’ of images. But any account of visual storytelling must be with the recognition that all storytelling beats with a deeply atavistic pulse: that is, a ‘good story’ relies upon formal patterns of plot and characterisation that have been embedded in the practice of storytelling over many generations.

F

Thousands of scripts arrive every week at the offices of the major film studios. But aspiring screenwriters really need to look no further for essential advice than the fourth-century BC Greek Philosopher Aristotle. He left some incomplete lecture notes on the art of telling stories in various literary and dramatic modes, a slim volume known as *The Poetics*. Though he can never have envisaged the popcorn-fuelled actuality of a multiplex cinema, Aristotle is almost prescient about the key elements required to get the crowds flocking to such a cultural hub. He analyzed the process with cool rationalism. When a story enchants us, we lose the sense of where we are; we are drawn into the story so thoroughly that we forget it is a story being told. This is, in Aristotle’s phrase, ‘the suspension of disbelief’.

G

We know the feeling. If ever we have stayed in our seats, stunned with grief, as the credits roll by, or for days after seeing that vivid evocation of horror have been nervous about taking a shower at home, then we have suspended disbelief. We have been caught, or captivated, in the storyteller’s web. Did it all really happen? We really thought so for a while. Aristotle must have witnessed often enough this suspension of disbelief. He taught at Athens, the city where theater developed as a primary form of civic ritual and recreation. Two theatrical types of storytelling, tragedy and comedy, caused Athenian audiences to lose themselves in sadness and laughter respectively. Tragedy, for Aristotle, was particularly potent in its capacity to enlist and then purge the emotions of those watching the story unfold on the stage, so he tried to identify those factors in the storyteller’s art that brought about such engagement. He had, as an obvious sample for analysis, not only the fifth-century BC masterpieces of Classical Greek tragedy written by Aeschylus, Sophocles and Euripides. Beyond them stood Homer, whose stories even then had canonical status: *The Iliad* and *The Odyssey* were already considered literary landmarks—stories by which all other stories should be measured. So what was the secret of Homer’s narrative art?

H

It was not hard to find. Homer created credible heroes. His heroes belonged to the past, they were mighty and magnificent, yet they were not, in the end, fantasy figures. He made his heroes sulk, bicker, cheat and cry. They were, in short, characters – protagonists of a story that an audience would care about, would want to follow,

would want to know what happens next. As Aristotle saw, the hero who shows a human side-some flaw or weakness to which mortals are prone-is intrinsically dramatic.d by logging.

Questions 14-18

The Reading Passage has eight paragraphs A-H

Which paragraph contains the following information?

Write the correct letter A-H, in boxes 14-18 on your answer sheet.

- 14 A misunderstanding of a modern way for telling stories
- 15 The typical forms mentioned for telling stories
- 16 The fundamental aim of storytelling
- 17 A description of reciting stories without any assistance
- 18 How to make story characters attractive

Questions 19-22

Classify the following information as referring to

- A adopted the writing system from another country
- B used organic materials to record stories
- C used tools to help to tell stories

Write the correct letter, A, B or C in boxes 19-22 on your answer sheet.

- 19 Egyptians
- 20 Ojibway
- 21 Polynesians
- 22 Greek

答案

- 14. E
- 15. G
- 16. A
- 17. B
- 18. H
- 19. B
- 20. B
- 21. C
- 22. A

READING PASSAGE 3

You should spend about 20 minutes on Questions 27-40 which are based on Reading Passage 3 below.

Paper of Computer?

A

Computer technology was supposed to replace paper. But that hasn't happened. Every country in the Western world uses more paper today, on a per-capita basis, than it did ten years ago. The consumption of uncoated free-sheet paper, for instance – the most common kind of office paper – rose almost fifteen per cent in the United States between 1995 and 2000. This is generally taken as evidence of how hard it is to eradicate old, wasteful habits and of how stubbornly resistant we are to the efficiencies offered by computerization. A number of cognitive psychologists and ergonomics experts, however, don't agree. Paper has persisted, they argue, for very good reasons: when it comes to performing certain kinds of cognitive tasks, the paper has many advantages over computers. The dismay people feel at the sight of a messy desk – or the spectacle of air-traffic controllers tracking flights through notes scribbled on paper strips – arises from a fundamental confusion about the role that paper plays in our lives.

B

The case for paper is made most eloquently in “The Myth of the Paperless Office”, by two social scientists, Abigail Sellen and Richard Harper. They begin their book with an account of a study they conducted at the International Monetary Fund, in Washington, D.C. Economists at the I.M.F. spend most of their time writing reports on complicated economic questions, work that would seem to be perfectly suited to sitting in front of a computer. Nonetheless, the I.M.F. is awash in paper, and Sellen and Harper wanted to find out why. Their answer is that the business of writing reports – at least at the I.M.F. – is an intensely collaborative process, involving the professional judgments and contributions of many people. The economists bring drafts of reports to conference rooms, spread out the relevant pages, and negotiate changes with one other. They go back to their offices and jot down comments in the margin, taking advantage of the freedom offered by the informality of the handwritten note. Then they deliver the annotated draft to the author in person, taking him, page by page, through the suggested changes. At the end of the process, the author spreads out all the pages with comments on his desk and starts to enter them on the computer – moving the pages around as he works, organizing and reorganizing, saving and discarding.

C

Without paper, this kind of collaborative and iterative work process would be much more difficult. According to Sellen and Harper, the paper has a unique set of “affordances” – that is, qualities that permit specific kinds of uses. Paper is tangible: we can pick up a document, flip through it, read little bits here and there, and quickly get a sense of it. Paper is spatially flexible, meaning that we can spread it out

and arrange it in the way that suits us best. And it's tailorable: we can easily annotate it, and scribble on it as we read, without altering the original text. Digital documents, of course, have their own affordances. They can be easily searched, shared, stored, accessed remotely, and linked to other relevant material. But they lack the affordances that really matter to a group of people working together on a report. Sellen and Harper write:

D

Paper enables a certain kind of thinking. Picture, for instance, the top of your desk. Chances are that you have a keyboard and a computer screen off to one side, and a clear space roughly eighteen inches square in front of your chair. What covers the rest of the desktop is probably piles – piles of papers, journals, magazines, binders, postcards, videotapes, and all the other artefacts of the knowledge economy. The piles look like a mess, but they aren't. When a group at Apple Computer studied piling behavior several years ago, they found that even the most disorderly piles usually make perfect sense to the piler, and that office workers could hold forth in great detail about the precise history and meaning of their piles. The pile closest to the cleared, eighteen-inch-square working area, for example, generally represents the most urgent business, and within that pile, the most important document of all is likely to be at the top. Piles are living, breathing archives. Over time, they get broken down and resorted, sometimes chronologically and sometimes thematically and sometimes chronologically and thematically; clues about certain documents may be physically embedded in the file by, say, stacking a certain piece of paper at an angle or inserting dividers into the stack.

E

But why do we pile documents instead of filing them? Because piles represent the process of active, ongoing thinking. The psychologists Alison Kidd, whose research Sellen and Harper refer to extensively, argues that “knowledge workers” use the physical space of the desktop to hold “ideas which they cannot yet categorize or even decide how they might use.” The messy desk is not necessarily a sign of disorganization. It may be a sign of complexity: those who deal with many unresolved ideas simultaneously cannot sort and file the papers on their desks, because they haven't yet sorted and filed the ideas in their head. Kidd writes that many of the people she talked to use the papers on their desks as contextual cues to “recover a complex set of threads without difficulty and delay” when they come in on a Monday morning, or after their work has been interrupted by a phone call. What we see when we look at the piles on our desks is, in a sense, the contents of our brains.

F

This idea that paper facilitates a highly specialized cognitive and social process is a far cry from the way we have historically thought about the stuff. Paper first began to proliferate in the workplace in the late nineteenth century as part of the move toward “systematic management.” To cope with the complexity of the industrial economy, managers were instituting company-wide policies and demanding monthly weekly, or even daily updates from their subordinates. Thus was born the monthly sales report, and the office manual and the internal company newsletter. The typewriter took off in

the eighteen-eighties, making it possible to create documents in a fraction of the time it had previously taken, and that was followed closely by the advent of carbon paper, which meant that a typist could create ten copies of that document simultaneously. Paper was important not to facilitate creative collaboration and thought but as an instrument of control.

Questions 27-32

The reading passage has seven paragraphs, A-F

Choose the correct heading for paragraphs A-F from the list below.

Write the correct number, i-x, in boxes 27-32 on your answer sheet.

List of headings

- i paper continued as a sharing or managing must
 - ii piles can be more inspiring rather than disorganising
 - iii Favorable situations that economists used paper pages
 - iv overview of an unexpected situation: paper survived
 - v comparison between efficiencies for using paper and using computer
 - vi IMF' s paperless office seemed to be a waste of papers
 - vii example of failure for the avoidance of paper record
 - viii There are advantages of using a paper in offices
 - ix piles reflect certain characteristics in people' s thought
 - x joy of having the paper square in front of a computer
- 27 Paragraph A
28 Paragraph B
29 Paragraph C
30 Paragraph D
31 Paragraph E
32 Paragraph F

答案

27. iv
28. iii
29. viii
30. ii
31. ix
32. i

Test 5

READING PASSAGE 1

You should spend about 20 minutes on Questions 1-13 which are based on Reading Passage 1 below.

What are you laughing at?

A

We like to think that laughing is the height of human sophistication. Our big brains let us see the humour in a strategically positioned pun, an unexpected plot twist or a clever piece of wordplay. But while joking and wit are uniquely human inventions, laughter certainly is not. Other creatures, including chimpanzees, gorillas and even rats, chuckle. Obviously, they don't crack up at Homer Simpson or titter at the boss's dreadful jokes, but the fact that they laugh in the first place suggests that sniggers and chortles have been around for a lot longer than we have. It points the way to the origins of laughter, suggesting a much more practical purpose than you might think.

B

There is no doubt that laughing typical involves groups of people. 'Laughter evolved as a signal to others – it almost disappears when we are alone,' says Robert Provine, a neuroscientist at the University of Maryland. Provine found that most laughter comes as a polite reaction to everyday remarks such as 'see you later', rather than anything particularly funny. And the way we laugh depends on the company we're keeping. Men tend to laugh longer and harder when they are with other men, perhaps as a way of bonding. Women tend to laugh more and at a higher pitch when men are present, possibly indicating flirtation or even submission.

C

To find the origins of laughter, Provine believes we need to look at the play. He points out that the masters of laughing are children, and nowhere is their talent more obvious than in the boisterous antics, and the original context plays,' he says. Well-known primate watchers, including Dian Fossey and Jane Goodall, have long argued that chimps laugh while at play. The sound they produce is known as a panting laugh. It seems obvious when you watch their behavior – they even have the same ticklish spots as we do. But remove the context, and the parallel between human laughter and a chimp's characteristic pant laugh is not so clear. When Provine played a tape of the pant laughs to 119 of his students, for example, only two guessed correctly what it was.

D

These findings underline how chimp and human laughter vary. When we laugh the sound is usually produced by chopping up a single exhalation into a series of shorter with one sound produced on each inward and outward breath. The question is: does this pant laughter have the same source as our own laughter? New research lends

weight to the idea that it does. The findings come from Elke Zimmerman, head of the Institute for Zoology in Germany, who compared the sounds made by babies and chimpanzees in response to tickling during the first year of their life. Using sound spectrographs to reveal the pitch and intensity of vocalizations, she discovered that chimp and human baby laughter follow broadly the same pattern. Zimmerman believes the closeness of baby laughter to chimp laughter supports the idea that laughter was around long before humans arrived on the scene. What started simply as a modification of breathing associated with enjoyable and playful interactions has acquired a symbolic meaning as an indicator of pleasure.

E

Pinpointing when laughter developed is another matter. Humans and chimps share a common ancestor that lived perhaps 8 million years ago, but animals might have been laughing long before that. More distantly related primates, including gorillas, laugh, and anecdotal evidence suggests that other social mammals may do too. Scientists are currently testing such stories with a comparative analysis of just how common laughter is among animals. So far, though, the most compelling evidence for laughter beyond primates comes from research done by Jaak Panksepp from Bowling Green State University, Ohio, into the ultrasonic chirps produced by rats during play and in response to tickling.

F

All this still doesn't answer the question of why we laugh at all. One idea is that laughter and tickling originated as a way of sealing the relationship between mother and child. Another is that the reflex response to tickling is protective, alerting us to the presence of crawling creatures that might harm us or compelling us to defend the parts of our bodies that are most vulnerable in hand-to-hand combat. But the idea that has gained most popularity in recent years is that laughter in response to tickling is a way for two individuals to signal and test their trust in one another. This hypothesis starts from the observation that although a little tickle can be enjoyable if it goes on too long it can be torture. By engaging in a bout of tickling, we put ourselves at the mercy of another individual, and laughing is a signal that we laughter is what makes it a reliable signal of trust according to Tom Flamson, a laughter researcher at the University of California, Los Angeles. 'Even in rats, laughter, tickle, play and trust are linked. Rats chirp a lot when they play,' says Flamson. 'These chirps can be aroused by tickling. And they get bonded to us as a result, which certainly seems like a show of trust.'

G

We'll never know which animal laughed the first laugh, or why. But we can be sure it wasn't in response to a prehistoric joke. The funny thing is that while the origins of laughter are probably quite serious, we owe human laughter and our language-based humor to the same unique skill. While other animals pant, we alone can control our breath well enough to produce the sound of laughter. Without that control, there would also be no speech – and no jokes to endure.

Questions 1-6

Look at the following research findings (questions 1-6) and the list of people below.

Match each finding with the correct person A, B, C or D.

Write the correct letter, A, B, C or D, in boxes 1-6 on your answer sheet.

NB You may use any letter more than once.

- A Tom Flamson
- B Elke Zimmerman
- C Robert Provine
- D Jaak Panksepp

- 1 Babies and chimps produce similar sounds of laughter.
- 2 Primates are not the only animals who produce laughter Pan
- 3 Laughter also suggests that we feel safe and easy with others.
- 4 Laughter is a response to a polite situation instead of humour.
- 5 Animal laughter evolved before human laughter
- 6 Laughter is a social activity.

答案

- 1. B
- 2. D
- 3. A
- 4. C
- 5. B
- 6. C

READING PASSAGE 2

You should spend about 20 minutes on Questions 14-26 which are based on Reading Passage 2 below.

Leaf-Cutting Ants and Fungus

A

The ants and their agriculture have been extensively studied over the years, but the recent research has uncovered intriguing new findings of the fungus they cultivate, how they domesticated it and how they cultivate it and preserve it from pathogens. For example, the fungus farms, which the ants were thought to keep free of pathogens, turn out to be vulnerable to a devastating mold, found nowhere else but in ants' nests. To keep the mold in check, the ants long ago made a discovery that would do credit to any pharmaceutical laboratory.

B

Leaf-cutting ants and their fungus farms are a marvel of nature and perhaps the best-known example of symbiosis, the mutual dependence of two species. The ant's achievement is remarkable – the biologist Edward O. Wilson has called it “one of the major breakthroughs in animal evolution” – because it allows them to eat, courtesy of their mushroom's digestive powers, the otherwise poisoned harvest of tropical forests whose leaves are laden with terpenoids, alkaloids and other chemicals designed to sicken browsers.

C

Fungus growing seems to have originated only once in evolution because all gardening ants belong to a single tribe, the descendants of the first fungus farmer. There are more than 200 known species of the attine ant tribe, divided into 12 groups, or genera. The leaf-cutters use fresh vegetation; the other groups, known as the lower attines because their nests are smaller and their techniques more primitive, feed their gardens with detritus like dead leaves, insects and faeces.

D

The leaf-cutters' fungus was indeed descended from a single strain, propagated clonally, or just by budding, for at least 23 million years. But the lower attine ants used different varieties of the fungus, and in one case a quite separate species, the four biologists discovered. The pure strain of fungus grown by the leaf-cutters, it seemed to Mr Currie, resembled the monocultures of various human crops, that are very productive for a while and then succumb to some disastrous pathogen, such as the Irish potato blight. Monocultures, which lack the genetic diversity to respond to changing environmental threats, are sitting ducks for parasites. Mr Currie felt there had to be a parasite in the ant-fungus system. But a century of ant research offered no support for the idea. Textbooks describe how leaf-cutter ants scrupulously weed their gardens of all foreign organisms. “People kept telling me, ‘You know the ants keep their gardens free of parasites, don't you?’ ” Mr Currie said of his efforts to find a hidden interloper.

E

But after three years of sifting through attine ant gardens, Mr Currie discovered they are far from free of infection. In last month's issue of the Proceedings of the National Academy of Sciences, he and two colleagues, Dr Mueller and David Mairoch, isolated several alien organisms, particularly a family of parasitic molds called Escovopsis.

F

Escovopsis turns out to be a highly virulent pathogen that can devastate a fungus garden in a couple of days. It blooms like a white cloud, with the garden dimly visible underneath. In a day or two, the whole garden is enveloped. "Other ants won't go near it and the ants associated with the garden just starve to death," Dr. Rehner said.

"They just seem to give up, except for those that have rescued their larvae." The deadly mold then turns greenish-brown as it enters its spore-forming stage.

G

Evidently, the ants usually manage to keep Escovopsis and other parasites under control. But with any lapse in control, or if the ants are removed, Escovopsis will quickly burst forth. Although new leaf-cutter gardens start off free of Escovopsis, within two years some 60 percent become infected. The discovery of Escovopsis's role brings a new level of understanding of the evolution of the attine ants. "In the last decade, evolutionary biologists have been increasingly aware of the role of parasites as driving forces in evolution," Dr Schultz said. There is now a possible reason to explain why the lower attine species keep changing the variety of fungus in their mushroom gardens, and occasionally domesticating new ones – to stay one step ahead of the relentless Escovopsis.

H

Interestingly, Mr. Currie found that the leaf-cutters had in general fewer alien molds in their gardens than the lower attines, yet they had more Escovopsis infections. It seems that the price they pay for cultivating a pure variety of fungus is a higher risk from Escovopsis. But the leaf-cutters may have a little alternative: they cultivate a special variety of fungus which, unlike those grown by the lower attines, produces nutritious swollen tips for the ants to eat.

I

Discovery of a third partner in the ant-fungus symbiosis raises the question of how the attine ants, especially the leaf-cutters, keep this dangerous interloper under control. Amazingly enough, Mr Currie has again provided the answer. "People have known for a hundred years that ants have a whitish growth on the cuticle," said Dr Mueller, referring to the insects' body surface. "People would say this is like a cuticular wax. But Cameron was the first one in a hundred years to put these things under a microscope. He was it was not inert wax. It is alive." Mr Currie discovered a specialized patch on the ants' cuticle that harbors a particular kind of bacterium, one well known to the pharmaceutical industry, because it is the source of half the antibiotics used in medicine. From each of 22 species of attine ant studied, Mr. Cameron and colleagues isolated a species of Streptomyces bacterium, they reported in Nature in April. The Streptomyces does not have much effect on ordinary

laboratory funguses. But it is a potent poisoner of Escovopsis, inhibiting its growth and suppressing spore formation. It also stimulates the growth of the ants' mushroom fungus. The bacterium is carried by virgin queens when they leave to establish new nests but is not found on male ants, playboys who take no responsibility in nest-making or gardening.

J

Because both the leaf-cutters and the lower attines use Streptomyces, the bacterium may have been part of their symbiosis for almost as long as the Escovopsis mold. If so, some Alexander Fleming of an ant discovered antibiotics millions of years before people did. Even now, the ants are accomplishing two feats beyond the powers of human technology. The leaf-cutters are growing a monocultural crop year after year without disaster, and they are using an antibiotic apparently so wisely and prudently that, unlike people, they are not provoking antibiotic resistance in the target pathogen.

Questions 14-19

Use the information in the passage to match the options (listed A-C) with activities or features of ants below.

Write the appropriate letters A-C in boxes 14-19 on your answer sheet.

NB You may use any letter more than once

A Leaf-cutting ants

B Lower attines

C Both

- 14 Build small nests and live with the different foreign fungus.
- 15 Use toxic leaves to feed the fungus.
- 16 Raise fungus which doesn't live with other foreigners.
- 17 Use substance to fight against Escovopsis.
- 18 Use dead vegetable to feed the fungus.
- 19 Are free of parasites explained previously?

Questions 20-24

The Reading Passage has ten paragraphs A-J.

Which paragraph contains the following information?

Write the correct letter A-J, in boxes 20-24 on your answer sheet.

- 20 Dangerous outcome of Escovopsis.
- 21 The disadvantage of growing single fungus.
- 22 comparison of features of two different nests.
- 23 Two achievements made by ants earlier than human.
- 24 Advantage of growing a new breed of fungus.

答案

14. B

15. A

16. A

-
- 17. C
 - 18. B
 - 19. A
 - 20. F
 - 21. H
 - 22. C
 - 23. J
 - 24. G
 - 25. A
 - 26. C

Test 6

READING PASSAGE 1

You should spend about 20 minutes on Questions 1-13 which are based on Reading Passage 1 below.

SOSUS: Listening to the Ocean

A

The oceans of Earth cover more than 70 percent of the planet's surface, yet, until quite recently, we knew less about their depths than we did about the surface of the Moon. Distant as it is, the Moon has been far more accessible to study because astronomers long have been able to look at its surface, first with the naked eye and then with the telescope—both instruments that focus light. And, with telescopes tuned to different wavelengths of light, modern astronomers can not only analyze Earth's atmosphere but also determine the temperature and composition of the Sun or other stars many hundreds of light-years away. Until the twentieth century, however, no analogous instruments were available for the study of Earth's oceans: Light, which can travel trillions of miles through the vast vacuum of space, cannot penetrate very far in seawater.

B

Curious investigators long have been fascinated by sound and the way it travels in water. As early as 1490, Leonardo da Vinci observed: "If you cause your ship to stop and place the head of a long tube in the water and place the outer extremity to your ear, you will hear ships at a great distance from you." In 1687, the first mathematical theory of sound propagation was published by Sir Isaac Newton in his *Philosophiae Naturalis Principia Mathematica*. Investigators were measuring the speed of sound in the air beginning in the mid-seventeenth century, but it was not until 1826 that Daniel Colladon, a Swiss physicist, and Charles Sturm, a French mathematician, accurately measured its speed in the water. Using a long tube to listen underwater (as da Vinci had suggested), they recorded how fast the sound of a submerged bell traveled across Lake Geneva. Their result—1,435 meters (1,569 yards) per second in the water of 1.8 degrees Celsius (35 degrees Fahrenheit) — was only 3 meters per second off from the speed accepted today. What these investigators demonstrated was that water — whether fresh or salt — is an excellent medium for sound, transmitting it almost five times faster than its speed in air.

C

In 1877 and 1878, the British scientist John William Strutt, third Baron Rayleigh, published his two-volume seminal work, *The Theory of Sound*, often regarded as marking the beginning of the modern study of acoustics. The recipient of the Nobel Prize for Physics in 1904 for his successful isolation of the element argon, Lord Rayleigh made key discoveries in the fields of acoustics and optics that are critical to

the theory of wave propagation in fluids. Among other things, Lord Rayleigh was the first to describe a sound wave as a mathematical equation (the basis of all theoretical work on acoustics) and the first to describe how small particles in the atmosphere scatter certain wavelengths of sunlight, a principle that also applies to the behavior of sound waves in water.

D

A number of factors influence how far sound travels underwater and how long it lasts. For one, particles in seawater can reflect, scatter, and absorb certain frequencies of sound – just as certain wavelengths of light may be reflected, scattered, and absorbed by specific types of particles in the atmosphere. Seawater absorbs 30 times the amount of sound absorbed by distilled water, with specific chemicals (such as magnesium sulfate and boric acid) damping out certain frequencies of sound. Researchers also learned that low-frequency sounds, whose long wavelengths generally pass over tiny particles, tend to travel farther without loss through absorption or scattering. Further work on the effects of salinity, temperature, and pressure on the speed of sound has yielded fascinating insights into the structure of the ocean. Speaking generally, the ocean is divided into horizontal layers in which sound speed is influenced more greatly by temperature in the upper regions and by pressure in the lower depths. At the surface is a sun-warmed upper layer, the actual temperature and thickness of which varies with the season. At mid-latitudes, this layer tends to be isothermal, that is, the temperature tends to be uniform throughout the layer because the water is well mixed by the action of waves, winds, and convection currents; a sound signal moving down through this layer tends to travel at an almost constant speed. Next comes a transitional layer called the thermocline, in which temperature drops steadily with depth; as the temperature falls, so does the speed of sound.

E

The U.S. Navy was quick to appreciate the usefulness of low-frequency sound and the deep sound channel in extending the range at which it could detect submarines. In great secrecy during the 1950s, the U.S. Navy launched a project that went by the code name Jezebel; it would later come to be known as the Sound Surveillance System (SOSUS). The system involved arrays of underwater microphones, called hydrophones, that were placed on the ocean bottom and connected by cables to onshore processing centers. With SOSUS deployed in both deep and shallow water along both coasts of North America and the British West Indies, the U.S. Navy not only could detect submarines in much of the northern hemisphere, it also could distinguish how many propellers a submarine had, whether it was conventional or nuclear, and sometimes even the class of sub.

F

The realization that SOSUS could be used to listen to whales also was made by Christopher Clark, a biological acoustician at Cornell University, when he first visited a SOSUS station in 1992. When Clark looked at the graphic representations of sound, scrolling 24 hours day, every day, he saw the voice patterns of blue, finback, minke, and humpback whales. He also could hear the sounds. Using a SOSUS receiver in the

West Indies, he could hear whales that were 1,770 kilometers (1,100 miles) away. Whales are the biggest of Earth's creatures. The blue whale, for example, can be 100 feet long and weigh as many tons. Yet these animals also are remarkably elusive. Scientists wish to observe blue whales and position them on a map. Moreover, they can track not just one whale at a time, but many creatures simultaneously throughout the North Atlantic and the eastern North Pacific. They also can learn to distinguish whale calls. For example, Fox and colleagues have detected changes in the calls of finback whales during different seasons and have found that blue whales in different regions of the Pacific Ocean have different calls. Whales firsthand must wait in their ships for the whales to surface. A few whales have been tracked briefly in the wild this way but not for very great distances, and much about them remains unknown. Using the SOSUS stations, scientists can track the whales in real-time and position them on a map. Moreover, they can track not just one whale at a time, but many creatures simultaneously throughout the North Atlantic and the eastern North Pacific. They also can learn to distinguish whale calls. For example, Fox and colleagues have detected changes in the calls of finback whales during different seasons and have found that blue whales in different regions of the Pacific Ocean have different calls.

G

SOSUS, with its vast reach, also has proved instrumental in obtaining information crucial to our understanding of Earth's weather and climate. Specifically, the system has enabled researchers to begin making ocean temperature measurements on a global scale – measurements that are keys to puzzling out the workings of heat transfer between the ocean and the atmosphere. The ocean plays an enormous role in determining air temperature – the heat capacity in only the upper few meters of the ocean is thought to be equal to all of the heat in the entire atmosphere. For sound waves traveling horizontally in the ocean, speed is largely a function of temperature. Thus, the travel time of a wave of sound between two points is a sensitive indicator of the average temperature along its path. Transmitting sound in numerous directions through the deep sound channel can give scientists measurements spanning vast areas of the globe. Thousands of sound paths in the ocean could be pieced together into a map of global ocean temperatures and, by repeating measurements along the same paths overtimes, scientists could track changes in temperature over months or years.

H

Researchers also are using other acoustic techniques to monitor climate. Oceanographer Jeff Nystuen at the University of Washington, for example, has explored the use of sound to measure rainfall over the ocean. Monitoring changing global rainfall patterns undoubtedly will contribute to understanding major climate change as well as the weather phenomenon known as El Niño. Since 1985, Nystuen has used hydrophones to listen to rain over the ocean, acoustically measuring not only the rainfall rate but also the rainfall type, from drizzle to thunderstorms. By using the sound of rain underwater as a “natural” rain gauge, the measurement of rainfall over the oceans will become available to climatologists.

Questions 5-8

The Reading Passage has seven paragraphs A-H

Which paragraph contains the following information?

Write the correct letter A-H, in boxes 5-8 on your answer sheet.

NB You may use any letter more than once

- 5 Elements affect sound transmission in the ocean.
- 6 Relationship between global climate and ocean temperature
- 7 Examples of how sound technology help people research ocean and creatures in it
- 8 Sound transmission underwater is similar to that of light in any condition.

答案

- 5. D
- 6. G
- 7. F
- 8. D

READING PASSAGE 2

You should spend about 20 minutes on Questions 14-26 which are based on Reading Passage 2 below.

Left-handed or Right-handed

Section A

The probability that two right-handed people would have a left-handed child is only about 9.5 percent. The chance rises to 19.5 percent if one parent is a lefty and 26 percent if both parents are left-handed: The preference, however, could also stem from an infant's imitation of his parents. To test genetic influence, starting in the 1970s British biologist Marian Annett of the University of Leicester hypothesized that no single gene determines handedness. Rather, during fetal development, a certain molecular factor helps to strengthen the brain's left hemisphere, which increases the probability that the right hand will be dominant because the left side of the brain controls the right side of the body, and vice versa. Among the minority of people who lack this factor, handedness develops entirely by chance.

Research conducted on twins complicates the theory, however. One in five sets of identical twins involves one right-handed and one left-handed person, despite the fact that their genetic material is the same. Genes, therefore, are not solely responsible for handedness.

Section B

The genetic theory is also undermined by results from Peter Hepper and his team at Queen's University in Belfast, Ireland. In 2004 the psychologists used ultrasound to show that by the 15th week of pregnancy, fetuses already have a preference as to which thumb they suck. In most cases, the preference continued after birth. At 15 weeks, though, the brain does not yet have control over the body's limbs. Hepper speculates that fetuses tend to prefer whichever side of the body is developing quicker and that their movements, in turn, influence the brain's development. Whether this early preference is temporary or holds up throughout development and infancy is unknown. Genetic predetermination is also contradicted by the widespread observation that children do not settle on either their right or left hand until they are two or three years old.

Section C

But even if these correlations were true, they did not explain what actually causes left-handedness. Furthermore, specialization on either side of the body is common among animals. Cats will favor one paw over another when fishing toys out from under the couch. Horses stomp more frequently with one hoof than the other. Certain crabs motion predominantly with the left or right claw. In evolutionary terms, focusing power and dexterity in one limb is more efficient than having to train two, four or even eight limbs equally. Yet for most animals, the preference for one side or the other is seemingly random. The overwhelming dominance of the right hand is associated only with humans. That fact directs attention toward the brain's two hemispheres and perhaps toward language.

Section D

Interest in hemispheres dates back to at least 1836. That year, at a medical conference, French physician Marc Dax reported on an unusual commonality among his patients. During his many years as a country doctor, Dax had encountered more than 40 men and women for whom speech was difficult, the result of some kind of brain damage. What was unique was that every individual suffered damage to the left side of the brain. At the conference, Dax elaborated on his theory, stating that each half of the brain was responsible for certain functions and that the left hemisphere controlled speech. Other experts showed little interest in the Frenchman's ideas.

Over time, however, scientists found more and more evidence of people experiencing speech difficulties following an injury to the left brain. Patients with damage to the right hemisphere most often displayed disruptions in perception or concentration. Major advancements in understanding the brain's asymmetry were made in the 1960s as a result of so-called split-brain surgery, developed to help patients with epilepsy. During this operation, doctors severed the corpus callosum – the nerve bundle that connects the two hemispheres. The surgical cut also stopped almost all normal communication between the two hemispheres, which offered researchers the opportunity to investigate each side's activity.

Section E

In 1949 neurosurgeon Juhn Wada devised the first test to provide access to the brain's functional organization of language. By injecting an anesthetic into the right or left carotid artery, Wada temporarily paralyzed one side of a healthy brain, enabling him to more closely study the other side's capabilities. Based on this approach, Brenda Milner and the late Theodore Rasmussen of the Montreal Neurological Institute published a major study in 1975 that confirmed the theory that country doctor Dax had formulated nearly 140 years earlier: in 96 percent of right-handed people, language is processed much more intensely in the left hemisphere. The correlation is not as clear in lefties, however. For two-thirds of them, the left hemisphere is still the most active language processor. But for the remaining third, either the right side is dominant or both sides work equally, controlling different language functions.

That last statistic has slowed acceptance of the notion that the predominance of right-handedness is driven by left-hemisphere dominance in language processing. It is not at all clear why language control should somehow have dragged the control of body movement with it. Some experts think one reason the left hemisphere reigns over language is that the organs of speech processing – the larynx and tongue – are positioned on the body's symmetry axis. Because these structures were centered, it may have been unclear, in evolutionary terms, which side of the brain should control them, and it seems unlikely that shared operation would result in smooth motor activity.

Language and handedness could have developed preferentially for very different reasons as well. For example, some researchers, including evolutionary psychologist Michael C. Corballis of the University of Auckland in New Zealand, think that the origin of human speech lies in gestures. Gestures predated words and helped language emerge. If the left hemisphere began to dominate speech, it would have dominated

gestures, too, and because the left brain controls the right side of the body, the right hand developed more strongly.

Section F

Perhaps we will know more soon. In the meantime, we can revel in what, if any, differences handedness brings to our human talents. Popular wisdom says right-handed, left-brained people excel at logical, analytical thinking. Left-handed, right-brained individuals are thought to possess more creative skills and maybe better at combining the functional features emergent on both sides of the brain. Yet some neuroscientists see such claims as pure speculation. Fewer scientists are ready to claim that left-handedness means greater creative potential. Yet lefties are prevalent among artists, composers and the generally acknowledged great political thinkers. Possibly if these individuals are among the lefties whose language abilities are evenly distributed between hemispheres, the intense interplay required could lead to unusual mental capabilities.

Section G

Or perhaps some lefties become highly creative because they must be more clever to get by in our right-handed world. This battle, which begins during the very early stages of childhood, may lay the groundwork for exceptional achievements.

Questions 14-18

The Reading Passage has seven paragraphs A-G

Which paragraph contains the following information?

Write the correct letter A-G, in boxes 14-18 on your answer sheet.

NB You may use any letter more than once.

- 14 The phenomenon of using one side of their body for animals.
- 15 Statistics on the rate of one-handedness born.
- 16 The age when the preference for using one hand is fixed.
- 17 great talents of occupations in the left-handed population.
- 18 The earliest record of researching hemisphere's function

Questions 19-22

Look at the following researchers (Questions 19-22) and the list of findings below.

Match each researcher with the correct finding

A Brenda Milner

B Marian Annett

C Peter Hepper

D Michale Corballis

- 19 Ancient language evolution is connected to body gesture and therefore influences handedness.
- 20 A child handedness is not determined by just biological factors.
- 21 Language process is generally undergoing in the left hemisphere of the brain.
- 22 The rate of development of one side of the body has an influence on hemisphere preference in the fetus.

答案

14. C

15. A

16. B

17. F

18. D

19. D

20. B

21. A

22. C

READING PASSAGE 3

You should spend about 20 minutes on Questions 27-40 which are based on Reading Passage 3 below.

The Power of Nothing

Geoff Watts, New Scientist (May 26th, 2001)

A

Want to devise a new form of alternative medicine? No problem. Here is the recipe. Be warm, sympathetic, reassuring and enthusiastic. Your treatment should involve physical contact, and each session with your patients should last at least half an hour. Encourage your patients to take an active part in their treatment and understand how their disorders relate to the rest of their lives. Tell them that their own bodies possess the true power to heal. Make them pay you out of their own pockets. Describe your treatment in familiar words, but embroidered with a hint of mysticism: energy fields, energy flows, energy blocks, meridians, forces, auras, rhythms and the like. Refer to the knowledge of an earlier age: wisdom carelessly swept aside by the rise and rise of blind, mechanistic science. Oh, come off it, you are saying. Something invented off the top of your head could not possibly work, could it?

B

Well yes, it could – and often well enough to earn you a living. A good living if you are sufficiently convincing, or better still, really believe in your therapy. Many illnesses get better on their own, so if you are lucky and administer your treatment at just the right time you will get the credit. But that's only part of it. Some of the improvement really would be down to you. Your healing power would be the outcome of a paradoxical force that conventional medicine recognizes but remains oddly ambivalent about: the placebo effect.

C

Placebos are treatments that have no direct effect on the body, yet still, work because the patient has faith in their power to heal. Most often the term refers to a dummy pill, but it applies just as much to any device or procedure, from a sticking plaster to a crystal to an operation. The existence of the placebo effect implies that even quackery may confer real benefits, which is why any mention of placebo is a touchy subject for many practitioners of complementary and alternative medicine, who are likely to regard it as tantamount to a charge of charlatanism. In fact, the placebo effect is a powerful part of all medical care, orthodox or otherwise, though its role is often neglected or misunderstood.

D

One of the great strengths of CAM may be its practitioners' skill in deploying the placebo effect to accomplish real healing. "Complementary practitioners are miles better at producing non-specific effects and good therapeutic relationships," says Edzard Ernst, professor of CAM at Exeter University. The question is whether CAM could be integrated into conventional medicines, as some would like, without losing much of this power.

E

At one level, it should come as no surprise that our state of mind can influence our physiology: anger opens the superficial blood vessels of the face; sadness pumps the tear glands. But exactly how placebos work their medical magic is still largely unknown. Most of the scant research done so far has focused on the control of pain because it's one of the commonest complaints and lends itself to experimental study. Here, attention has turned to the endorphins, morphine-like neurochemicals known to help control pain.

F

But exactly how placebos work their medical magic is still largely unknown. Most of the scant research to date has focused on the control of pain because it's one of the commonest complaints and lends itself to experimental study. Here, attention has turned to the endorphins, natural counterparts of morphine that are known to help control pain. "Any of the neurochemicals involved in transmitting pain impulses or modulating them might also be involved in generating the placebo response," says Don Price, an oral surgeon at the University of Florida who studies the placebo effect in dental pain.

G

"But endorphins are still out in front." That case has been strengthened by the recent work of Fabrizio Benedetti of the University of Turin, who showed that the placebo effect can be abolished by a drug, naloxone, which blocks the effects of endorphins. Benedetti induced pain in human volunteers by inflating a blood-pressure cuff on the forearm. He did this several times a day for several days, using morphine each time to control the pain. On the final day, without saying anything, he replaced the morphine with a saline solution. This still relieved the subjects' pain: a placebo effect. But when he added naloxone to the saline the pain relief disappeared. Here was direct proof that placebo analgesia is mediated, at least in part, by these natural opiates.

H

Still, no one knows how belief triggers endorphin release, or why most people can't achieve placebo pain relief simply by willing it. Though scientists don't know exactly how placebos work, they have accumulated a fair bit of knowledge about how to trigger the effect. A London rheumatologist found, for example, that red dummy capsules made more effective painkillers than blue, green or yellow ones. Research on American students revealed that blue pills make better sedatives than pink, a colour more suitable for stimulants. Even branding can make a difference: if Aspro or Tylenol is what you like to take for a headache, their chemically identical generic equivalents may be less effective.

I

It matters, too, how the treatment is delivered. Decades ago, when the major tranquilliser chlorpromazine was being introduced, a doctor in Kansas categorised his colleagues according to whether they were keen on it, openly skeptical of its benefits, or took a "let's try and see" attitude. His conclusion: the more enthusiastic the doctor, the better the drug performed. And this year Ernst surveyed published studies that compared doctors' bedside manners. The studies turned up one consistent finding:

“Physicians who adopt a warm, friendly and reassuring manner,” he reported, “are more effective than those whose consultations are formal and do not offer reassurance.”

J

Warm, friendly and reassuring are precisely CAM’s strong suits, of course. Many of the ingredients of that opening recipe – the physical contact, the generous swathes of time, the strong hints of supernormal healing power – are just the kind of thing likely to impress patients. It’s hardly surprising, then, that complementary practitioners are generally best at mobilising the placebo effect, says Arthur Kleinman, professor of social anthropology at Harvard University.

Questions 27-32

Use the information in the passage to match the deed (listed A-H) with people below. Write the appropriate letters A-H in boxes 27-32 on your answer sheet.

NB You may use any letter more than once

- | | |
|---|---|
| A | Should easily be understood |
| B | Should improve by itself |
| C | Should not involve any mysticism |
| D | Ought to last a minimum length of time. |
| E | Needs to be treated at the right time. |
| F | Should give more recognition. |
| G | Can earn valuable money. |
| H | Do not rely on any specific treatment |
- 27 Appointments with an alternative practitioner
28 An alternative practitioner’s description of the treatment
29 An alternative practitioner who has faith in what he does
30 the illness of patients convinced of alternative practice
31 Improvements of patients receiving alternative practice
32 Conventional medical doctors (who is aware of placebo)

答案

27. D
28. A
29. G
30. B
31. H
32. F

Test 7

READING PASSAGE 2

You should spend about 20 minutes on Questions 14-26 which are based on Reading Passage 2 below.

Griffith and American films

Movies are key cultural artefacts that offer a window into American cultural and social history. A mixture of art, business, and popular entertainment, the popular entertainment, the movies provide a host of insights into Americans' shifting ideas, fantasies, and preoccupations

A

Many films of the early silent era dealt with gender relations. Before 1905, as Kathy Peiss has argued, movie screens were filled with salacious sexual imagery and risqué humor, drawn from burlesque halls and vaudeville theaters. Early films offered many glimpses of women disrobing or of passionate kisses. As the movies' female audience grew, sexual titillation and voyeurism persisted. But an ever-increasing number of the film dealt with the changing work and sexual roles of women in a more sophisticated manner. While D.W. Griffith's films presented an idealized picture of the frail Victorian child-woman and showed an almost obsessive preoccupation with female honor and chastity, other silent movies presented quite different images of femininity. These ranged from the exotic, sexually aggressive vamp to the athletic, energetic "serial queen"; the street smart urban working gal, who repels the sexual advances of her lascivious boss; and cigarette-smoking, alcohol drinking chorus girls or burlesque queens.

B

In early 1910, director D.W. Griffith was sent by the Biograph Company to the west coast with his acting troupe, consisting of actors Blanche Sweet, Lillian Gish, Mary Pickford, Lionel Barrymore, and others. While there, the company decided to explore new territories, traveling several miles north to Hollywood, a little village that was friendly and enjoyed the movie company filming there. By focusing the camera on particular actors and actresses, Griffith inadvertently encouraged the development of the star system. As early as 1910, newspapers were deluged with requests for actors' names. But most studios refused to divulge their identities, fearing the salary demands of popular performers. As one industry observer put it, "In the 'star' your producer gets not only a 'production' value ... but a 'trademark' value, and an 'insurance' value which are ... very potent in guaranteeing the sale of this product." As the star system emerged, salaries soared. In the course of just two years, the salary of actress Mary Pickford rose from less than \$400 a week in 1914 to \$10,000 a week in 1916. This action made Griffith believe the big potential in the movie industry. Thus many competitors completely copy the same system as Griffith used, for the considerable

profits. Additionally, they also study the theory and methods which Griffith suggested.

C

From the moment America entered the war, Hollywood feared that the industry would be subject to heavy-handed government censorship. But the government itself wanted no repeat of World War I, when the Committee on Public Information had whipped up anti-German hysteria and oversold the war as “a Crusade not merely to re-win the tomb of Christ, but to bring back to earth the rule of right, the peace, goodwill to men and gentleness he taught.”

D

The formation of the movie trust ushered in a period of rationalization within the film industry. Camera and projecting equipment were standardized; film rental fees were fixed; theaters were upgraded; which improved the quality of movies by removing damaged prints from circulation. This was also a period of intense artistic and technical innovation, as pioneering directors like David Wark Griffith and others created a new language of film and revolutionized screen narrative.

E

With just six months of film experience, Griffith, a former stage actor, was hired as a director by the Biograph Company and promised \$50 a week and one-twentieth of a cent for every foot of film sold to a rental exchange. Each week, Griffith turned out two or three one-reelers. While earlier directors had used such cinematic devices as close-ups, slow motion, fade-ins and fade-outs, lighting effects, and editing before, Griffith’s great contribution to the movie industry was to show how these techniques could be used to create a wholly new style of storytelling, distinct from the theater. Griffith’s approach to movie storytelling has been aptly called “photographic realism.” This is not to say that he merely wished to record a story accurately; rather he sought to convey the illusion of realism. He demanded that his performers act less in a more lifelike manner, avoiding the broad, exaggerated gestures and pantomiming of emotions that characterized the nineteenth-century stage. He wanted his performers to take on a role rather than directly addressing the camera.

Above all, he used close-ups, lighting, editing, and other cinematic techniques convey suspense and other emotions and to focus the audience’s attention on individual performers.

F

During the 1920s and 1930s, a small group of film companies consolidated their control. Known as the “Big Five” – Paramount, Warner Brothers, RKO, 20th Century-Fox, and Lowe’s (MGM) and the “Little Three” – Universal, Columbia, and United Artists, they formed fully integrated companies. The old film company’s opposition was shocked by new tycoons. The confusion of tongues in the foreign version of American films deepened when American directors themselves embarked on the shooting of the new version. They did not usually speak Spanish (or the given target language) and, at that time, there were only a few translators at the studio’s disposal. For this reason, it was more general to contract Spanish directors, actors, and screenwriters to produce American films in Spanish for Latin American audiences and for the public in the Iberian Peninsula. Hollywood had depended on overseas markets

for as much as 40 percent of its revenue. But in an effort to nurture their own film industries and prevent an excessive outflow of dollars, Britain, France, and Italy imposed stiff import tariffs and restrictive quotas on imported American movies.

G

A basic problem facing today's Hollywood is the rapidly rising cost of making and marketing a movie: an average of \$40 million today. The immense cost of producing movies has led the studios to seek guaranteed hits: blockbuster loaded with high-tech special effects, sequels, and remakes of earlier movies, foreign films, and even old TV shows. Hollywood has also sought to cope with rising costs by focusing ever more intently on its core audiences. Since the mid-1980s, the movie-going audience has continued to decrease in size. Ticket sales fell from 1.2 billion in 1983 to 950 million in 1992, with the biggest drop occurring among adults. And since over half of Hollywood's profits are earned overseas, the target market has to be changed due to the increasing costs and salary of making a film. The industry has concentrated much of its energy on crude action films easily understood by an international audience, featuring stars like Arnold Schwarzenegger and Sylvester Stallone.

Questions 14-19

The Reading Passage 2 has six paragraphs A-F

Choose the correct heading for each paragraph from the list of headings below.

Write the correct number, i-vii, in boxes 14-19 on your answer sheet.

List of Headings

- i Detailed description for a film system
 - ii Griffith's contribution to American films
 - iii The gender in the development of American film
 - iv Change the view of the American movie
 - v People's reaction to making movies in the war period
 - vi The increasing market of the film in society
 - vii Griffith improved gender recognition in society
- 14 Paragraph A
 - 15 Paragraph B
 - 16 Paragraph C
 - 17 Paragraph D
 - 18 Paragraph E
 - 19 Paragraph F

Questions 20-23

Use the information in the passage to match the companies (listed A-C) with opinions or deeds below.

Write the appropriate letters A, B, C or D in boxes 20-23 on your answer sheet.

- A old company' s opposition
 - B huge drop happens among adults
 - C the pressure to change its market
 - D completely copy his system
- 20 Griffith' s successful in the 1910s, led his rivals
- 21 The growing costs and salary in Hollywood which shows it has
- 22 The increasing new movie industries have a big impact on
- 23 In 1992, ticket sales declined dramatically, due to

答案

- 14. iii
- 15. i
- 16. v
- 17. iv
- 18. ii
- 19. vi
- 20. D
- 21. C
- 22. A
- 23. B

Test 8

READING PASSAGE 1

You should spend about 20 minutes on Questions 1-13 which are based on Reading Passage 1 below.

Origin of Species&Continent Formation

A

THE FACT THAT there was once a Pangean supercontinent, a Panthalassa Ocean, and a Tethys Ocean, has profound implications for the evolution of multicellular life on Earth. These considerations were unknown to the scientists of the 19th century – making their scientific deductions even more remarkable. Quite independently of each other, Charles Darwin and his young contemporary Alfred Russel Wallace reached the conclusion that life had evolved by natural selection. Wallace later wrote in *My Life* of his own inspiration:

B

Why do some species die and some live? The answer was clearly that on the whole the best fitted lived. From the effects of disease the most healthy escaped; from enemies the strongest, the swiftest or the most cunning from famine the best hunters – then it suddenly flashed on me that this self-acting process would improve the race, because in every generation the inferior would inevitably be killed off and the superior would remain, that is, the fittest would survive.

C

Both Darwin's and Wallace's ideas about natural selection had been influenced by the essays of Thomas Malthus in his *Principles of Population*. Their conclusion, however, had been the direct result of their personal observation of animals and plants in widely separated geographic locations: Darwin from his experiences during the voyage of the *Beagle*, and particularly during the ship's visit to the Galapagos Islands in the East Pacific in 1835; Wallace during his years of travel in the Amazon Basin and in the Indonesia-Australian Archipelago in the 1850s.

D

Darwin had been documenting his ideas on natural selection for many years when he received a paper on this selfsame subject from Wallace, who asked for Darwin's opinion and help in getting it published. In July 1858, Charles Lyell and J. D Hooker, close friends of Darwin, pressed Darwin to present his conclusions so that he would not lose priority to an unknown naturalist. Presiding over the hastily called but now historic meeting of the Linnean Society in London, Lyell and Hooker explained to the distinguished members how "these two gentlemen" (who were absent: Wallace was abroad and Darwin chose not to attend), had "independently and unknown to one another, conceived the same very ingenious theory,"

E

Both Darwin and Wallace had realized that the anomalous distribution of species in particular regions had profound evolutionary significance. Subsequently, Darwin spent the rest of his days in almost total seclusion thinking and writing mainly about the origin of species. In contrast, Wallace applied himself to the science of biogeography, the study of the pattern and distribution of species, and its significance, resulting in the publication of a massive two-volume work the *Geographical Distribution of Animals* in 1876.

F

Wallace was a gentle and modest man, but also persistent and quietly courageous. He spent years working in the most arduous possible climates and terrains, particularly in the Malay archipelago, he made patient and detailed zoological observations and collected a huge number of specimens for museums and collectors-which is how he made a living. One result of his work was the conclusion that there is a distinct faunal boundary, called “Wallace’s line,” between an Asian realm of animals in Java, Borneo and the Philippines and an Australian realm in New Guinea and Australia. In essence, this boundary posed a difficult question: How on Earth did plants and animals with a clear affinity to the Northern Hemisphere meet with their Southern Hemispheric counterparts along such a distinct Malaysian demarcation zone? Wallace was uncertain about demarcation on one particular island-Celebes, a curiously shaped place that is midway between the two groups. Initially, he assigned its flora-fauna to the Australian side of the line, but later he transferred it to the Asian side. Today we know the reason for his dilemma. 200MYA East and West Celebes were islands with their own natural history lying on opposite sides of the Tethys Ocean. They did not collide until about 15 MYA. The answer to the main question is that Wallace’s Line categorizes Laurasia-derived flora-fauna (the Asian) and Gondwana-derived flora-fauna (the Australian), fauna that had evolved on opposing shores of the Tethys. The closure of the Tethys Ocean today is manifested by the ongoing collision of Australia/New Guinea with Indochina/Indonesia and the continuing closure of the Mediterranean Sea – a remnant of the Western Tethys Ocean.

G

In his origin of continents and oceans, Wegener quoted at length from Wallace’s *Geographical Distribution of Animals*. According to Wegener’s reading, Wallace had identified three clear divisions of Australian animals, which supported his own theory of continental displacement. Wallace had shown that animals long established in southwestern Australia had an affinity with animals in South Africa, Madagascar, India, and Ceylon, but did not have an affinity with those in Asia. Wallace also showed that Australian marsupials and monotremes are clearly related to those in South America, the Moluccas, and various Pacific islands and that none are found in neighboring Indonesia. From this and related data, Wegener concluded that the then broadly accepted “landbridge” theory could not account for this distribution of animals and that only this theory of continental drift could explain it.

H

The theory that Wegener dismissed in preference to his own proposed that plants and animals had once migrated across now-submerged intercontinental landbridges. In

1885, one of Europe's leading geologists, Eduard Suess, theorized that as the rigid Earth cools, its upper-crust shrinks and wrinkles like the withering skin of an aging apple. He suggested that the planet's seas and oceans now fill the wrinkles between once-contiguous plateaus.

I

Today, we know that we live on a dynamic Earth with shifting, colliding and separating tectonic plates, not a “withering skin”, and the main debate in the field of biogeography has shifted. The discussion now concerns “dispersalism” versus “vicarianism”: unrestricted radiation of species on the one hand and the development of barriers to migration on the other. Dispersion is a short-term phenomenon – the daily or seasonal migration of species and their radiation to the limits of their natural environment on an extensive and continuous landmass. Vicarian evolution, however, depends upon the separation and isolation of a variety of species within the confines of natural barriers in the form of islands, lakes, or shallow seas – topographical features that take a long time to develop.

Questions 1-5

Use the information in the passage to match the people (listed A-E) with opinions or deeds below.

Write the appropriate letters A-E in boxes 1-5 on your answer sheet.

- | | |
|---|--------------------|
| A | Suess |
| B | Wallace |
| C | Darwin and Wallace |
| D | Wegener |
| E | Lyell and Hooker |
- 1 urged Darwin to publish his scientific findings
 - 2 Depicted physical feature of earth's crust.
 - 3 believed in continental drift theory while rejecting another one
 - 4 Published works about wildlife distribution in a different region.
 - 5 Evolution of species is based on selection by nature.

答案

1. E
2. A
3. D
4. B
5. C

READING PASSAGE 2

You should spend about 20 minutes on Questions 14-26 which are based on Reading Passage 2 below.

Western Immigration of Canada

A

By the mid-1870s Canada wanted an immigrant population of agricultural settlers established in the West. No urban centres existed on the prairies in the 1870s, and rural settlement was the focus of the federal government's attention. The western rural settlement was desired, as it would provide homesteads for the sons and daughters of eastern farmers, as eastern agricultural land filled to capacity. As well, eastern farmers and politicians viewed western Canada, with its broad expanses of unpopulated land, as a prime location for expanding Canada's agricultural output, especially in terms of wheat production to serve the markets of eastern Canada.

B

To bolster Canada's population and agricultural output, the federal government took steps to secure western land. The Dominion of Canada purchased Rupert's Land from the Hudson's Bay Company in 1870. In 1872, the federal government enacted the Dominion Lands Act. This act enabled settlers to acquire 160 acres of free land, as long as settlers remained on their land for a period of three years, made certain minor improvements to the land, and paid a \$10.00 registration fee. The Canadian government also created a Mounted Police Force in 1873. The Mounties journeyed west to secure the area for future settlers. By 1876 the NWMP had established themselves in the West. The major posts included Swan River, Fort Saskatchewan, Fort Calgary, Fort Walsh and Fort Macleod. All of these initiatives attracted a number of eastern-Canadian settlers, as well as European and American immigrants, to Canada's West, and particularly to the area of Manitoba.

C

The surest way to protect Canadian territory, and to achieve the secondary goal for joining British Columbia to the rest of the country, was to import large numbers of Eastern Canadian and British settlers. Settling the West also made imperative the building of a transcontinental railway. The railway would work to create an east-west economy, in which western Canada would feed the growing urban industrial population of the east, and in return become a market for eastern Canadian manufactured goods.

D

Winnipeg became the metropolis of the West during this period. Winnipeg's growth before 1900 was the result of a combination of land speculation, growth of housing starts, and the federal government's solution in 1881 of Winnipeg as a major stop along the CPR. This decision culminated in a land boom between 1881 and 1883 which resulted in the transformation of hamlets like Portage la Prairie and Brandon into towns, and a large increase in Manitoba's population. Soon, Winnipeg stood at the junction of three transcontinental railway lines which employed thousands in rail

yards. Winnipeg also became the major processor of agricultural products for the surrounding hinterland.

E

The majority of settlers to Winnipeg, and the surrounding countryside, during this early period, were primarily Protestant English-speaking settlers from Ontario and the British Isles. These settlers established Winnipeg upon a British-Ontarian ethos which came to dominate the society's social, political, and economic spirit. This British-Ontarian ethnic homogeneity, however, did not last very long. Increasing numbers of foreign immigrants, especially from Austria-Hungary and Ukraine soon added a new ethnic element to the recent British, the older First Nation Métis, and Selkirk's settler population base. Settling the West with (in particular) Eastern Canadians and British immigrant offered the advantage of safeguarding the 49th parallel from the threat of American take-over, had not the Minnesota legislature passed a resolution which provided for the annexation of the Red River district. The Red River in 1870 was the most important settlement on the Canadian prairies. It contained 11,963 inhabitants of whom 9,700 were Métis and First Nations. But neighbouring Minnesota already had a population of over 100,000.

F

Not all of the settlers who came to western Canada in the 1880s, however, desired to remain there. In the 1870s and 1880s, economic depression kept the value of Canada's staple exports low, which discouraged many from permanent settlement in the West. Countries including Brazil, Argentina, Australia, New Zealand and the United States competed with Canada for immigrants. Many immigrants and thousands of Canadians chose to settle in the accessible and attractive American frontier. Canada before 1891 has been called "a huge demographic railway station" where thousands of men, women, and children were constantly going and coming, and where the number of departures invariably exceeded that of arrivals."

G

By 1891 Eastern Canada had its share of both large urban centres and problems associated with city life. While the booming economic centres of Toronto and Montreal were complete with electricity and telephones in the cities' wealthiest areas by the turn of the century, slum conditions characterised the poorest areas like the district known as 'the Ward' in Toronto. Chickens and pigs ran through the streets; privy buckets spilled onto backyards and lanes creating cesspools in urban slums. These same social reformers believed that rural living, in stark contrast to urban, would lead to a healthy, moral, and charitable way of life. Social reformers praised the ability of fresh air, hard work, and open spaces for 'Canadianizing' immigrants. Agricultural pursuits were seen as especially fitting for attaining this 'moral' and family-oriented way of life, in opposition to the single male-dominated atmosphere of the cities. Certainly, agriculture played an important part in the Canadian economy in 1891. One-third of the workforce worked on farms.

H

The Canadian government presented Canada's attractions to potential overseas migrants in several ways. The government offered free or cheap land to potential

agriculturists. As well, the government established agents and/or agencies for the purpose of attracting emigrants overseas. Assisted passage schemes, bonuses and commissions to agents and settlers and pamphlets also attracted some immigrants to Canada. The most influential form of attracting others to Canada, however, remained the letters home written by emigrants already in Canada. Letters from trusted friends and family members. Letters home often contained exaggerations of the ‘wonder of the new world.’ Migrant workers and settlers already in Canada did not want to disappoint, or worry, their family and friends at home. Embellished tales of good fortune and happiness often succeeded in encouraging others to come.

Questions 14-20

The Reading Passage has eight paragraphs A-H

Choose the correct heading for paragraphs A-H from the list below.

Write the correct number, i-xii, in boxes 14-20 on your answer sheet

List of Headings

- i Not all would stay in Canada forever
- ii Government’s safeguard in the West
- iii Eastern Canada is full
- iv Built-up to the new infrastructure
- v An exclusive British domination in Ontario established ever since
- vi Ethnic and language make-up
- vii Pursuing a pure life
- viii Police recruited from mid-class families
- ix Demand of western immigration
- x Early major urban development of the West
- xi Attracting urban environment
- xii Advertising of Western Canada

Example: Paragraph A ix

- 14 Paragraph B
- 15 Paragraph C
- 16 Paragraph D
- 17 Paragraph E
- 18 Paragraph F
- 19 Paragraph G
- 20 Paragraph H

答案

- 14. ii
- 15. iv
- 16. x
- 17. vi
- 18. i
- 19. vii
- 20. xii

Test 9

READING PASSAGE 1

You should spend about 20 minutes on Questions 1-13 which are based on Reading Passage 1 below.

Museum Blockbuster

A

Since the 1980s, the term “blockbuster” has become the fashionable word for the special spectacular museum, art gallery or science centre exhibitions. These exhibitions have the ability to attract large crowds and often large corporate sponsors. Here is one of some existing definitions of a blockbuster: Put by Elsen (1984), a blockbuster is a “... large scale loan exhibition that people who normally don’t go to museums will stand in line for hours to see ...” James Rosenfield, writing in *Direct Marketing* in 1993, has described a successful blockbuster exhibition as a “... triumph of both curatorial and marketing skills ...” My own definition for a blockbuster is “a popular, high profile exhibition on display for a limited period, that attracts the general public, who are prepared to both stand in line and pay a fee in order to partake in the exhibition.” What both Elsen and Rosenfield omit in their descriptions of a blockbuster, is that people are prepared to pay a fee to see a blockbuster and that the term blockbuster can just as easily apply to a movie or a museum exhibition.

B

Merely naming an exhibition or movie a blockbuster, however, does not make it a blockbuster. The term can only apply when the item in question has had an overwhelmingly successful response from the public. However, in literature from both the UK and USA the other words that also start to appear in descriptions of a blockbuster are “less scholarly”, “non-elitist” and “popularist”. Detractors argue that blockbusters are designed to appeal to the lowest common denominator, while others extol the virtues of encouraging scholars to cooperate on projects and to provide exhibitions that cater for a broad selection of the community rather than an elite sector.

C

Maintaining and increasing visitor levels is paramount in the new museology. This requires continued product development. Not only the creation or hiring of blockbuster exhibitions but regular exhibition changes and innovations. In addition, the visiting public has become customers rather than visitors, and the skills that are valued in museums, science centres and galleries to keep the new customers coming through the door have changed. High on the list of requirements are commercial, business, marketing and entrepreneurial skills. Curators are now administrators. Being a director of an art gallery no longer requires an Arts Degree. As succinctly

summarised in the Economist in 1994 “business nous and public relation skills” were essential requirements for a director, and the ability to compete with other museums to stage travelling exhibitions which draw huge crowds.

D

The new museology has resulted in the convergence of museums, the heritage industry, and tourism, profit-making and pleasure-giving. This has given rise to much debate about the appropriateness of adopting the activities of institutions so that they more closely reflect the priorities of the market place and whether it is appropriate to see museums primarily as tourist attractions. At many institutions, you can now hold office functions in the display areas, or have dinner with the dinosaurs. Whatever commentators may think, managers of museums, art galleries and science centres worldwide are looking for artful ways to blend culture and commerce, and blockbuster exhibitions are at the top of the list. But while blockbusters are all part of the new museology, there is proof that you don't need a museum, science centre or art gallery to benefit from the drawing power of a blockbuster or to stage a blockbuster.

E

But do blockbusters held in public institutions really create a surplus to fund other activities? If the bottom line is profit, then according to the accounting records of many major museums and galleries, blockbusters do make money. For some museums overseas, it may be the money that they need to update parts of their collections or to repair buildings that are in need of attention. For others in Australia, it may be the opportunity to illustrate that they are attempting to pay their way, by recovering part of their operating costs or funding other operating activities with off-budget revenue. This makes the economic rationalists cheerful. However, not all exhibitions that are hailed to be blockbusters will be blockbusters, and some will not make money. It is also unlikely that the accounting systems of most institutions will recognise the real cost of either creating or hiring a blockbuster.

F

Blockbusters require large capital expenditure, and draw on resources across all branches of an organisation; however, the costs don't end there. There is a Human Resource Management cost in addition to a measurable 'real' dollar cost. Receiving a touring exhibition involves large expenditure as well, and draws resources from across functional management structures in project management style. everyone from a general labourer to a building servicing unit, the front of the house, technical, promotion, education and administration staff, are required to perform additional tasks. Furthermore, as an increasing number of institutions in Australia try their hand at increasing visitor numbers, memberships (and therefore revenue), by staging blockbuster exhibitions, it may be less likely that blockbusters will continue to provide a surplus to subsidise other activities due to the competitive nature of the market. There are only so many consumer dollars to go around, and visitors will need to choose between blockbuster products.

G

Unfortunately, when the bottom-line is the most important objective to the mounting of blockbuster exhibitions, this same objective can be hard to maintain. Creating,

mounting or hiring blockbusters is exhausting for staff, with the real costs throughout an institution difficult to calculate. Although the direct aims may be financial, creating or hiring a blockbuster has many positive spin-offs; by raising their profile through a popular blockbuster exhibition, a museum will be seen in a more favorable light at budget time. Blockbusters mean crowds, and crowds are good for the local economy, providing increased employment for shops, hotels, restaurants, the transport industry and retailers. Blockbusters expose staff to the vagaries and pressures of the market place and may lead to creative excellence. Either the success or failure of a blockbuster may highlight the need for managers and policymakers to rethink their strategies. However, the new museology and the apparent trend towards blockbusters make it likely that museums, art galleries and particularly science centres will be seen as part of the entertainment and tourism industry, rather than as cultural icons deserving of government and philanthropic support.

H

Perhaps the best pathway to take is one that balances both blockbusters and regular exhibitions. However, this easy middle ground may only work if you have enough space, and have alternate sources of funding to continue to support the regular less exciting fare. Perhaps the advice should be to make sure that your regular activities and exhibitions are more enticing, and find out what your local community wants from you. The question (trend) now at most museums and science centres, is “What blockbusters can we tour to overseas venues and will it be cost-effective?”

Questions 1-4

The Reading Passage has seven paragraphs A-H

Which paragraphs contains the following information?

Write the correct letter A-H, in boxes 1-4 on your answer sheet.

NB You may use any letter more than once.

- 1 A reason for changing the exhibition programs.
- 2 The time people have to wait in a queue in order to enjoy exhibitions.
- 3 Terms people used when referring to the blockbuster
- 4 There was some controversy over confining target groups of a blockbuster.

答案

1. C
2. A
3. B
4. B

READING PASSAGE 2

You should spend about 20 minutes on Questions 14-26 which are based on Reading Passage 2 below.

The Lost City

Thanks to modern remote-sensing techniques, a ruined city in Turkey is slowly revealing itself as one of the greatest and most mysterious cities of the ancient world. Sally Palmer uncovers more.

A

The low granite mountain, known as Kerkenes Dag, juts from the northern edge of the Cappadocian plain in Turkey. Sprawled over the mountainside are the ruins of an enormous city, contained by crumbling defensive walls seven kilometers long. Many respected archaeologists believe these are the remains of the fabled city of Pteria, the sixth-century BC stronghold of the Medes that the Greek historian Herodotus described in his famous work *The Histories*. The short-lived city came under Median control and only fifty years later was sacked, burned and its strong stone walls destroyed.

B

British archaeologist Dr Geoffrey Summers has spent ten years studying the site. Excavating the ruins is a challenge because of the vast area they cover. The 7 km perimeter walls run around a site covering 271 hectares. Dr Summers quickly realised it would take far too long to excavate the site using traditional techniques alone. So he decided to use modern technology as well to map the entire site, both above and beneath the surface, to locate the most interesting areas and priorities to start digging.

C

In 1993, Dr Summers hired a special hand-held balloon with a remote-controlled camera attached. He walked over the entire site holding the balloon and taking photos. The one afternoon, he rented a hot-air balloon and floated over the site, taking yet more pictures. By the end of the 1994 season, Dr Summers and his team had a jigsaw of aerial photographs of the whole site. The next stage was to use remote sensing, which would let them work out what lay below the intriguing outlines and ruined walls. "Archaeology is a discipline that lends itself very well to remote sensing because it revolves around space," says Scott Branting, an associate director of the project. He started working with Dr Summers in 1995.

D

The project used two main remote-sensing techniques. The first is magnetometry, which works on the principle that magnetic fields at the surface of the Earth are influenced by what is buried beneath. It measures localised variations in the direction and intensity of this magnetic field. "The Earth's magnetic field can vary from place to place, depending on what happened there in the past," says Branting. "if something containing iron oxide was heavily burnt, by natural or human actions, the iron particles in it can be permanently reoriented, like a compass needle, to align with the Earth's magnetic field present at that point in time and space.' The magnetometer

detects differences in the orientations and intensities of these iron particles from the present-day magnetic field and uses them to produce an image of what lies below ground.

E

Kirkenes Dag lends itself particularly well to magnetometry because it was all burnt at once in a savage fire. In places, the heat was sufficient to turn sandstone to glass and to melt granite. The fire was so hot that there were strong magnetic signatures set to the Earth's magnetic field from the time – around 547 BC – resulting in extremely clear pictures. Furthermore, the city was never rebuilt. “if you have multiple layers confusing picture because you have different walls from different periods giving signatures that all go in different directions,” says Branting. “We only have one going down about 1.5 meters, so we can get a good picture of this fairly short-lived city.”

F

The other main sub-surface mapping technique, which is still being used at the site, is resistivity. This technique measures the way electrical pulses are conducted through sub-surface soil. It's done by shooting pulses into the ground through a thin metal probe. Different materials have different electrical conductivity. For example, stone and mudbrick are poor conductors, but looser, damp soil conducts very well. By walking around the site and taking about four readings per metre, it is possible to get a detailed idea of what is where beneath the surface. The teams then build up pictures of walls, hearths and other remains. “It helps a lot if it has rained because the electrical pulse can get through more easily,” says Branting. “Then if something is more resistant, it really shows up.” This is one of the reasons that the project has a spring season when most of the resistivity work is done. Unfortunately, testing resistivity is a lot slower than magnetometry. “If we did resistivity over the whole site it would take about 100 years,” says Branting. Consequently, the team is concentrating on areas where they want to clarify pictures from the magnetometry.

G

Remote sensing does not reveal everything about Kerkenes Dag, but it shows the most interesting sub-surface areas of the site. The archaeologists can then excavate these using traditional techniques. One surprise came when they dug out one of the gates in the defensive walls. “Our observations in early seasons led us to assume that wall, such as would be found at most other cities in the Ancient Near East,” says Dr Summers. “When we started to excavate we were staggered to discover that the walls were made entirely from stone and that the gate would have stood at least ten metres high. After ten years of study, Pteris is gradually giving up its secrets.”

Questions 14-17

The Reading Passage has seven paragraphs A-G

Which paragraph contains the following information?

Write the correct letter A-G, in boxes 14-17 on your answer sheet.

- 14 The reason why various investigative methods are introduced.
- 15 An example of an unexpected discovery.
- 16 The methods to survey the surface of the site from above.

17 The reason why experts want to study the site.

答案

14. B

15. G

16. C

17. A

READING PASSAGE 3

You should spend about 20 minutes on Questions 27-40 which are based on Reading Passage 3 below.

Knowledge in medicine

A

What counts as knowledge? What do we mean when we say that we know something? What is the status of different kinds of knowledge? In order to explore these questions, we are going to focus on one particular area of knowledge – medicine.

B

How do you know when you are ill? This may seem to be an absurd question. You know you are ill because you feel ill; your body tells you that you are ill. You may know that you feel pain or discomfort but knowing you are ill is a bit more complex. At times, people experience the symptoms of illness, but in fact, they are simply tired or over-worked or they may just have a hangover. At other times, people may be suffering from a disease and fail to be aware of the illness until it has reached a late stage in its development. So how do we know we are ill, and what counts as knowledge?

C

Think about this example. You feel unwell. You have a bad cough and always seem to be tired. Perhaps it could be stress at work, or maybe you should give up smoking. You feel worse. You visit the doctor who listens to your chest and heart, takes your temperature and blood pressure, and then finally prescribes antibiotics for your cough.

D

Things do not improve but you struggle on thinking you should pull yourself together, perhaps things will ease off at work soon. A return visit to your doctor shocks you. This time the doctor, drawing on years of training and experience, diagnoses pneumonia. This means that you will need bed rest and a considerable time off work. The scenario is transformed. Although you still have the same symptoms, you no longer think that these are caused by pressure at work. You now have proof that you are ill. This is the result of the combination of your own subjective experience and the diagnosis of someone who has the status of a medical expert. You have a medically authenticated diagnosis and it appears that you are seriously ill; you know you are ill and have the evidence upon which to base this knowledge.

E

This scenario shows many different sources of knowledge. For example, you decide to consult the doctor in the first place because you feel unwell – this is personal knowledge about your own body. However, the doctor's expert diagnosis is based on experience and training, with sources of knowledge as diverse as other experts, laboratory reports, medical textbooks and years of experience.

F

One source of knowledge is the experience of our own bodies; the personal knowledge we have of changes that might be significant, as well as the subjective

experiences are mediated by other forms of knowledge such as the words we have available to describe our experience, and the common sense of our families and friends as well as that drawn from popular culture. Over the past decade, for example, Western culture has seen a significant emphasis on stress-related illness in the media. Reference to being ‘stressed out’ has become a common response in daily exchanges in the workplace and has become part of popular common-sense knowledge. It is thus not surprising that we might seek such an explanation of physical symptoms of discomfort.

G

We might also rely on the observations of others who know us. Comments from friends and family such as ‘you do look ill’ or ‘that’ s a bad cough’ might be another source of knowledge. Complementary health practices, such as holistic medicine, produce their own sets of knowledge upon which we might also draw in deciding the nature and degree of our ill health and about possible treatments.

H

Perhaps the most influential and authoritative source of knowledge is the medical knowledge provided by the general practitioner. We expect the doctor to have access to expert knowledge. This is socially sanctioned. It would not be acceptable to notify our employer that we simply felt too unwell to turn up for work or that our faith healer, astrologer, therapist or even our priest thought it was not a good idea. We need an expert medical diagnosis in order to obtain the necessary certificate if we need to be off work for more than the statutory self-certification period. The knowledge of the medical sciences is privileged in this respect in contemporary Western culture. Medical practitioners are also seen as having the required expert knowledge that permits them legally to prescribe drugs and treatment to which patients would not otherwise have access. However, there is a range of different knowledge upon which we draw when making decisions about our own state of health.

I

However, there is more than existing knowledge in this little story; new knowledge is constructed within it. Given the doctor’ s medical training and background, she may hypothesize ‘is this now pneumonia?’ and then proceed to look for evidence about it. She will use observations and instruments to assess the evidence and – critically – interpret it in light of her training and experience. This results in new knowledge and new experience both for you and for the doctor. This will then be added to the doctor’ s medical knowledge and may help in the future diagnosis of pneumonia.

Questions 33-40

The Reading Passage has nine paragraphs A-I

Which paragraph contains the following information?

Write the correct letter A-I, in boxes 33-40 on your answer sheet.

33 the contrast between the nature of personal judgment and the nature of doctor’ s diagnosis

34 a reference of culture about pressure

-
- 35 sick leave will not be permitted without the professional diagnosis
36 how doctors' opinions are regarded in society
37 the illness of patients can become part of new knowledge
38 a description of knowledge drawn from non-specialized sources other than personal knowledge
39 an example of collective judgment from personal experience and professional doctor
40 a reference that some people do not realize they are ill

答案

33. E
34. F
35. H
36. H
37. I
38. G
39. D
40. B

Test 10

READING PASSAGE 1

You should spend about 20 minutes on Questions 1-13 which are based on Reading Passage 1 below.

Traditional Farming System in Africa

A

By tradition land in Luapula is not owned by individuals, but as in many other parts of Africa is allocated by the headman or head woman of a village to people of either sex, according to need. Since land is generally prepared by hand, one ulupwa cannot take on a very large area; in this sense, the land has not been a limiting resource over large parts of the province. The situation has already changed near the main townships, and there has long been a scarcity of land for cultivation in the Valley. In these areas registered ownership patterns are becoming prevalent.

B

Most of the traditional cropping in Luapula, as in the Bemba area to the east, is based on citemene, a system whereby crops are grown on the ashes of tree branches. As a rule, entire trees are not felled but are pollarded so that they can regenerate. Branches are cut over an area of varying size early in the dry season and stacked to dry over a rough circle about a fifth to a tenth of the pollarded area. The wood is fired before the rains and in the first year planted with the African cereal finger millet (*Eleusine coracane*).

C

During the second season, and possibly for a few seasons more the area is planted to variously mixed combinations of annuals such as maize, pumpkins (*Telfiria occidentalis*) and other cucurbits, sweet potatoes, groundnuts, *Phaseolus* beans and various leafy vegetables, grown with a certain amount of rotation. The diverse sequence ends with vegetable cassava, which is often planted into the developing last-but-one crop as a relay.

D

Richards (1969) observed that the practice of citemene entails a definite division of labour between men and women. A man stakes out a plot in an unobtrusive manner since it is considered provocative towards one's neighbours to mark boundaries in an explicit way. The dangerous work of felling branches is the men's province and involves much pride. Branches are stacked by the women and fired by the men. Formerly women and men cooperated in the planting work, but the harvesting was always done by the women. At the beginning of the cycle little weeding is necessary, since the firing of the branches effectively destroys weeds. As the cycle progresses weeds increase and nutrients eventually become depleted to a point where further effort with annual crops is judged to be not worthwhile: at this point the cassava is

planted, since it can produce a crop on nearly exhausted soil. Thereafter the plot is abandoned, and a new area pollarded for the next citemene cycle.

E

When the forest is not available – this is increasingly the case nowadays – various ridging systems (ibala) are built on small areas, to be planted with combinations of maize, beans, groundnuts and sweet potatoes, usually relayed with cassava. These plots are usually tended by women, and provide subsistence. Where their roots have year-round access to water tables mango, guava and oil-palm trees often grow around houses, forming a traditional agroforestry system. In season some of the fruit is sold by the roadside or in local markets.

F

The margins of dambos are sometimes planted to local varieties of rice during the rainy season, and areas adjacent to vegetables irrigated with water from the dambo during the dry season. The extent of cultivation is very limited, no doubt because the growing of crops under dambo conditions calls for a great deal of skill. Near towns, some of the vegetable produce is sold in local markets.

G

Fishing has long provided a much-needed protein supplement to the diet of Luapulans, as well as being the one substantial source of cash. Much fish has dried for sale to areas away from the main waterways. The Mweru and Bangweulu Lake Basins are the main areas of year-round fishing, but the Luapula River is also exploited during the latter part of the dry season. Several previously abundant and desirable species, such as the Luapula salmon or mpumbu (*Labeo altivelis*) and pale (*Sarotherodon machochir*) have all but disappeared from Lake Mweru, apparently due to mismanagement.

H

Fishing has always been a far more remunerative activity in Luapula than crop husbandry. A fisherman may earn more in a week than a bean or maize grower in a whole season. I sometimes heard claims that the relatively high earnings to be obtained from fishing induced an ‘easy come, easy go’ outlook among Luapulan men. On the other hand, someone who secures good but erratic earnings may feel that their investment in economically productive activity is not worthwhile because Luapulans fail to cooperate well in such activities. Besides, a fisherman with spare cash will find little in the way of working equipment to spend his money on. Better spend one’s money in the bars and have a good time!

I

Only small numbers of cattle or oxen are kept in the province owing to the prevalence of the tsetse fly. For the few herds, the dambos provide subsistence grazing during the dry season. The absence of animal draft power greatly limits peoples’ ability to plough and cultivate land: a married couple can rarely manage to prepare by hand-hoeing. Most people keep freely roaming chickens and goats. These act as a reserve for bartering, but may also be occasionally slaughtered for ceremonies or for entertaining important visitors. These animals are not a regular part of most peoples’ diet.

J

Citemene has been an ingenious system for providing people with seasonal production of high-quality cereals and vegetables in regions of acid, heavily leached soils. Nutritionally, the most serious deficiency was that of protein. This could at times be alleviated when fish was available, provided that cultivators lived near the Valley and could find the means of bartering for dried fish. The citemene/fishing system was well adapted to the ecology of the miombo regions and sustainable for long periods, but only as long as the human population densities stayed at low levels. Although population densities are still much lower than in several countries of South-East Asia, neither the fisheries nor the forests and woodlands of Luapula are capable, with unmodified traditional practices, of supporting the people in a sustainable manner.

Overall, people must learn to intensify and diversify their productive systems while yet ensuring that these systems will remain productive in the future when even more people will need food. Increasing overall production of food, though a vast challenge in itself, will not be enough, however. At the same time, storage and distribution systems must allow everyone accesses to at least a moderate share of the total.

Questions 5-8

Classify the following items with the correct description.

Write your answers in boxes 5-8 on your answer sheet.

- | | |
|---|-------|
| A | fish |
| B | oxen |
| C | goats |
- 5 be used in some unusual occasions, such as celebrations.
6 cannot thrive for being affected by the pests.
7 be the largest part of creating profit.
8 be sold beyond the local area.

答案

5. C
6. B
7. A
8. A

READING PASSAGE 2

You should spend about 20 minutes on Questions 14-26 which are based on Reading Passage 2 below.

Can we call it “ART” Life-Casting and Art

A

When these life-castings were made in the 19th century, no one thought of them as art. But, if critics today can hail Tracey Emin’s unmade bed and the lights going off and on in a gallery as masterpieces of some kind, then shouldn’t these more skillful and profoundly strange works have a greater claim on our attention?

B

Art changes over time; what is art changes, too. Objects intended for devotional, ritualistic or recreational use are re-categorised, by latecomers from another civilisation who no longer respond to these original purposes. Where would New Yorker cartoon be without Lascaux gags in which one bison-painter makes anachronistically “artistic” remarks to another? What also happens is that techniques and crafts judged non-artistic at the time are reassessed?

C

In the 19th century, life-casting was to sculpture what photography was to painting; and both were viewed as cheating short-cuts by the senior arts. Their virtues – of speed and unwavering realism – also implied their limitations; they left little or no room for the imagination. For many, life-casting was an insult to the sculptor’s creative gesture; in a famous lawsuit of 1834, a moulder whose mask of the dying Napoleon had been reproduced and sold without his permission, was judged to have no rights in the image – in other words, he was specifically held not to be an artist. Rodin said of life-casting: “It happens fast, but it doesn’t make art.” Others feared that the whole canon of aesthetics might be blown off course if too much nature was allowed in, it would lead art away from its proper pursuit of the ideal.

D

Gauguin, at the end of the century, worried about future developments in photography: if ever the process went into colour, what painter would labour away at a likeness with a brush made from squirrel-tail? But painting has proved robust. Photography changed it, of course, just as the novel had to reassess narrative after the arrival of the cinema. But the gap between the senior and junior arts was always narrower than the die-hards implied: painters have always used technical back-up – studio assistants to do the boring bits, cameras Lucida and Obscura; while apparently lesser crafts involve great skill, thought, preparation, choice, and – depending on how we define it – imagination. Life-casting was complex, technical work, as Benjamin Robert Haydon discovered when he poured 250 litres of plaster over his black model Wilson and nearly killed him.

E

Time changes our view in another way, too. Each new art movement implies a reassessment of what has gone before; what is done now alters what was done before.

In some cases, this is merely self-serving, with the new is using the old to justify itself. Look how all of that points to this; aren't we clever to be the culmination of all that has gone before? But usually it is a matter of re-alerting the sensibility, reminding us not to take things for granted; every so often we need the aesthetic equivalent of a cataract operation. So there are many items in this show – innocent bit-players back in the last half of the 19th century – which would sit happily nowadays in a commercial or public gallery. Many curators would probably put in for the stunning cast of the hand of a giant from Barnum's circus.

F

The initial impact is on the eye, in the contradiction (which Mueck constantly exploits) between unexpected size and extreme verisimilitude. Next, the human element kicks in: you note that the nails are dirt-encrusted – unless this is the caster's decorative addition – and the paddy fingertips extend far beyond them. (Was the giant an anxious gnawer, or does giantism mean that the flesh simply outgrows the nails?) Then you take in the element of choice, arrangement, art if you like – the neat, pleated, buttoned sleeve end that gives the item balance and variation of texture. This is just a moulded hand, yet the part stands utterly for the whole: and, as an item on public display, it reminds us, slyly, poignantly, of the full-size original who in his time was just as much a victim of gawping. We are not a long way from Degas's *La Petite Danseuse* (which, after all, one critic said should be in the Dupuytren pathology museum); though we are nearer to contemporary art that lazily gets called cutting-edge.

G

Barthes proclaimed the death of the author, the liberation of the text from authorial intention, and the consequent empowerment of the reader; he announced this, needless to say, in a text written with a particular intention in order to communicate something very specific to a reader. An own goal of Keith Weller proportions. But what doesn't work for literature works much better for art. Pictures do float free of their creators' intentions; over time, the "reader" does become more powerful. Few of us can look at a medieval altarpiece as its painter "intended", we believe too little and aesthetically know too much, so we recreate, we find new fields of pleasure in the work. Equally, the lack of artistic intention of Paul Richer and other forgotten craftsmen who brushed oil on to flesh, who moulded, cast, decorated and primped a century and more ago is now irrelevant.

H

What counts is the surviving object and our living response to it. The tests are simple: does it interest the eye, excite the brain, move the mind to reflection, and involve the heart; further, is an apparent level of skill involved? Much currently fashionable art bothers only the eye and briefly the brain, but it fails to engage the mind or the heart. It may, to use the old dichotomy, be beautiful, but it is rarely true to any significant depth. One of the constant pleasures of art is its ability to come at us from an unexpected angle and stop us short in wonder. That is what many of the objects in this show do. The *Ataxic Venus* doesn't make Ron Mueck's *Dead Dad* any less intense and moving an image; but she does offer herself as a companion, precursor, and, yes,

rival.

Questions 14-18

The Reading Passage has seven paragraphs A-H

Which paragraph contains the following information?

Write the correct letter A-H, in boxes 14-18 on your answer sheet.

- 14 Technicians do boring work
- 15 A trial on a famous figure's mask in the 19th century
- 16 The intention from the author is claimed matters in Art
- 17 How to assess an art
- 18 Detailed depiction of an earlier work

答案

- 14. D
- 15. C
- 16. G
- 17. H
- 18. F

READING PASSAGE 3

You should spend about 20 minutes on Questions 27-40 which are based on Reading Passage 3 below.

Honey bees in trouble Can native pollinators fill the gap?

A

Recently, ominous headlines have described a mysterious ailment, colony collapse disorder (CCD), that is wiping out the honeybees that pollinate many crops. Without honeybees, the story goes, fields will be sterile, economies will collapse, and food will be scarce.

B

But what few accounts acknowledge is that what's at risk is not itself a natural state of affairs. For one thing, in the United States, where CCD was first reported and has had its greatest impacts, honeybees are not a native species. Pollination in modern agriculture isn't alchemy, it's industry. The total number of hives involved in the U.S. pollination industry has been somewhere between 2.5 million and 3 million in recent years. Meanwhile, American farmers began using large quantities of organophosphate insecticides, planted large-scale crop monocultures, and adopted "clean farming" practices that scrubbed native vegetation from field margins and roadsides. These practices killed many native bees outright – they're as vulnerable to insecticides as an agricultural pest – and made the agricultural landscape inhospitable to those that remained. Concern about these practices and their effects on pollinators isn't new – in her 1962 ecological alarm cry *Silent Spring*, Rachel Carson warned of a 'Fruitless Fall' that could result from the disappearance of insect pollinators.

C

If that 'Fruitless Fall' has not-yet-occurred, it may be largely thanks to the honeybee, which farmers turned to as the ability of wild pollinators to service crops declined. The honeybee has been semi-domesticated since the time of the ancient Egyptians, but it wasn't just familiarity that determined this choice: the bees' biology is in many ways suited to the kind of agricultural system that was emerging. For example, honeybee hives can be closed up and moved out of the way when pesticides are applied to a field. The bees are generalist pollinators, so they can be used to pollinate many different crops. And although they are not the most efficient pollinator of every crop, honeybees have strength in numbers, with 20,000 to 100,000 bees living in a single hive. "Without a doubt, if there was one bee you wanted for agriculture, it would be the honeybee," says Jim Cane, of the U.S. Department of Agriculture. The honeybee, in other words, has become a crucial cog in the modern system of industrial agriculture. That system delivers more food, and more kinds of it, to more places, more cheaply than ever before. But that system is also vulnerable, because making a farm field into the photosynthetic equivalent of a factory floor, and pollination into a series of continent-long assembly lines, also leaches out some of the resilience characteristics of natural ecosystems.

D

Breno Freitas, an agronomist, pointed out that in nature such a high degree of specialization usually is a very dangerous game: it works well while all the rest is in equilibrium, but runs quickly to extinction at the least disbalance. In effect, by developing an agricultural system that is heavily reliant on a single pollinator species, we humans have become riskily overspecialized. And when the human-honeybee relationship is disrupted, as it has been by colony collapse disorder, the vulnerability of that agricultural system begins to become clear.

E

In fact, a few wild bees are already being successfully managed for crop pollination.

“The problem is trying to provide native bees inadequate numbers on a reliable basis in a fairly short number of years in order to service the crop,” Jim Cane says. “You’re talking millions of flowers per acre in a two-to three-week time frame, or less, for a lot of crops.” On the other hand, native bees can be much more efficient pollinators of certain crops than honeybees, so you don’t need as many to do the job. For example, about 750 blue orchard bees (*Osmia lignaria*) can pollinate a hectare of apples or almonds, a task that would require roughly 50,000 to 150,000 honeybees. There are bee tinkerers engaged in similar work in many corners of the world. In Brazil, Breno Freitas has found that *Centris tarsata*, the native pollinator of wild cashew, can survive in commercial cashew orchards if growers provide a source of floral oils, such as by interplanting their cashew trees with a Caribbean cherry.

F

In certain places, native bees may already be doing more than they’re getting credit for. Ecologist Rachael Winfree recently led a team that looked at pollination of four summer crops (tomato, watermelon, peppers, and muskmelon) at 29 farms in the region of New Jersey and Pennsylvania. Winfree’s team identified 54 species of wild bees that visited these crops, and found that wild bees were the most important pollinators in the system: even though managed honeybees were present on many of the farms, wild bees were responsible for 62 percent of flower visits in the study. In another study focusing specifically on watermelon, Winfree and her colleagues calculated that native bees alone could provide sufficient pollination at 90 percent of the 23 farms studied. By contrast, honeybees alone could provide sufficient pollination at only 78 percent of farms.

G

“The region I work in is not typical of the way most food is produced,” Winfree admits. In the Delaware Valley, most farms and farm fields are relatively small, each farmer typically grows a variety of crops, and farms are interspersed with suburbs and other types of land use which means there are opportunities for homeowners to get involved in bee conservation, too. The landscape is a bee-friendly patchwork that provides a variety of nesting habitat and floral resources distributed among different kinds of crops, weedy field margins, fallow fields, suburban neighborhoods, and semi-natural habitat like old woodlots, all at a relatively small scale. In other words, “pollinator-friendly” farming practices would not only aid pollination of agricultural crops, but also serve as a key element in the overall conservation strategy for wild pollinators, and often aid other wild species as well.

H

Of course, not all farmers will be able to implement all of these practices. And researchers are suggesting a shift to a kind of polyglot agricultural system. For some small-scale farms, native bees may indeed be all that's needed. For larger operations, a suite of managed bees – with honeybees filling the generalist role and other, native bees pollinating specific crops – could be augmented by free pollination services from resurgent wild pollinators. In other words, they're saying, we still have an opportunity to replace a risky monoculture with something diverse, resilient, and robust.

Questions 36-40

Complete each sentence with the correct ending, A-F, below.

Write the correct letter, A-F, in boxes 36-40 on your answer sheet

- 36 The headline of colony collapse disorder states that
37 Viewpoints of Freitas manifest that
38 Examples of blue orchard bees have shown that
39 *Centris tarsata* is mentioned to exemplify that
40 One finding of the research in Delaware Valley is that
- A native pollinators can survive when a specific plant is supplied.
B it would cause severe consequences both to commerce and agriculture.
C honey bees can not be bred.
D some agricultural landscapes are favorable in supporting wild bees.
E a large scale of honey bees are needed to pollinate.
F an agricultural system is fragile when relying on a single pollinator

答案

36. B
37. F
38. E
39. A
40. D

Test 11

READING PASSAGE 3

You should spend about 20 minutes on Questions 27-40 which are based on Reading Passage 3 below.

How should reading be taught? By Keith Rayner and Barbara R

Foorman

A

Learning to speak is automatic for almost all children, but learning to read requires elaborate instruction and conscious effort. Well aware of the difficulties, educators have given a great deal of thought to how they can best help children learn to read. No single method has triumphed. Indeed, heated arguments about the most appropriate form of reading instruction continue to polarize the teaching community.

B

Three general approaches have been tried. In one, called whole-word instruction, children learn by rote how to recognise at a glance a vocabulary of 50 to 100 words. Then they gradually acquire other words, often through seeing them used over and over again in the context of a story.

Speakers of most languages learn the relationship between letters and the sounds associated with them (phonemes). That is, children are taught how to use their knowledge of the alphabet to sound out words. This procedure constitutes a second approach to teaching reading – phonics.

Many schools have adopted a different approach: the whole-language method. The strategy here relies on the child's experience with the language. For example, students are offered engaging books and are encouraged to guess the words that they do not know by considering the context of the sentence or by looking for clues in the storyline and illustrations, rather than trying to sound them out.

Many teachers adopted the whole-language approach because of its intuitive appeal. Making reading fun promises to keep children motivated, and learning to read depends more on what the student does than on what the teacher does. The presumed benefits of whole-language instruction – and the contrast to the perceived dullness of phonics – led to its growing acceptance across America during the 1990s and a movement away from phonics.

C

However, many linguists and psychologists objected strongly to the abandonment of phonics in American schools. Why was this so? In short, because research had clearly demonstrated that understanding how letters related to the component sounds in words is critically important in reading. This conclusion rests, in part, on knowledge

of how experienced readers make sense of words on a page. Advocates of whole-language instruction have argued forcefully that people often derive meanings directly from print without ever determining the sound of the word. Some psychologists today accept this view, but most believe that reading is typically a process of rapidly sounding out words mentally. Compelling evidence for this comes from experiments which show that subjects often confuse homophones (words that sound the same, such as Jrose and 'rows5). This supports the idea that readers convert strings of letters to sounds.

D

In order to evaluate different approaches to teaching reading, a number of experiments have been carried out, firstly with college students, then with school pupils. Investigators trained English-speaking college students to read using unfamiliar symbols such as Arabic letters (the phonics approach), while another group learned entire words associated with certain strings of Arabic letters (whole-word). Then both groups were required to read a new set of words constructed from the original characters. In general, readers who were taught the rules of phonics could read many more new words than those trained with a whole-word procedure.

Classroom studies comparing phonics with either whole-word or whole-language instruction are also quite illuminating. One particularly persuasive study compared two programmes used in 20 first-grade classrooms. Half the students were offered traditional reading instruction, which included the use of phonics drills and applications. The other half were taught using an individualised method that drew from their experiences with languages; these children produce their own booklets of stories and developed sets of words to be recognised (common components of the whole-language approach). This study found that the first group scored higher at year's end on tests of reading and comprehension.

E

If researchers are so convinced about the need for phonics instruction, why does the debate continue? Because the controversy is enmeshed in the philosophical differences between traditional and progressive (or new) approaches, differences that have divided educators for years. The progressive challenge the results of laboratory tests and classroom studies on the basis of a broad philosophical skepticism about the values of such research. They champion student-centred learning and teacher empowerment. Sadly, they fail to realise that these very admirable educational values are equally consistent with the teaching of phonics.

F

If schools of education insisted that would-be reading teachers learned something about the vast research in linguistics and psychology that bears on reading, their graduates would be more eager to use phonics and would be prepared to do so effectively. They could allow their pupils to apply the principles of phonics while reading for pleasure. Using whole-language activities to supplement phonics instruction certainly helps to make reading fun and meaningful for children, so no one would want to see such tools discarded. Indeed, recent work has indicated that the combination of literature-based instruction and phonics is more powerful than either

method used alone.

Teachers need to strike a balance. But in doing so, we urge them to remember that reading must be grounded in a firm understanding of the connections between letters and sounds. Educators who deny this reality are neglecting decades of research. They are also neglecting the needs of their students.

Questions 27-31

Reading Passage 3 has six sections, A-F.

Choose the correct heading for sections B-F from the list of headings below.

Write the correct number, i-ix, in boxes 27-31 on your answer sheet.

List of Headings

- i Disagreement about the reading process
- ii The roots of the debate
- iii A combined approach
- iv Methods of teaching reading
- v A controversial approach
- vi Inconclusive research
- vii Research with learners
- viii Allowing teachers more control
- ix A debate amongst educators

Example

Section A ix

- 27 Section B
- 28 Section C
- 29 Section D
- 30 Section E
- 31 Section F

答案

- 27. iv
- 28. i
- 29. vii
- 30. ii
- 31. iii

Test 12

READING PASSAGE 1

You should spend about 20 minutes on Questions 1-13 which are based on Reading Passage 1 below.

Going Bananas

A

The world's favourite fruit could disappear forever in 10 years time. The banana is among the world's oldest crops. Agricultural scientists believe that the first edible banana was discovered around ten thousand years ago. It has been at an evolutionary standstill ever since it was first propagated in the jungles of South-East Asia at the end of the last ice age. Normally the wild banana, a giant jungle herb called *Musa acuminata*, contains a mass of hard seeds that make the fruit virtually inedible. But now and then, hunter-gatherers must have discovered rare mutant plants that produced seed-less, edible fruits. Geneticists now know that the vast majority of these soft-fruited plants resulted from genetic accidents that gave their cells three copies of each chromosome instead of the usual two. This imbalance prevents seeds and pollen from developing normally, rendering the mutant plants sterile. And that is why some scientists believe the world's most popular fruit could be doomed. It lacks the genetic diversity to fight off pests and diseases that are invading the banana plantations of Central America and the small-holdings of Africa and Asia alike.

B

In some ways, the banana today resembles the potato before blight brought famine to Ireland a century and a half ago. But "it holds a lesson for other crops, too", says Emile Frison, top banana at the International Network for the Improvement of Banana and Plantain in Montpellier, France. "The state of the banana", Frison warns, "can teach a broader lesson the increasing standardisation of food crops around the world is threatening their ability to adapt and survive."

C

The first Stone Age plant breeders cultivated these sterile freaks by replanting cuttings from their stems. And the descendants of those original cuttings are the bananas we still eat today. Each is a virtual clone, almost devoid of genetic diversity. And that uniformity makes it ripe for a disease like no other crop on Earth. Traditional varieties of sexually reproducing crops have always had a much broader genetic base, and the genes will recombine in new arrangements in each generation. This gives them much greater flexibility in evolving responses to disease – and far more genetic resources to draw on in the face of an attack. But that advantage is fading fast, as growers increasingly plant the same few, high-yielding varieties. Plant breeders work feverishly to maintain resistance in these standardized crops. Should these efforts falter, yields of even the most productive crop could swiftly crash. "When some pest

or disease comes along, severe epidemics can occur,” says Geoff Hawtin, director of the Rome-based International Plant Genetic Resources Institute.

D

The banana is an excellent case in point. Until the 1950s, one variety, the Gros Michel, dominated the world’s commercial banana business. Found by French botanists in Asia in the 1820s, the Gros Michel was by all accounts a fine banana, richer and sweeter than today’s standard banana and without the latter’s bitter aftertaste when green. But it was vulnerable to a soil fungus that produced wilt known as Panama disease. “Once the fungus gets into the soil it remains there for many years. There is nothing farmers can do. Even chemical spraying won’t get rid of it,” says Rodomiro Ortiz, director of the International Institute for Tropical Agriculture in Ibadan, Nigeria. So plantation owners played a running game, abandoning infested fields and moving to “clean” land – until they ran out of clean land in the 1950s and had to abandon the Gros Michel. Its successor and still the reigning commercial king is the Cavendish banana, a 19th-century British discovery from southern China. The Cavendish is resistant to Panama disease and, as a result, it literally saved the international banana industry. During the 1960s, it replaced the Gros Michel on supermarket shelves. If you buy a banana today, it is almost certainly a Cavendish. But even so, it is a minority in the world’s banana crop.

E

Half a billion people in Asia and Africa depend on bananas. Bananas provide the largest source of calories and are eaten daily. Its name is synonymous with food. But the day of reckoning may be coming for the Cavendish and its indigenous kin. Another fungal disease, black Sigatoka, has become a global epidemic since its first appearance in Fiji in 1963. Left to itself, black Sigatoka – which causes brown wounds on leaves and premature fruit ripening – cuts fruit yields by 50 to 70 per cent and reduces the productive lifetime of banana plants from 30 years to as little as 2 or 3. Commercial growers keep Sigatoka at bay by a massive chemical assault. Forty sprayings of fungicide a year is typical. But despite the fungicides, diseases such as black Sigatoka are getting more and more difficult to control. “As soon as you bring in a new fungicide, they develop resistance,” says Frison. “One thing we can be sure of is that the Sigatoka won’t lose in this battle.” Poor farmers, who cannot afford chemicals, have it even worse. They can do little more than watch their plants die. “Most of the banana fields in Amazonia have already been destroyed by the disease,” says Luadir Gasparotto, Brazil’s leading banana pathologist with the government research agency EMBRAPA. Production is likely to fall by 70 per cent as the disease spreads, he predicts. The only option will be to find a new variety.

F

But how? Almost all edible varieties are susceptible to diseases, so growers cannot simply change to a different banana. With most crops, such a threat would unleash an army of breeders, scouring the world for resistant relatives whose traits they can breed into commercial varieties. Not so with the banana. Because all edible varieties are sterile, bringing in new genetic traits to help cope with pests and diseases is nearly impossible. Nearly, but not totally. Very rarely, a sterile banana will experience a

genetic accident that allows an almost normal seed to develop, giving breeders a tiny window for improvement. Breeders at the Honduran Foundation of Agricultural Research have tried to exploit this to create disease-resistant varieties. Further backcrossing with wild bananas yielded a new seedless banana resistant to both black Sigatoka and Panama disease.

G

Neither Western supermarket consumers nor peasant growers like the new hybrid. Some accuse it of tasting more like an apple than a banana. Not surprisingly, the majority of plant breeders have till now turned their backs on the banana and got to work on easier plants. And commercial banana companies are now washing their hands of the whole breeding effort, preferring to fund a search for new fungicides instead. “We supported a breeding programme for 40 years, but it wasn’t able to develop an alternative to Cavendish. It was very expensive and we got nothing back,” says Ronald Romero, head of research at Chiquita, one of the Big Three companies that dominate the international banana trade.

H

Last year, a global consortium of scientists led by Frison announced plans to sequence the banana genome within five years. It would be the first edible fruit to be sequenced. Well, almost edible. The group will actually be sequencing inedible wild bananas from East Asia because many of these are resistant to black Sigatoka. If they can pinpoint the genes that help these wild varieties to resist black Sigatoka, the protective genes could be introduced into laboratory tissue cultures of cells from edible varieties. These could then be propagated into new, resistant plants and passed on to farmers.

I

It sounds promising, but the big banana companies have, until now, refused to get involved in GM research for fear of alienating their customers. “Biotechnology is extremely expensive and there are serious questions about consumer acceptance,” says David McLaughlin, Chiquita’s senior director for environmental affairs. With scant funding from the companies, the banana genome researchers are focusing on the other end of the spectrum. Even if they can identify the crucial genes, they will be a long way from developing new varieties that smallholders will find suitable and affordable. But whatever biotechnology’s academic interest, it is the only hope for the banana. Without banana production worldwide will head into a tailspin. We may even see the extinction of the banana as both a lifesaver for hungry and impoverished Africans and as the most popular product on the world’s supermarket shelves.

Questions 4-10

Look at the following statements (Questions 4-10) and the list of people below
Match each statement with the correct person, A-F.

Write the correct letter: A-F, in boxes 4-10 on your answer sheet.

NB You may use any letter more than once.

- 4 Pest invasion may seriously damage the banana industry.
- 5 The effect of fungal infection in the soil is often long-lasting.
- 6 A commercial manufacturer gave up on breeding bananas for disease-resistant

species.

- 7 Banana disease may develop resistance to chemical sprays.
- 8 A banana disease has destroyed a large number of banana plantations.
- 9 Consumers would not accept the genetically altered crop.
- 10 Lessons can be learned from bananas for other crops.

List of People

- A Rodomiro
- B David McLaughlin
- C Emile Frison
- D Ronald Romero
- E Luadir Gasparotto
- F Geoff Hawtin

答案

- 4. F
- 5. A
- 6. D
- 7. C
- 8. E

READING PASSAGE 2

You should spend about 20 minutes on Questions 14-26 which are based on Reading Passage 2 below.

Can we call it “ART” (2) Life-Casting and Art

Julian Barnes explores the questions posed by Life-Casts, an exhibition of plaster moulds of living people and objects which were originally used for scientific purposes

A

Art changes over time and our idea of what art is changing too. For example, objects originally intended for devotional, ritualistic or recreational purposes may be recategorized as art by members of other later civilisations, such as our own, which no longer respond to these purposes.

B

What also happens is that techniques and crafts which would have been judged inartistic at the time they were used are reassessed. Life-casting is an interesting example of this. It involved making a plaster mould of a living person or thing. This was complex, technical work, as Benjamin Robert Haydon discovered when he poured 250 litres of plaster over his human model and nearly killed him. At the time, the casts were used for medical research and, consequently, in the nineteenth-century life-casting was considered inferior to sculpture in the same way that, more recently, photography was thought to be a lesser art than painting. Both were viewed as unacceptable shortcuts by the ‘senior’ arts. Their virtues of speed and unwavering realism also implied their limitations; they left little or no room for the imagination.

C

For many, life-casting was an insult to the sculptor’s creative genius. In an infamous lawsuit of 1834, a moulder whose mask of the dying French emperor Napoleon had been reproduced and sold without his permission was judged to have no rights to the image. In other words, he was specifically held not to be an artist. This judgement reflects the view of established members of the nineteenth-century art world such as Rodin, who commented that life-casting ‘happens fast but it doesn’t make Art’. Some even feared that ‘if too much nature was allowed in, it would lead Art away from its proper course of the Ideal.

D

The painter Gauguin, at the end of the nineteenth century, worried about future developments in photography. If ever the process went into colour, what painter would labour away at a likeness with a brush made from squirrel-tail? But painting has proved robust. Photography has changed it, of course, just as the novel had to reassess narrative after the arrival of the cinema. But the gap between the senior and junior arts was always narrower than the traditionalists implied. Painters have always used technical back-up such as studio assistants to do the boring bits, while apparently lesser crafts involve great skill, thought, preparation and, depending on how we define it, imagination.

E

Time changes our view in another way, too. Each new movement implies a reassessment of what has gone before? What is done now alters what was done before. In some cases, this is merely self-serving, with the new art using the old to justify itself. It seems to be saying, look at how all of that points to this! Aren't we clever to be the culmination of all that has gone before? But usually, it is a matter of re-alerting the sensibility, reminding us not to take things for granted. Take, for example, the cast of the hand of a giant from a circus, made by an anonymous artist around 1889, an item that would now sit happily in any commercial or public gallery. The most significant impact of this piece is on the eye, in the contradiction between unexpected size and verisimilitude. Next, the human element kicks in, you note that the nails are dirt-encrusted, unless this is the caster's decorative addition, and the fingertips extend far beyond them. Then you take in the element of choice, arrangement, art if you like, in the neat, pleated, buttoned sleeve-end that gives the item balance and variation of texture. This is just a moulded hand, yet the part stands utterly for the whole. It reminds us slyly, poignantly, of the full-size original.

F

But is it art? And, if so, why? These are old tediously repeated questions to which artists have often responded, 'It is art because I am an artist and therefore what I do is art. However, what doesn't work for literature works much better for artworks of art do float free of their creators' intentions. Over time the "reader" does become more powerful. Few of us can look at a medieval altarpiece as its painter intended. We believe too little and aesthetically know too much, so we recreate and find new fields of pleasure in the work. Equally, the lack of artistic intention of Paul Richer and other forgotten craftsmen who brushed oil onto flesh, who moulded, cast and decorated in the nineteenth century is now irrelevant. What counts is the surviving object and our response to it. The tests are simple: does it interest the eye, excite the brain, move the mind to reflection and involve the heart. It may, to use the old dichotomy, be beautiful but it is rarely true to any significant depth. One of the constant pleasures of art is its ability to come at us from an unexpected angle and stop us short in wonder.

Questions 14-18

The Reading Passage has six paragraphs A-F

Which paragraph contains the following information?

- 14 an example of a craftsman's unsuccessful claim to ownership of his work
- 15 an example of how trends in the art can change attitudes to an earlier work
- 16 the original function of a particular type of art
- 17 ways of assessing whether or not an object is an art
- 18 how artists deal with the less interesting aspects of their work

答案

- 14. D
- 15. C
- 16. G
- 17. H
- 18. F

Test 13

READING PASSAGE 1

You should spend about 20 minutes on Questions 1-13 which are based on Reading Passage 1 below.

The coming back of the “Extinct” Grass in Britain

A

It's Britain's dodo, called interrupted brome because of its gappy seed-head, this unprepossessing grass was found nowhere else in the world. Sharp-eyed Victorian botanists were the first to notice it, and by the 1920s the odd-looking grass had been found across much of southern England. Yet its decline was just as dramatic. By 1972 it had vanished from its last stronghold—two hay fields at Pampisford, near Cambridge. Even the seeds stored at the Cambridge University Botanic Garden as an insurance policy were dead, having been mistakenly kept at room temperature. Botanists mourned: a unique living entity was gone forever.

B

Yet reports of its demise proved premature. Interrupted brome has come back from the dead, and not through any fancy genetic engineering. Thanks to one green-fingered botanist, interrupted brome is alive and well and living as a pot plant. Britain's dodo is about to become a phoenix, as conservationists set about relaunching its career in the wild.

Questions 8-13

Use the information in the passage to match the people (listed A-F) with opinions or deeds below.

Write the appropriate letters A-F in boxes 8-13 on your answer sheet.

NB You may use any letter more than once

- A George Claridge Druce
- B Nathaniel Fiennes
- C Professor Hackel
- D A. M. Barnard
- E Philip Smith
- J Joan Thirsk

Choose the people who

- 8 reestablished the British unique plants
- 9 identified the interrupted brome as just to its parent brome
- 10 gave an independent taxonomy place to interrupted brome
- 11 discovered and picked the first sample of interrupted brome
- 12 recorded the first ‘show up’ of sainfoin plants in Britain

13 collected the last seeds just before its extinction

答案

8. E

9. C

10. A

11. D

12. F

13. E

READING PASSAGE 3

You should spend about 20 minutes on Questions 27-40 which are based on Reading Passage 3 below.

Book review on Musicophilia

Norman M. Weinberger reviews the latest work of Oliver Sacks on music.

A

Music and the brain are both endlessly fascinating subjects, and as a neuroscientist specialising in auditory learning and memory, I find them especially intriguing. So I had high expectations of *Musicophilia*, the latest offering from neurologist and prolific author Oliver Sacks. And I confess to feeling a little guilty reporting that my reactions to the book are mixed.

B

Sacks himself is the best part of *Musicophilia*. He richly documents his own life in the book and reveals highly personal experiences. The photograph of him on the cover of the book—which shows him wearing headphones, eyes closed, clearly enchanted as he listens to Alfred Brendel perform Beethoven’s *Pathétique Sonata*—makes a positive impression that is borne out by the contents of the book. Sacks’ voice throughout is steady and erudite but never pontifical. He is neither self-conscious nor self-promoting.

C

The preface gives a good idea of what the book will deliver. In it, Sacks explains that he wants to convey the insights gleaned from the “enormous and rapidly growing body of work on the neural underpinnings of musical perception and imagery, and the complex and often bizarre disorders to which these are prone.” He also stresses the importance of “the simple art of observation” and “the richness of the human context.” He wants to combine “observation and description with the latest in technology,” he says, and to imaginatively enter into the experience of his patients and subjects. The reader can see that Sacks, who has been practicing neurology for 40 years, is torn between the “old-fashioned” path of observation and the new-fangled, high-tech approach: He knows that he needs to take heed of the latter, but his heart lies with the former.

D

The book consists mainly of detailed descriptions of cases, most of them involving patients whom Sacks has seen in his practice. Brief discussions of contemporary neuroscientific reports are sprinkled liberally throughout the text. Part, “*Haunted by Music*,” begins with the strange case of Tony Cicoria, a nonmusical, middle-aged surgeon who was consumed by a love of music after being hit by lightning. He suddenly began to crave listening to piano music, which he had never cared for in the past. He started to play the piano and then to compose music, which arose spontaneously in his mind in a “torrent” of notes. How could this happen? Was the cause psychological? (He had had a near-death experience when the lightning struck him.) Or was it the direct result of a change in the auditory regions of his cerebral

cortex? Electroencephalography (EEG) showed his brain waves to be normal in the mid-1990s, just after his trauma and subsequent “conversion” to music. There are now more sensitive tests, but Cicoria, has declined to undergo them; he does not want to delve into the causes of his musicality. What a shame!

E

Part II, “A Range of Musicality,” covers a wider variety of topics, but unfortunately, some of the chapters offer little or nothing that is new. For example, chapter 13, which is five pages long, merely notes that the blind often has better hearing than the sighted. The most interesting chapters are those that present the strangest cases. Chapter 8 is about “amusia,” an inability to hear sounds like music, and “dysharmonia,” a highly specific impairment of the ability to hear harmony, with the ability to understand melody left intact. Such specific “dissociations” are found throughout the cases Sacks recounts.

F

To Sacks’ credit, part III, “Memory, Movement and Music,” brings us into the underappreciated realm of music therapy. Chapter 16 explains how “melodic intonation therapy” is being used to help expressive aphasic patients (those unable to express their thoughts verbally following a stroke or other cerebral incident) once again become capable of fluent speech. In chapter 20, Sacks demonstrates the near-miraculous power of music to animate Parkinson’s patients and other people with severe movement disorders, even those who are frozen into odd postures. Scientists cannot yet explain how music achieves this effect

G

To readers who are unfamiliar with neuroscience and music behavior, Musicophilia may be something of a revelation. But the book will not satisfy those seeking the causes and implications of the phenomena Sacks describes. For one thing, Sacks appears to be more at ease discussing patients than discussing experiments. And he tends to be rather uncritical in accepting scientific findings and theories.

H

It’s true that the causes of music-brain oddities remain poorly understood. However, Sacks could have done more to draw out some of the implications of the careful observations that he and other neurologists have made and of the treatments that have been successful. For example, he might have noted that the many specific dissociations among components of music comprehension, such as loss of the ability to perceive harmony but not melody, indicate that there is no music center in the brain. Because many people who read the book are likely to believe in the brain localisation of all mental functions, this was a missed educational opportunity.

I

Another conclusion one could draw is that there seem to be no “cures” for neurological problems involving music. A drug can alleviate a symptom in one patient and aggravate it in another or can have both positive and negative effects in the same patient. Treatments mentioned seem to be almost exclusively antiepileptic medications, which “damp down” the excitability of the brain in general; their effectiveness varies widely.

J

Finally, in many of the cases described here the patient with music-brain symptoms is reported to have “normal” EEG results. Although Sacks recognises the existence of new technologies, among them far more sensitive ways to analyze brain waves than the standard neurological EEG test, he does not call for their use. In fact, although he exhibits the greatest compassion for patients, he conveys no sense of urgency about the pursuit of new avenues in the diagnosis and treatment of music-brain disorders. This absence echoes the book’s preface, in which Sacks expresses fear that “the simple art of observation may be lost” if we rely too much on new technologies. He does call for both approaches, though, and we can only hope that the neurological community will respond.

Questions 37-40

Complete each sentence with the correct ending, A-F, below.

Write the correct letter, A-F, in boxes 37-40 on your answer sheet.

- 37 The content covered dissociations in understanding between harmony and melody
- 38 The study of treating musical disorders
- 39 The EEG scans of Sacks’ patients
- 40 Sacks believes testing based on new technologies
- A show no music-brain disorders.
- B indicates that medication can have varied results.
- C is key for the neurological community to unravel the mysteries.
- D should not be used in isolation.
- E indicate that not everyone can receive a good education.
- F show a misconception that there is a function centre localized in the brain

答案

37. F
38. B
39. A
40. D

Test 14

READING PASSAGE 1

You should spend about 20 minutes on Questions 1-13 which are based on Reading Passage 1 below.

Can We Hold Back the Flood?

A

LAST winter's floods on the rivers of central Europe were among the worst since the Middle Ages, and as winter storms return, the spectre of floods is returning too. Just weeks ago, the river Rhône in south-east France burst its banks, driving 15,000 people from their homes, and worse could be on the way. Traditionally, river engineers have gone for Plan A: get rid of the water fast, draining it off the land and down to the sea in tall-sides rivers re-engineered as high-performance drains. But however big they dig city drains, however wide and straight they make the rivers, and however high they build the banks, the floods keep coming back to haunt them, from the Mississippi to the Danube. And when the floods come, they seem to be worse than ever.

B

No wonder engineers are turning to Plan B: sap the water's destructive strength by dispersing it into fields, forgotten lakes, flood plains and aquifers. Back in the days when rivers took a more tortuous path to the sea, floodwaters lost impetus and volume while meandering across flood plains and idling through wetlands and inland deltas. But today the water tends to have an unimpeded journey to the sea. And this means that when it rains in the uplands, the water comes down all at once. Worse, whenever we close off more flood plain, the river's flow farther downstream becomes more violent and uncontrollable. Dykes are only as good as their weakest link – and the water will unerringly find it.

C

Today, the river has lost 7 per cent of its original length and runs up to a third faster. When it rains hard in the Alps, the peak flows from several tributaries coincide in the main river, where once they arrived separately. And with four-fifths of the Lower Rhine's flood plain barricaded off, the waters rise ever higher. The result is more frequent flooding that does ever-greater damage to the homes, offices and roads that sit on the flood plain. Much the same has happened in the US on the mighty Mississippi, which drains the world's second-largest river catchment into the Gulf of Mexico.

D

The European Union is trying to improve rain forecasts and more accurately model how intense rains swell rivers. That may help cities prepare, but it won't stop the floods. To do that, say, hydrologists, you need a new approach to engineering, not just Agency – country £1 billion – puts it like this: “The focus is now on working

with the forces of nature. Towering concrete walls are out, and new wetlands are in.” to help keep London’s upstream and reflooding 10 square kilometres outside Oxford. Nearer to London it has spent £100 million creating new wetlands and a relief channel across 16 kilometres.

E

The same is taking place on a much grander scale in Austria, in one of Europe’s largest river restorations to date. Engineers are regenerating flood plains along 60 kilometres of the river Drave as it exits the Alps. They are also widening the river bed and channeling it back into abandoned meanders, oxbow lakes and backwaters overhung with willows. The engineers calculate that the restored flood plain can now store up to 10 million cubic metres of floodwaters and slow storm surges coming out of the Alps by more than an hour, protecting towns as far downstream as Slovenia and Croatia.

F

“ Rivers have to be allowed to take more space. They have to be turned from flood-chutes into flood-foilers,” says Nienhuis. And the Dutch, for whom preventing floods is a matter of survival, have gone furthest. A nation built largely on drained marshes and seabed had the fright of its life in 1993 when the Rhine almost overwhelmed it. The same happened again in 1995 when a quarter of a million people were evacuated from the Netherlands. But a new breed of “soft engineers” wants our cities to become porous, and Berlin is their governed by tough new rules to prevent its drains from becoming overloaded after heavy rains. Herald Kraft, an architect working in the city, says: “We now see rainwater as giant Potsdamer Platz, a huge new commercial redevelopment by DaimlerChrysler in the heart of the city.

G

Los Angeles has spent billions of dollars digging huge drains and concreting river beds to carry away the water from occasional intense storms. “In LA we receive half the water we need in rainfall, and we throw it away. Then we spend hundreds of millions to import water,” says Andy Lipkis, an LA environmentalist who kick-started the idea of the porous city by showing it could work on one house. Lipkis, along with citizens groups like Friends of the Los Angeles River and Unpaved LA, want to beat the urban flood hazard and fill the taps by holding onto the city’s floodwater. And it’s not just a pipe dream. The authorities this year launched a \$100 million scheme to road-test the porous city in one flood-hit community in Sun Valley. The plan is to catch the rain that falls on thousands of driveways, parking lots and rooftops in the valley. Trees will soak up water from parking lots. Homes and public buildings will capture roof water to irrigate gardens and parks. And road drains will empty into old gravel pits and other leaky places that should recharge the city’s underground water reserves. Result: less flooding and more water for the city. Plan B says every city should be porous, every river should have room to flood naturally and every coastline should be left to build its own defences. It sounds expensive and utopian, until you realise how much we spend trying to drain cities and protect our watery margins – and how bad we are at it.

Questions 1-6

The Reading Passage has seven paragraphs A-G.

Which paragraph contains the following information?

Write the correct letter A-G, in boxes 1-6 on your answer sheet.

- 1 A new approach carried out in the UK.
- 2 Reasons why the twisty path and dykes failed
- 3 Illustration of an alternative Plan in LA which seems much unrealistic
- 4 The traditional way of tackling flood
- 5 The effort made in the Netherlands and Germany
- 6 One project on a river benefits three nations

答案

1. D
2. B
3. G
4. A
5. F
6. E

READING PASSAGE 3

You should spend about 20 minutes on Questions 27-40 which are based on Reading Passage 3 below.

TV Addiction 2

A

Excessive cravings do not necessarily involve physical substances. Gambling can become compulsive; sex can become obsessive. One activity, however, stands out for its prominence and ubiquity – the world’s most popular pastime, television. Most people admit to having a love-hate relationship with it. They complain about the “boob tube” and “couch potatoes,” then they settle into their sofas and grab the remote control. Parents commonly fret about their children’s viewing (if not their own). Even researchers who study TV for a living marvel at the medium’s hold on them personally. Percy Tannenbaum of the University of California at Berkeley has written: “Among life’s, more embarrassing moments have been countless occasions when I am engaged in conversation in a room while a TV set is on, and I cannot for the life of me stop from periodically glancing over to the screen. This occurs not only during dull conversations but during reasonably interesting ones just as well.”

B

Scientists have been studying the effects of television for decades, generally focusing on whether watching violence on TV correlates with being violent in real life. Less attention has been paid to the basic allure of the small screen – the medium, as opposed to the message.

C

The term “TV addiction” is imprecise and laden with value judgments, but it captures the essence of a very real phenomenon. Psychologists and psychiatrists formally define substance dependence as a disorder characterized by criteria that include spending a great deal of time using the substance; using it more often than one intends; thinking about reducing use or making repeated unsuccessful efforts to reduce use; giving up important social, family or occupational activities to use it; and reporting withdrawal symptoms when one stops using it.

D

All these criteria can apply to people who watch a lot of television. That does not mean that watching television, in itself, is problematic. Television can teach and amuse; it can reach aesthetic heights; it can provide much-needed distraction and escape. The difficulty arises when people strongly sense that they ought not to watch as much as they do and yet find themselves strangely unable to reduce their viewing. Some knowledge of how the medium exerts its pull may help heavy viewers gain better control over their lives.

E

The amount of time people spend watching television is astonishing. On average, individuals in the industrialized world devote three hours a day to the pursuit – fully half of their leisure time, and more than on any single activity save work and

sleep. At this rate, someone who lives to 75 would spend nine years in front of the tube. To some commentators, this devotion means simply that people enjoy TV and make a conscious decision to watch it. But if that is the whole story, why do so many people experience misgivings about how much they view? In Gallup polls in 1992 and 1999, two out of five adult respondents and seven out of 10 teenagers said they spent too much time watching TV. Other surveys have consistently shown that roughly 10 percent of adults calls themselves TV addicts.

F

What is it about TV that has such a hold on us? In part, the attraction seems to spring from our biological ‘orienting response.’ First described by Ivan Pavlov in 1927, the orienting response is our instinctive visual or auditory reaction to any sudden or novel stimulus. It is part of our evolutionary heritage, a built-in sensitivity to movement and potential predatory threats.

G

In 1986 Byron Reeves of Stanford University, Esther Thorson of the University of Missouri and their colleagues began to study whether the simple formal features of television – cuts, edits, zooms, pans, sudden noises – activate the orienting response, thereby keeping attention on the screen. By watching how brain waves were affected by formal features, the researchers concluded that these stylistic tricks can indeed trigger involuntary responses and ‘derive their attentional value through the evolutionary significance of detecting movement ... It is the form, not the content, of television that is unique.’

H

The orienting response many partly explain common viewer remarks such as: “If a television is on, I just can’t keep my eyes off it,” “I don’t want to watch as much as I do, but I can’t help it,” and “I feel hypnotized when I watch television.” In the years since Reeves and Thorson published their pioneering work, researchers have delved deeper. Annie Lang’s research team at Indiana University has shown that heart rate decreases for four to six seconds after an orienting stimulus. In ads, action sequences and music videos, formal features frequently come at a rate of one per second, thus activating the orienting response continuously.

I

Lang and her colleagues have also investigated whether formal features affect people’s memory of what they have seen. In one of their studies, participants watched a program and then filled out a score sheet. Increasing the frequency of edits (defined here as a change from one camera angle to another in the same visual scene) improved memory recognition, presumably because it focused attention on the screen. Increasing the frequency of cuts – changes to a new visual scene-had a similar effect but only up to a point. If the number of cuts exceeded 10 in two minutes, recognition dropped off sharply.

J

Producers of educational television for children have found that formal features can help to learn. But increasing the rate of cuts and edits eventually overloads the brain. Music videos and commercials that use rapid intercutting of unrelated scenes are

designed to hold attention more than they are to convey information. People may remember the name of the product or band, but the details of the ad itself float in one ear and out the other. The orienting response is overworked. Viewers still attend to the screen, but they feel tired and worn out, with little compensation psychological reward. Our ESM findings show much the same thing.

K

Sometimes the memory of the product is very subtle. Many ads today are deliberately oblique: they have an engaging storyline, but it is hard to tell what they are trying to sell. Afterwards, you may not remember the product consciously. Yet advertisers believe that if they have gotten your attention when you later go to the store you will feel better or more comfortable with a given product because you have a vague recollection of having heard of it.

Questions 34-37

Look at the following researchers (Questions 34-37) and the list of statements below. Match each researcher with the correct statements.

Write the correct letter A-G in boxes 34-37 on your answer sheets.

- 34 Percy Tannenbaum
- 35 Ivan Pavlov
- 36 Byron Reeves and Esther Thorson
- 37 Annie Lang

List of Statements

- A It is the specific media formal characteristic that counts.
- B TV distraction shows a human physical reaction to a new and prompted stimulus
- C Conveying information is the most important thing.
- D It is hard to ignore the effects of TV.
- E Whether people can remember deeper of the content relates to the format.
- F The heart rate remains stable when watching.
- G Clinically reliance on TV does not meet the criteria of addiction

答案

- 34. D
- 35. B
- 36. A
- 37. E

Test 15

READING PASSAGE 1

You should spend about 20 minutes on Questions 1-13 which are based on Reading Passage 1 below.

William Gilbert and Magnetism

A

16th and 17th centuries saw two great pioneers of modern science: Galileo and Gilbert. The impact of their findings is eminent. Gilbert was the first modern scientist, also the accredited father of the science of electricity and magnetism, an Englishman of learning and a physician at the court of Elizabeth. Prior to him, all that was known of electricity and magnetism was what the ancients knew, nothing more than that the lodestone possessed magnetic properties and that amber and jet, when rubbed, would attract bits of paper or other substances of small specific gravity. However, he is less well-known than he deserves.

B

Gilbert's birth predated Galileo. Born in an eminent local family in Colchester county in the UK, on May 24, 1544, he went to grammar school, and then studied medicine at St. John's College, Cambridge, graduating in 1573. Later he traveled in the continent and eventually settled down in London.

C

He was a very successful and eminent doctor. All this culminated in his election to the president of the Royal Science Society. He was also appointed the personal physician to the Queen (Elizabeth I) and later knighted by the Queen. He faithfully served her until her death. However, he didn't outlive the Queen for long and died on December 10, 1603, only a few months after his appointment as personal physician to King James.

D

Gilbert was first interested in chemistry but later changed his focus due to the large portion of the mysticism of alchemy involved (such as the transmutation of metal). He gradually developed his interest in physics after the great minds of the ancient, particularly about the knowledge the ancient Greeks had about lodestones, strange minerals with the power to attract iron. In the meantime, Britain became a major seafaring nation in 1588 when the Spanish Armada was defeated, opening the way to the British settlement of America. British ships depended on the magnetic compass, yet no one understood why it worked. Did the pole star attract it, as Columbus once speculated; or was there a magnetic mountain at the pole, as described in *Odyssey* which ships would never approach because the sailors thought its pull would yank out all their iron nails and fittings? For nearly 20 years William Gilbert conducted ingenious experiments to understand magnetism. His works include *On the Magnet*

and Magnetic Bodies, Great Magnet of the Earth.

E

Gilbert's discovery was so important to modern physics. He investigated the nature of magnetism and electricity. He even coined the word "electric". Though the early beliefs of magnetism were also largely entangled with superstitions such as that rubbing garlic on lodestone can neutralize its magnetism, one example being that sailors even believed the smell of garlic would even interfere with the action of the compass, which is why helmsmen were forbidden to eat it near a ship's compass. Gilbert also found that metals can be magnetized by rubbing materials such as fur, plastic or the like on them. He named the ends of a magnet "north pole" and "south pole". The magnetic poles can attract or repel, depending on polarity. In addition, however, ordinary iron is always attracted to a magnet. Though he started to study the relationship between magnetism and electricity, sadly he didn't complete it. His research of static electricity using amber and jet only demonstrated that objects with electrical charges can work like magnets attracting small pieces of paper and stuff. It is a French guy named du Fay that discovered that there are actually two electrical charges, positive and negative.

F

He also questioned the traditional of astronomical beliefs. Though a Copernican, he didn't express in his quintessential beliefs whether the earth is at the center of the universe or in orbit around the sun. However, he believed that stars are not equidistant from the earth, but have their own earth-like planets orbiting around them. The earth is itself like a giant magnet, which is also why compasses always point north. They spin on an axis that is aligned with the earth's polarity. He even likened the polarity of the magnet to the polarity of the earth and built an entire magnetic philosophy on this analogy. In his explanation, magnetism was the soul of the earth. Thus a perfectly spherical lodestone, when aligned with the earth's poles, would wobble all by itself in 24 hours. Further, he also believed that stars and other stars wobble just like the earth does around a crystal core, and speculated that moon might also be a magnet caused to orbit by its magnetic attraction to the earth. This was perhaps the first proposal that a force might cause a heavenly orbit.

G

His research method was revolutionary in that he used experiments rather than pure logic and reasoning like the ancient Greek philosophers did. It was a new attitude toward the scientific investigation. Until then, scientific experiments were not in fashion. It was because of this scientific attitude, together with his contribution to our knowledge of magnetism, that a unit of magnetomotive force, also known as magnetic potential, was named Gilbert in his honor. His approach of careful observation and experimentation rather than the authoritative opinion or deductive philosophy of others had laid the very foundation for modern science.

Questions 1-7

Reading passage 1 has seven paragraphs A-G.

Choose the correct heading for each paragraph from the list of headings below

Write the correct number i-x in boxes on your answer sheet.

List of Headings

- i Early years of Gilbert
- ii What was new about his scientific research method
- iii The development of chemistry
- iv Questioning traditional astronomy
- v Pioneers of the early science
- vi Professional and social recognition
- vii Becoming the president of the Royal Science Society
- viii The great works of Gilbert
- ix His discovery about magnetism
- x His change of focus
- 1 Paragraph A
- 2 Paragraph B
- 3 Paragraph C
- 4 Paragraph D
- 5 Paragraph E
- 6 Paragraph F
- 7 Paragraph G

答案

READING PASSAGE 2

You should spend about 20 minutes on Questions 14-26 which are based on Reading Passage 2 below.

Twin Study: Two of a kind

A

THE scientific study of twins goes back to the late 19th century, when Francis Galton, an early geneticist, realised that they came in two varieties: identical twins born from one egg and non-identical twins that had come from two. That insight turned out to be key, although it was not until 1924 that it was used to formulate what is known as the twin rule of pathology, and twin studies really got going.

B

The twin rule of pathology states that any heritable disease will be more concordant (that is, more likely to be jointly present or absent) in identical twins than in non-identical twins – and, in turn, will be more concordant in non-identical twins than in non-siblings. Early work, for example, showed that the statistical correlation of skin-mole counts between identical twins was 0.4, while non-identical twins had a correlation of only 0.2. (A score of 1.0 implies a perfect correlation, while a score of zero implies no correlation.) This result suggests that moles are heritable, but it also implies that there is an environmental component to the development of moles, otherwise, the correlation in identical twins would be close to 1.0.

C

Twin research has shown that whether or not someone takes up smoking is determined mainly by environmental factors, but once he does so, how much he smokes is largely down to his genes. And while a person's religion is clearly a cultural attribute, there is a strong genetic component to religious fundamentalism. Twin studies are also unraveling the heritability of various aspects of human personality. Traits from neuroticism and anxiety to thrill – and novelty-seeking all have large genetic components. Parenting matters, but it does not determine personality in the way that some had thought.

D

More importantly, perhaps, twin studies are helping the understanding of diseases such as cancer, asthma, osteoporosis, arthritis and immune disorders. And twins can be used, within ethical, for medical experiments. A study that administered vitamin C to one twin and a placebo to the other found that it had no effect on the common cold. The lesson from all of today's twin studies is that most human traits are at least partially influenced by genes. However, for the most part, the age-old dichotomy between nature and nurture is not very useful. Many genetic programs are open to input from the environment, and genes are frequently switched on or off by environmental signals. It is also possible that genes themselves influence their environment. Some humans have an innate preference for participation in sports. Others are drawn to novelty. Might people also be drawn to certain kinds of friends

and types of experience? In this way, a person's genes might shape the environment they act in as much as the environment shapes the actions of the genes.

E

In the past, such research has been controversial. Josef Mengele, a Nazi doctor working at the Auschwitz extermination camp during the second world war, was fascinated by twins. He sought them out among arrivals at the camp and preserved them from the gas-chambers for a series of brutal experiments. After the war, Cyril Burt, a British psychologist who worked on the heredity of intelligence, tainted twin research with results that appear, in retrospect, to have been rather too good. Some of his data on identical twins who had been reared apart were probably faked. In any case, the prevailing ideology in the social sciences after the war was Marxist and disliked suggestions that differences in human potential might have underlying genetic causes. Twin studies were thus viewed with suspicion.

F

The ideological pendulum has swung back; however, as the human genome project and its aftermath have turned genes from abstract concepts to real pieces of DNA. The role of genes in sensitive areas such as intelligence is acknowledged by all but a few die-hards. The interesting questions now concern how nature and nurture interact to produce particular bits of biology, rather than which of the two is more important. Twin studies, which are a good way to ask these questions, are back in fashion, and many twins are enthusiastic participants in this research.

G

Research at the Twinsburg festival began in a small way, with a single stand in 1979. Gradually, news spread and more scientists began turning up. This year, half a dozen groups of researchers were lodged in a specially pitched research tent. In one corner of this tent, Paul Breslin, who works at the Monell Institute in Philadelphia, watched over several tables where twins sat sipping clear liquids from cups and making notes. It was the team's third year at Twinsburg. Dr Breslin and his colleagues want to find out how genes influence human perception, particularly the senses of smell and taste and those (warmth, cold, pain, tingle, itch and so on) that result from stimulation of the skin. Perception is an example of something that is probably influenced by both genes and experience. Even before birth, people are exposed to flavours such as chocolate, garlic, mint and vanilla that pass intact into the bloodstream, and thus to the fetus. Though it is not yet clear whether such pre-natal exposure shapes taste-perception, there is evidence that it shapes preferences for foods encountered later in life.

H

However, there are clearly genetic influences at work, as well – for example in the ability to taste quinine. Some people experience this as intensely bitter, even when it is present at very low levels. Others, whose genetic endowment is different, are less bothered by it. Twin studies make this extremely clear. Within a pair of identical twins, either both, or neither, will find quinine hard to swallow. Non-identical twins will agree less frequently.

I

On the other side of the tent Dennis Drayna, from the National Institute on Deafness and Other Communication Disorders, in Maryland, was studying hearing. He wants to know what happens to sounds after they reach the ear. It is not clear, he says, whether the sound is processed into sensation mostly in the ear or in the brain. Dr Drayna has already been involved in a twin study which revealed that the perception of musical pitch is highly heritable. At Twinsburg, he is playing different words, or parts of words, into the left and right ears of his twinned volunteers. The composite of the two sounds that an individual reports hearing depends on how he processes this diverse information and that, Dr Drayna believes, may well be influenced by genetics.

J

Elsewhere in the marquee, Peter Miraldi, of Kent State University in Ohio, was trying to find out whether genes affect an individual's motivation to communicate with others. A number of twin studies have shown that personality and sociability are heritable, so he thinks this is fertile ground. And next to Mr Miraldi was a team of dermatologists from Case Western Reserve University in Cleveland. They are looking at the development of skin disease and male-pattern baldness. The goal of the latter piece of research is to find the genes responsible for making men's hair fall out.

K

The busiest part of the tent, however, was the queue for forensic-science research into fingerprints. The origins of this study are shrouded in mystery. For many months, the festival's organisers have been convinced that the Secret Service – the American government agency responsible for, among other things, the safety of the president – is behind it. When The Economist contacted the Secret Service for more information, we were referred to Steve Nash, who is chairman of the International Association for Identification (IAI) and is also a detective in the scientific investigations section of the Marin County Sheriff's Office in California. The IAI, based in Minnesota, is an organisation of forensic scientists from around the world. Among other things, it publishes the Journal of Forensic Identification.

Questions 14-18

The Reading Passage has seven paragraphs A-K

Which paragraph contains the following information?

Write the correct letter A-K, in boxes 14-18 on your answer sheet.

NB You may use any letter more than once.

- 14 Mentioned research conducted in Ohio
- 15 Medical contribution to the researches for twins.
- 16 Research situation under life-threatening conditions
- 17 Data of similarities of identical twins
- 18 Reasons that make one study unconvincing

答案

14. J

15. D

16. E

17. B 18. E

READING PASSAGE 3

You should spend about 20 minutes on Questions 27-40 which are based on Reading Passage 3 below.

Compliance or Noncompliance for Children

A

Many Scientists believe that socialization takes a long process, while compliance is the outset of it. Accordingly, compliance for the education of children is the priority. Motivationally distinct forms of child compliance, mutually positive affect, and maternal control, observed in 3 control contexts in 103 dyads of mothers and their 26-41-month-old children, were examined as correlates of internalization, assessed using observations of children while alone with prohibited temptations and maternal ratings. One form of compliance (committed compliance), when the child appeared committed wholeheartedly to the maternal agenda and eager to endorse and accept it, was emphasized. Mother-child mutually positive affect was both a predictor and a concomitant of committed compliance. Children who shared positive effect with their mothers showed a high level of committed compliance and were also more internalized. Differences and similarities between children's compliance with requests and prohibitions ("Do" vs. "Don't" demand contexts) were also explored. Maternal "Dos" appeared more challenging to toddlers than the "Don'ts." Some individual coherence of behavior was also found across both demand contexts. The implication of committed compliance for emerging internalized regulators of conduct is discussed.

B

A number of parents were not easy to be aware of the compliance, some even overlooked their children's noncompliance. Despite good education, these children did not follow the words from their parents on several occasion, especially boys in certain ages. Fortunately, this rate was acceptable, some parents could be patient with the noncompliance. Someone held that noncompliance is probably not a wrong thing. In order to determine the effects of different parental disciplinary techniques on young children's compliance and noncompliance, mothers were trained to observe emotional incidents involving their own toddler-aged children. Reports of disciplinary encounters were analyzed in terms of the types of discipline used (reasoning, verbal prohibition, physical coercion, love withdrawal, and combinations thereof) and children's responses to that discipline (compliance/ noncompliance and avoidance). The relation between compliance/ noncompliance and type of misdeed (harm to persons, harm to property, and lapses of self-control) was also analyzed. Results indicated that love withdrawal combined with other techniques was most effective in securing children's compliance and that its effectiveness was not a function of the type of technique with which it was combined. Avoidant responses and affective reunification with the parent were more likely to follow love withdrawal than any other technique. Physical coercion was somewhat less effective than love withdrawal, while reasoning and verbal prohibition were not at all effective except when both

were combined with physical coercion.

C

“Noncompliant Children sometimes prefer to say no directly as they were younger, they are easy to deal with the relationship with contemporaries. When they are growing up. During the period that children are getting older, who may learn to use more advanced approaches for their noncompliance. They are more skillful to negotiate or give reasons for refusal rather than show their opposite idea to parents directly.” Said Henry Porter, a scholar working in Psychology Institute of UK. He indicated that noncompliance means growth in some way, may have benefit for children. Many Experts held different viewpoints in recent years, they tried drilling compliance into children. His collaborator Wallace Friesen believed that Organizing a child’s daily activities so that they occur in the same order each day as much as possible. This first strategy for defiant children is ultimately the most important. Developing a routine helps a child to know what to expect and increases the chances that he or she will comply with things such as chores, homework, and hygiene requests. When undesirable activities occur in the same order at optimal times during the day, they become habits that are not questioned but done without thought.

Chances are that you have developed some type of routine for yourself in terms of showering, cleaning your house, or doing other types of work. You have an idea in your mind when you will do these things on a regular basis and this helps you to know what to expect. In fact, you have probably already been using most of these compliance strategies for yourself without realizing it. For children, without setting these expectations on a daily basis by making them part of a regular routine, they can become very upset. Just like adults, children think about what they plan to do that day and expect to be able to do what they want. So, when you come along and ask them to do something they weren’t already planning to do that day, this can result in automatic refusals and other undesirable defiant behaviors. However, by using this compliance strategy with defiant children, these activities are done almost every day in the same general order and the child expects to already do them.

D

Doctor Steven Walson addressed that organizing fun activities to occur after frequently refused activities. This strategy also works as a positive reinforcer when the child complies with your requests. By arranging your day so that things often refused to occur right before highly preferred activities, you are able to eliminate defiant behavior and motivate your child’s behavior of doing the undesirable activity. This is not to be presented in a way that the preferred activity is only allowed if a defiant child does the non-preferred activity. However, you can word your request in a way so that your child assumes that you have to do the non-preferred activity before moving on to the next preferred activity. For example, you do not want to say something such as, “If you clean your room we can play a game.” Instead of the word your request like this, “As soon as you are done cleaning your room we will be able to play that really fun game you wanted to play.”

E

Psychologist Paul Edith insisted praise is the best way to make children comply with.

This is probably a common term you are used to hearing by now. If you praise your child's behavior, he or she will be more likely to do that behavior. So, it is essential to use praise when working with defiant children. It also provides your child with positive attention. However, it is important to know how to praise children in a way that encourages future automatic reinforcement for your child when doing a similar behavior.

Questions 32-35

Look at the following people and list of statements below.

Match each person with the correct statement.

Write the correct letter A-G in boxes 32-35 on your answer sheet.

- 32 Henry Porter
- 33 Wallace Friesen
- 34 Steven Walson
- 35 Paul Edith

List of statements

- A children of all ages will indirectly show noncompliance
- B elder children tend to negotiate rather than show noncompliance
- C converse behavior means noncompliance
- D organizing fun activities to occur after frequently refused activities
- E organizing child's daily activities in the same order as much as possible.
- F use praise in order to make children compliant
- G take the children to school at an early age

答案

- 32. B
- 33. E
- 34. D
- 35. F

Test 16

READING PASSAGE 1

You should spend about 20 minutes on Questions 1-13 which are based on Reading Passage 1 below.

Food advertising on children

This review was commissioned by the Food Standards Agency to examine the current research evidence on:

- the extent and nature of food promotion to children
- the effect, if any, that this promotion has on their food knowledge, preferences and behaviour.

A

Children's food promotion is dominated by television advertising, and the great majority of this promotes the so-called 'Big Four' of pre-sugared breakfast cereals, soft drinks, confectionery and savoury snacks. In the last ten years advertising for fast food, outlets have rapidly increased. There is some evidence that the dominance of television has recently begun to wane. The importance of strong, global branding reinforces a need for multi-faceted communications combining television with merchandising, 'tie-ins' and point of sale activity. The advertised diet contrasts sharply with that recommended by public health advisors, and themes of fun and fantasy or taste, rather than health and nutrition, are used to promote it to children. Meanwhile, the recommended diet gets little promotional support.

B

There is plenty of evidence that children notice and enjoy food promotion. However, establishing whether this actually influences them is a complex problem. The review tackled it by looking at studies that had examined possible effects on what children know about food, their food preferences, their actual food behaviour (both buying and eating), and their health outcomes (eg. Obesity or cholesterol levels). The majority of studies examined food advertising, but a few examined other forms of food promotion. In terms of nutritional knowledge, food advertising seems to have little influence on children's general perceptions of what constitutes a healthy diet, but, in certain contexts, it does have an effect on more specific types of nutritional knowledge. For example, seeing soft drink and cereal adverts reduced primary aged children's ability to determine correctly whether or not certain products contained real fruit.

C

The review also found evidence that food promotion influences children's food preferences and purchase behaviour. A study of primary school children, for instance, found that exposure to advertising influenced which foods they claimed to like; and

another showed that labelling and signage on a vending machine had an effect on what was bought by secondary school pupils. A number of studies have also shown that food advertising can influence what children eat. One, for example, showed that advertising influenced a primary class' s choice of daily snack at playtime.

D

The next step, of trying to establish whether or not a link exists between food promotion and diet or obesity, is extremely difficult as it requires research to be done in real-world settings. A number of studies have attempted this by using the amount of television viewing as a proxy for exposure to television advertising. They have established a clear link between television viewing and diet, obesity, and cholesterol levels. It is impossible to say, however, whether this effect is caused by the advertising, the sedentary nature of television viewing or snacking that might take place whilst viewing. One study resolved this problem by taking a detailed diary of children' s viewing habits. This showed that the more food adverts they saw, the more snacks and calories they consumed.

E

Thus the literature does suggest food promotion is influencing children' s diet in a number of ways. This does not amount to proof; as noted above with this kind of research, incontrovertible proof simply isn' t attainable. Nor do all studies point to this conclusion; several have not found an effect. In addition, very few studies have attempted to measure how strong these effects are relative to other factors influencing children' s food choices. Nonetheless, many studies have found clear effects and they have used sophisticated methodologies that make it possible to determine that i) these effects are not just due to chance; ii) they are independent of other factors that influence diets, such as parents' eating habits or attitudes; and iii) they occur at a brand and category level.

F

Furthermore, two factors suggest that these findings actually downplay the effect that food promotion has on children. First, the literature focuses principally on television advertising; the cumulative effect of this combined with other forms of promotion and marketing is likely to be significantly greater. Second, the studies have looked at the direct effects of individual children, and understate indirect influences. For example, promotion for fast food outlets may not only influence the child but also encourage parents to take them for meals and reinforce the idea that this is normal and desirable behaviour.

G

This does not amount to proof of an effect, but in our view does provide sufficient evidence to conclude that an effect exists. The debate should now shift to what action is needed, and specifically to how the power of commercial marketing can be used to bring about improvements in young people' s eating.

Questions 1-7

Reading Passage 1 has seven paragraphs, A-G.

Choose the most suitable heading for paragraphs A-G from the list of headings below.

Write the appropriate number, i-x, in boxes 1-7 on your answer sheet.

List of Headings

- i General points of agreements and disagreements of researchers
 - ii How much children really know about food
 - iii Need to take action
 - iv Advertising effects of the “Big Four”
 - v Connection of advertising and children’s weight problems
 - vi Evidence that advertising affects what children buy to eat
 - vii How parents influence children’s eating habits
 - viii Advertising’s focus on unhealthy options
 - ix Children often buy what they want
 - x Underestimating the effects advertising has on children
- 1 Paragraph A
 - 2 Paragraph B
 - 3 Paragraph C
 - 4 Paragraph D
 - 5 Paragraph E
 - 6 Paragraph F
 - 7 Paragraph G

答案

- 1. viii
- 2. ii
- 3. vi
- 4. v
- 5. i
- 6. x
- 7. iii

READING PASSAGE 2

You should spend about 20 minutes on Questions 14-26 which are based on Reading Passage 2 below.

Finding Our Way

A

“Drive 200 yards, and then turn right,” says the car’s computer voice. You relax in the driver’s seat, follow the directions and reach your destination without error. It’s certainly nice to have the Global Positioning System (GPS) to direct you to within a few yards of your goal. Yet if the satellite service’s digital maps become even slightly outdated, you can become lost. Then you have to rely on the ancient human skill of navigating in three-dimensional space. Luckily, your biological finder has an important advantage over GPS: it does not go awry if only one part of the guidance system goes wrong, because it works in various ways. You can ask questions of people on the sidewalk. Or follow a street that looks familiar. Or rely on a navigational rubric: “If I keep the East River on my left, I will eventually cross 34th Street.” The human positioning system is flexible and capable of learning. Anyone who knows the way from point A to point B – and from A to C – can probably figure out how to get from B to C, too.

B

But how does this complex cognitive system really work? Researchers are looking at several strategies people use to orient themselves in space: guidance, path integration and route following. We may use all three or combinations thereof. And as experts learn more about these navigational skills, they are making the case that our abilities may underlie our powers of memory and logical thinking. Grand Central, Please Imagine that you have arrived in a place you have never visited – New York City. You get off the train at Grand Central Terminal in midtown Manhattan. You have a few hours to explore before you must return for your ride home. You head uptown to see popular spots you have been told about: Rockefeller Center, Central Park, the Metropolitan Museum of Art. You meander in and out of shops along the way. Suddenly, it is time to get back to the station. But how?

C

If you ask passersby for help, most likely you will receive information in many different forms. A person who orients herself by a prominent landmark would gesture southward: “Look down there. See the tall, broad MetLife Building? Head for that – the station is right below it.” Neurologists call this navigational approach “guidance,” meaning that landmark visible from a distance serves as the marker for one’s destination.

D

Another city dweller might say: “What places do you remember passing? ... Okay. Go toward the end of Central Park, then walk down to St. Patrick’s Cathedral. A few more blocks and Grand Central will be off to your left.” In this case, you are pointed toward the most recent place you recall, and you aim for it. Once there you head for

the next notable place and so on, retracing your path. Your brain is adding together the individual legs of your trek into a cumulative progress report. Researchers call this strategy “path integration.” Many animals rely primarily on path integration to get around, including insects, spiders, crabs and rodents. The desert ants of the genus *Cataglyphis* employ this method to return from foraging as far as 100 yards away. They note the general direction they came from and retrace their steps, using the polarization of sunlight to orient themselves even under overcast skies. On their way back they are faithful to this inner homing vector. Even when a scientist picks up an ant and puts it in a totally different spot, the insect stubbornly proceeds in the originally determined direction until it has gone “back” all of the distance it wandered from its nest. Only then does the ant realize it has not succeeded, and it begins to walk in successively larger loops to find its way home.

E

Whether it is trying to get back to the anthill or the train station, any animal using path integration must keep track of its own movements so it knows, while returning, which segments it has already completed. As you move, your brain gathers data from your environment – sights, sounds, smells, lighting, muscle contractions, a sense of time passing – to determine which way your body has gone. The church spire, the sizzling sausages on that vendor’s grill, the open courtyard, and the train station – all represent snapshots of memorable junctures during your journey.

F

In addition to guidance and path integration, we use a third method for finding our way. An office worker you approach for help on a Manhattan street corner might say:

“Walk straight down Fifth, turn left on 47th, turn right on Park, go through the walkway under the Helmsley Building, then cross the street to the MetLife Building into Grand Central.” This strategy, called route following, uses landmarks such as buildings and street names, plus directions – straight, turn, go through – for reaching intermediate points. Route following is more precise than guidance or path integration, but if you forget the details and take a wrong turn, the only way to recover is to backtrack until you reach a familiar spot, because you do not know the general direction or have a reference landmark for your goal. The route-following navigation strategy truly challenges the brain. We have to keep all the landmark and intermediate directions in our head. It is the most detailed and therefore most reliable method, but it can be undone by routine memory lapses. With path integration, our cognitive memory is less burdened; it has to deal with only a few general instructions and the homing vector. Path integration works because it relies most fundamentally on our knowledge of our body’s general direction of movement, and we always have access to these inputs. Nevertheless, people often choose to give route-following directions, in part because saying “Go straight that way!” just does not work in our complex, man-made surroundings.

G

Road Map or Metaphor? On your next visit to Manhattan, you will rely on your memory to get around. Most likely you will use guidance, path integration and route following in various combinations. But how exactly do these constructs deliver

concrete direction? Do we humans have, as an image of the real world, a kind of road map in our heads – with symbols for cities, train stations and churches; thick lines for highways; narrow lines for local streets? Neurobiologists and cognitive psychologists do call the portion of our memory that controls navigation a “cognitive map.” The map metaphor is obviously seductive: maps are the easiest way to present geographic information for convenient visual inspection. In many cultures, maps were developed before writing, and today they are used in almost every society. It is even possible that maps derive from a universal way in which our spatial-memory networks are wired.

H

Yet the notion of a literal map in our heads may be misleading; a growing body of research implies that the cognitive map is mostly a metaphor. It may be more like a hierarchical structure of relationships. To get back to Grand Central, you first envision the large scale – that is, you visualize the general direction of the station. Within that system, you then imagine the route to the last place you remember. After that, you observe your nearby surroundings to pick out a recognizable storefront or street corner that will send you toward that place. In this hierarchical, or nested, scheme, positions and distances are relative, in contrast with a road map, where the same information is shown in a geometrically precise scale.

Questions 14-18

Use the information in the passage to match the category of each navigation method (listed A-C) with the correct statement.

Write the appropriate letters A-C in boxes 14-18 on your answer sheet.

NB You may use any letter more than once

A Guidance

B Path integration

C Route following

14 Using basic direction from starting point and light intensity to move on.

15 Using a combination of place and direction heading for the destination.

16 Using an iconic building near your destination as orientation.

17 Using a retrace method from a known place if a mistake happens.

18 Using a passing spot as reference for a new integration.

答案

14. B

15. C

16. A

17. C

18. B

Test 17

READING PASSAGE 2

You should spend about 20 minutes on Questions 14-27 which are based on Reading Passage 2 below.

Stress of Workplace

A

How busy is too busy? For some it means having to miss the occasional long lunch; for others, it means missing lunch altogether. For a few, it is not being able to take a “sickie” once a month. Then there is a group of people for whom working every evening and weekend is normal, and frantic is the tempo of their lives. For most senior executives, workloads swing between extremely busy and frenzied. The vice-president of the management consultancy AT Kearney and its head of telecommunications for the Asia-Pacific region, Neil Plumridge, says his work weeks vary from a “manageable” 45 hours to 80 hours, but average 60 hours.

B

Three warning signs alert Plumridge about his workload: sleep, scheduling and family. He knows he has too much on when he gets less than six hours of sleep for three consecutive nights; when he is constantly having to reschedule appointments; “and the third one is on the family side”, says Plumridge, the father of a three-year-old daughter, and expecting a second child in October. “If I happen to miss a birthday or anniversary, I know things are out of control.” Being “too busy” is highly subjective. But for any individual, the perception of being too busy over a prolonged period can start showing up as stress: disturbed sleep, and declining mental and physical health. National workers’ compensation figures show stress causes the most lost time of any workplace injury. Employees suffering stress are off work an average of 16.6 weeks. The effects of stress are also expensive. Comcare, the Federal Government insurer, reports that in 2003-04, claim costs. Experts say the key to dealing with stress is not to focus on relief – a game of golf or a massage – but to reassess workloads. Neil Plumridge says he makes it a priority to work out what has to change; that might mean allocating extra resources to a job, allowing more time or changing expectations. The decision may take several days. He also relies on the advice of colleagues, saying his peers’ coach each other with business problems. “Just a fresh pair of eyes over an issue can help,” he says.

C

Executive stress is not confined to big organisations. Vanessa Stoykov has been running her own advertising and public relations business for seven years, specializing in work for financial and professional services firms. Evolution Media has grown so fast that it debuted on the BRW Fast 100 list of fastest-growing small enterprises last year – just after Stoykov had her first child. Stoykov thrives on the

mental stimulation of running her own business. “Like everyone, I have the occasional day when I think my head’s going to blow off,” she says. Because of the growth phase, the business is in, Stoykov has to concentrate on short-term stress relief – weekends in the mountains, the occasional “mental health” day – rather than delegating more work. She says: “We’re hiring more people, but you need to train them, teach them about the culture and the clients, so it’s actually more work rather than less.”

D

Identify the causes: Jan Elsner, Melbourne psychologist who specialises in executive coaching, says thriving on a demanding workload is typical of senior executives and other high-potential business people. She says there is no one-size-fits-all approach to stress: some people work best with high-adrenalin periods followed by quieter patches, while others thrive under sustained pressure. “We could take urine and blood hormonal measures and pass judgement of whether someone’s physiologically stressed or not,” she says. “But that’s not going to give us an indicator of what their experience of stress is, and what the emotional and cognitive impacts of stress are going to be.”

E

Elsner’s practice is informed by a movement known as positive psychology, a school of thought that argues “positive” experiences – feeling engaged, challenged, and that one is making a contribution to something meaningful – do not balance out negative ones such as stress; instead, they help people increase their resilience over time. Good stress, or positive experiences of being challenged and rewarded, is thus cumulative in the same way as bad stress. Elsner says many of the senior business people she coaches are relying more on regulating bad stress through methods such as meditation and yoga. She points to research showing that meditation can alter the biochemistry of the brain and actually help people “retrain” the way their brains and bodies react to stress. “Meditation and yoga enable you to shift the way that your brain reacts, so if you get proficient at it you’re in control.”

F

The Australian vice-president of AT Kearney, Neil Plumridge, says: “Often stress is caused by our setting unrealistic expectations of ourselves. I’ll promise a client I’ll do something tomorrow, and the [promise] another client the same thing, when I really know it’s not going to happen. I’ve put stress on myself when I could have said to the clients: ‘Why don’t I give that to you in 48 hours?’ The client doesn’t care.” Overcommitting is something people experience as an individual problem. We explain it as the result of procrastination or Parkinson’s law: that work expands to fill the time available. New research indicates that people may be hard-wired to do it.

G

A study in the February issue of the Journal of Experimental Psychology shows that people always believe they will be less busy in the future than now. This is a misapprehension, according to the authors of the report, Professor Gal Zauberman, of the University of North Carolina, and Professor John Lynch, of Duke University. “On average, an individual will be just as busy two weeks or a month from now as he or

she is today. But that is not how it appears to be in everyday life,” they wrote.

“People often make commitments long in advance that they would never make if the same commitments required immediate action. That is, they discount future time investments relatively steeply.” Why do we perceive a greater “surplus” of time in the future than in the present? The researchers suggest that people underestimate completion times for tasks stretching into the future and that they are bad at imagining the future competition for their time.

Questions 14-18

Use the information in the passage to match the people (listed A-D) with opinions or deeds below.

Write the correct letter A-D, in boxes 14-18 on your answer sheet.

NB You may use any letter more than once.

A Jan Elsnera

B Vanessa Stoykov

C Gal Zauberman

D Neil Plumridge

14 Work stress usually happens in the high level of a business.

15 More people's ideas involved would be beneficial for stress relief

16 Temporary holiday sometimes doesn't mean less work.

17 Stress leads to the wrong direction when trying to satisfy customers.

18 It is not correct that stress in the future will be eased more than now.

答案

14. A

15. D

16. B

17. D

18. C

Test 18

READING PASSAGE 2

You should spend about 20 minutes on Questions 14-27 which are based on Reading Passage 2 below.

Monkeys and Forests

AS AN EAST WIND blasts through a gap in the Cordillera de Tilarán, a rugged mountain range that splits northern Costa Rica in half, a female mantled howler monkey moves through the swaying trees of the forest canopy.

A

Ken Glander, a primatologist from Duke University, gazes into the canopy, tracking the female's movements. Holding a dart gun, he waits with infinite patience for the right moment to shoot. With great care, Glander aims and fires. Hit in the rump, the monkey wobbles. This howler belongs to a population that has lived for decades at Hacienda La Pacifica, a working cattle ranch in Guanacaste province. Other native primates – white-faced capuchin monkeys and spider monkeys – once were common in this area, too, but vanished after the Pan-American Highway was built nearby in the 1950s. Most of the surrounding land was clear-cut for pasture.

B

Howlers persist at La Pacifica, Glander explains, because they are leaf-eaters. They eat fruit when it's available but, unlike capuchin and spider monkeys, do not depend on large areas of fruiting trees. "Howlers can survive anywhere you have half a dozen trees because their eating habits are so flexible," he says. In forests, life is an arms race between trees and the myriad creatures that feed on leaves. Plants have evolved a variety of chemical defenses, ranging from bad-tasting tannins, which bind with plant-produced nutrients, rendering them indigestible, to deadly poisons, such as alkaloids and cyanide.

C

All primates, including humans, have some ability to handle plant toxins. "We can detoxify a dangerous poison known as caffeine, which is deadly to a lot of animals," Glander says. For leaf-eaters, long-term exposure to a specific plant toxin can increase their ability to defuse the poison and absorb the leaf nutrients. The leaves that grow in regenerating forests, like those at La Pacifica, are actually more howler friendly than those produced by the undisturbed, centuries-old trees that survive farther south, in the Amazon Basin. In younger forests, trees put most of their limited energy into growing wood, leaves and fruit, so they produce much lower levels of the toxin than do well-established, old-growth trees.

D

The value of maturing forests to primates is a subject of study at Santa Rosa National Park, about 35 miles northwest of Hacienda La Pacifica. The park hosts populations

not only of mantled howlers but also of white-faced capuchins and spider monkeys. Yet the forests there are young, most of them less than 50 years old. Capuchins were the first to begin using the reborn forests when the trees were as young as 14 years. Howlers, larger and heavier than capuchins, need somewhat older trees, with limbs that can support their greater body weight. A working ranch at Hacienda La Pacifica also explains their population boom in Santa Rosa. “Howlers are more resilient than capuchins and spider monkeys for several seasons,” Fedigan explains. “They can live within a small home range, as long as the trees have the right food for them. Spider monkeys, on the other hand, occupy a huge home range, so they can’t make it in fragmented habitat.”

E

Howlers also reproduce faster than do other monkey species in the area. Capuchins don’t bear their first young until about 7 years old, and spider monkeys do so even later, but howlers give birth for the first time at about 3.5 years of age. Also, while a female spider monkey will have a baby about once every four years, well-fed howlers can produce an infant every two years.

F

The leaves howlers eat hold plenty of water, so the monkeys can survive away from open streams and water holes. This ability gives them a real advantage over capuchin and spider monkeys, which have suffered during the long, ongoing drought in Guanacaste.

G

Growing human population pressures in Central and South America have led to persistent destruction of forests. During the 1990s, about 1.1 million acres of Central American forest were felled yearly. Alejandro Estrada, an ecologist at Estacion de Biologia Los Tuxtlas in Veracruz, Mexico, has been exploring how monkeys survive in a landscape increasingly shaped by humans. He and his colleagues recently studied the ecology of a group of mantled howler monkeys that thrive in a habitat completely altered by humans: a cacao plantation in Tabasco, Mexico. Like many varieties of coffee, cacao plants need shade to grow, so 40 years ago the landowners planted fig, monkeypod and other tall trees to form a protective canopy over their crop. The howlers moved in about 25 years ago after nearby forests were cut. This strange habitat, a hodgepodge of cultivated native and exotic plants, seems to support about as many monkeys as would a same-sized patch of wild forest. The howlers eat the leaves and fruit of the shade trees, leaving the valuable cacao pods alone, so the farmers tolerate them.

H

Estrada believes the monkeys bring underappreciated benefits to such farms, dispersing the seeds of fig and other shade trees and fertilizing the soil with faeces. He points out that howler monkeys live in shade coffee and cacao plantations in Nicaragua and Costa Rica as well as in Mexico. Spider monkeys also forage in such plantations, though they need nearby areas of forest to survive in the long term. He hopes that farmers will begin to see the advantages of associating with wild monkeys, which includes potential ecotourism projects.

“Conservation is usually viewed as a conflict between agricultural practices and the need to preserve nature,” Estrada says. “We’ re moving away from that vision and beginning to consider ways in which agricultural activities may become a tool for the conservation of primates in human-modified landscapes.”

Questions 14-19

The Reading Passage has eight paragraphs A-H

Which paragraph contains the following information?

Write the correct letter A-H, in boxes 14-19 on your answer sheet.

- 14 a reference of reduction in Forest inhabitant
- 15 Only one species of monkey survived while the other two species were vanished.
- 16 a reason for Howler Monkey of choosing new leaves
- 17 mention to Howler Monkey’ s nutrient and eating habits
- 18 a reference of asking farmers’ changing attitude toward wildlife
- 19 the advantage for Howler Monkey’ s flexibility living in a segmented habitat

Questions 20-22

Look at the following places and the list of descriptions below.

Match each description with the correct place, A-E.

Write the correct letter, A-E, in boxes 20-22 on your answer sheet.

List of places

- A Hacienda La Pacifica
- B Santa Rosa National Park
- C a cacao plantation in Tabasco, Mexico
- D Estacion de Biologia Los Tuxtlas in Veracruz, Mexico
- E Amazon Basin
- 20 howler Monkey’ s benefit to the local region’ s agriculture
- 21 The original home for all three native monkeys
- 22 A place where Capuchins monkey comes for a better habitat

答案

- 14. G
- 15. A
- 16. C
- 17. B
- 18. G
- 19. D
- 20. C
- 21. A
- 22. B

Test 19

READING PASSAGE 2

You should spend about 20 minutes on Questions 14-27 which are based on Reading Passage 2 below.

The reconstruction of community in Talbot Park, Auckland

A

An architecture of disguise is almost complete at Talbot Park in the heart of Auckland's Glen Innes. The place was once described as a state housing ghetto, rife with crime, vandalism and other social problems. But today after a \$48 million urban renewal makeover, the site is home to 700 residents – 200 more than before – and has people regularly inquiring whether they can buy or rent there. “It doesn't look like social housing,” Housing New Zealand housing services manager Dene Busby says of the tidy brick and weatherboard apartments and townhouses which would look just as much at home in “there is no reason why public housing should look cheap in my view,” says Design Group architect Neil of the eight three-bedroom terrace houses his firm designed.

B

Talbot Park is a triangle of government-owned land bounded by Apirana Ave, Pilkington Rd and Point England Rd. in the early 1960s, it was developed for state housing built around a linear park that ran through the middle. Initially, there was a strong sense of a family-friendly community. Former residents recall how the Talbot Park reserve played a big part in their childhoods – a place where the kids in the block came together to play softball, cricket, tiggy, leapfrog and bullrush. Sometimes they'd play “Maoris against Pakehas” but without any animosity. “It was all just good fun”, says Georgie Thompson in Ben Schrader's *We Call it Home: A History of State Housing in New Zealand*. “We had respect for our neighbours and addressed them by title Mr. and Mrs. so-and-so,” she recalls.

C

Quite what went wrong with Talbot Park is not clear. We call it Home Records that the community began to change in the late 1970s as more Pacific Islanders and Europeans moved in. The new arrivals didn't readily integrate with the community, a “them and us” mentality developed, and residents interact with their neighbours less. What was clear was the buildings were deteriorating and becoming dilapidated, petty crime was on the rise and the reserve – the focus of fond childhood memories – had become a wasteland and was considered unsafe.

D

But it wasn't until 2002 that Housing New Zealand decided the properties needed upgrading. The master renewal plan didn't take advantage of the maximum accommodation density allowable (one unit per 100 sq metres) but did increase

density to one unit per 180 sq m by refurbishing all 108 star flat units, removing the multis and building 111 new home. The Talbot strategy can be summed up as mix, match and manage. Mix up the housing with various plans from a mix of architects, match house styles to what's built by the private sector, match tenants to the mix, and manage their occupancy. Inevitably cost comes into the equation. "If you're going to build low-cost homes, you've got to keep them simple and you can't afford a fancy bit on them." Says Michael Thompson of Architectus which designed the innovative three-level Atrium apartments lining two sides of a covered courtyard. At \$300,000 per two-bedroom unit, the building is more expensive but provides for independent disabled accommodation as well as offering solar hot water heating and rainwater collection for toilet cisterns and outside taps.

E

The renewal project budget at \$1.5 million which will provide park pathways, planting, playgrounds, drinking fountains, seating, skateboard rails, a half-size basketball hard court, and a pavilion. But if there was any doubt this is a low socio-economic area, the demographics for the surrounding Tamaki area are sobering. Of the 5000 households there, 55 per cent are statehouses, 28 per cent privately owned (compared to about 65 per cent nationally) and 17 per cent are private rental. The area has a high concentration of households with incomes in the \$5000 to \$15,000 range and very few with an income of over \$70,000. That's in sharp contrast to the more affluent suburbs like Kohimarama and St John's that surround the area.

F

"The design is for people with different culture background," says architect James Lunday of Common Ground which designed the 21 large family homes.

"Architecturally we decided to be relatively conservative – a nice house in its own garden with a bit of space and good indoor-outdoor flow." There's a slight reflection of the whare and a Pacific fale, but not overplayed "The private sector is way behind in urban design and sustainable futures," says Bracey. "Redesigning streets and parks is a big deal and very difficult to do. The private sector won't do it, because it's so hard."

G

There's no doubt good urban design and good architecture play a significant part in the scheme. But probably more important is a new standard of social control. Housing New Zealand calls it "intensive tenancy management". Others view it as social engineering. "It's a model that we are looking at going forward," according to Housing New Zealand's central Auckland regional manager Graham Bodman. "The focus is on frequent inspections, helping tenants to get to know each other and trying to create an environment of respect for neighbours," says Bodman. That includes some strict rules – no loud parties after 10 pm, no dogs, no cats in the apartments, no washing hung over balcony rails and a requirement to mow lawns and keep the property tidy. Housing New Zealand has also been active in organising morning teas and street barbecues for residents to meet their neighbours. "It's all based on the intensification," says Community Renewal project manager Stuart Bracey. "We acknowledge if you are going to put more people living closer together, you have to

actually help them to live closer together because it creates tension – especially for people that aren't used to it."

Questions 14-20

The Reading Passage has seven paragraphs A-G

Choose the correct heading for paragraphs, A-G, from the list below.

Write the correct number, i-x, in boxes 14-20 on your answer sheet.

List of Headings

- i Financial hardship of community
- ii A good tendency of strengthening the supervision
- iii Details of plans for the community's makeover and upgrade
- iv Architecture suits families of various ethnic origins
- v Problems arise then the mentality of alienation developed later
- vi Introduction of a social housing community with unexpected high standard
- vii A practical design and need assist and cooperate in future
- viii closer relationship among neighbors in the original site
- ix different need from a makeup of a low financial background should be considered
- x How to make the community feel safe
- xi a plan with details for the house structure
- 14 Paragraph A
- 15 Paragraph B
- 16 Paragraph C
- 17 Paragraph D
- 18 Paragraph E
- 19 Paragraph F
- 20 Paragraph G

Questions 21-23

Use the information in the passage to match the people (listed A-E) with opinions or deeds below.

Write the appropriate letters, A-E, in boxes 21-23 on your answer sheet.

List of people

- A Michael Thompson
- B Graham Bodman
- C Stuart Bracey
- D James Lunday
- E Dene Busby
- 21 Design should meet the need of mix-raced cultural background
- 22 for a better living environment, regulations and social control should be imperative
- 23 organising more community's activities helps to strengthen the relationship in the community

答案

- 14. vi
- 15. viii
- 16. v
- 17. iii
- 18. ix
- 19. vii
- 20. ii
- 21. D
- 22. B
- 23. C

READING PASSAGE 3

You should spend about 20 minutes on Questions 28-40 which are based on Reading Passage 3 below.

Alzheimer

A

While it may not be possible to completely age-proof our brains, a brave new world of anti-aging research shows that our gray matter may be far more flexible than we thought. So no one, no matter how old, has to lose their mind. The brain has often been called the three-pound universe. It's our most powerful and mysterious organ, the seat of the self, laced with as many billions of neurons as the galaxy has stars. No wonder the mere notion of an aging, failing brain – and the prospect of memory loss, confusion, and the unraveling of our personality – is so terrifying. As Mark Williams, M.D., author of The American Geriatrics Society's Complete Guide to Aging and Health, says, "The fear of dementia is stronger than the fear of death itself." Yet the degeneration of the brain is far from inevitable. "Its design features are such that it should continue to function for a lifetime," says Zaven Khachaturian, Ph. D., director of the Alzheimer's Association's Ronald and Nancy Reagan Research Institute. "There's no reason to expect it to deteriorate with age, even though many of us are living longer lives." In fact, scientists' view of the brain's potential is rapidly changing, according to Stanford University neuroscientist Robert Sapolsky, Ph.D. "Thirty-five years ago we thought Alzheimer's disease was a dramatic version of normal aging. Now we realize it's a disease with a distinct pathology. In fact, some people simply don't experience any mental decline, so we've begun to study them." Antonio Damasio, M.D., Ph.D., head of the Department of Neurology at the University of Iowa and author of Descartes' Error, concurs. "Older people can continue to have extremely rich and healthy mental lives."

B

The seniors were tested in 1988 and again in 1991. Four factors were found to be related to their mental fitness: levels of education and physical activity, lung function, and feelings of self-efficacy. "Each of these elements alters the way our brain functions," says Marilyn Albert, Ph.D., of Harvard Medical School, and colleagues from Yale, Duke, and Brandeis Universities and the Mt. Sinai School of Medicine, who hypothesizes that regular exercise may actually stimulate blood flow to the brain and nerve growth, both of which create more densely branched neurons, rendering the neurons stronger and better able to resist disease. Moderate aerobic exercise, including long brisk walks and frequently climbing stairs, will accomplish this.

C

Education also seems to enhance brain function. People who have challenged themselves with at least a college education may actually stimulate the neurons in their brains. Moreover, native intelligence may protect our brains. It's possible that smart people begin life with a greater number of neurons, and therefore have a greater

reserve to fall back on if some begin to fail. “If you have a lot of neurons and keep them busy, you may be able to tolerate more damage to your brain before in shows,” says Peter Davies, M.D., of the Albert Einstein College of Medicine in the Bronx, New York. Early linguistic ability also seems to help our brains later in life. A recent study in the New England Journal of Medicine looked at 93 elderly nuns and examined the autobiographies they had written 60 years earlier, just as they were joining a convent. The nuns whose essays were complex and dense with ideas remained sharp into their eighties and nineties.

D

Finally, personality seems to play an important role in protecting our mental prowess. A sense of self-efficacy may protect our brain, buffeting it from the harmful effects of stress. According to Albert, there’ s evidence that elevated levels of stress hormones may harm brain cells and cause the hippocampus – a small seahorse-shaped organ that’ s a crucial moderator of memory – to atrophy. A sense that we can effectively chart our own course in the world may retard the release of stress hormones and protect us as we age. “It’ s not a matter of whether you experience stress or not,” Albert concludes, “it’ s your attitude toward it.” Reducing stress by meditating on a regular basis may buffer the brain as well. It also increases the activity of the brain’ s pineal gland, the source of the antioxidant hormone melatonin, which regulates sleep and may retard the aging process. Studies at the University of Massachusetts Medical Center and the University of Western Ontario found that people who meditated regularly had higher levels of melatonin than those who took 5-milligram supplements. Another study, conducted jointly by Maharishi International University, Harvard University, and the University of Maryland, found that seniors who meditated for three months experienced dramatic improvements in their psychological well-being, compared to their non-meditative peers.

E

Animal studies confirm that both mental and physical activity boost brain fitness. At the Beckman Institute for Advanced Science and Technology in Urbana, Illinois, psychologist William Greenough, Ph.D., let some rats play with a profusion of toys. These rodents developed about 25 percent more connections between their neurons than did rats that didn’ t get any mentally stimulating recreation. In addition, rats that exercised on a treadmill developed more capillaries in specific parts of their brains than did their sedentary counterparts. This increased the blood flow to their brains.

“Clearly the message is to do as many different flyings as possible,” Greenough says.

F

It’ s not just scientists who are catching the anti-aging fever. Walk into any health food store, and you’ ll find nutritional formulas – with names like Brainstorm and Smart ALEC – that claim to sharpen the mental ability. The book Smart Drugs & Nutrients, by Ward Dean, M.D., and John Morgenthaler, was self-published in 1990 and has sold over 120,000 copies worldwide. It has also spawned an underground network of people tweaking their own brain chemistry with nutrients and drugs – the latter sometimes obtained from Europe and Mexico. Sales of ginkgo – an

extract from the leaves of the 200-million-year-old ginkgo tree, which has been shown in published studies to increase oxygen in the brain and ameliorate symptoms of Alzheimer's disease – are up by 22 percent in the last six months alone, according to Paddy Spence, president of SPINS, a San Francisco-based market research firm. Indeed, products that increase and preserve mental performance are a small but emerging segment of the supplements industry, says Linda Gilbert, president of HealthFocus, a company that researches consumer health trends. While neuroscientists like Khachaturian liken the use of these products to the superstition of tossing salt over your shoulder, the public is nevertheless gobbling up nutrients that promise cognitive enhancement.

Questions 32-39

Use the information in the passage to match the people (listed A-G) with opinions or deeds below.

Write the appropriate letters A-G in boxes 32-39 on your answer sheet.

NB You may use any letter more than once

- A Zaven Khachaturian
- B William Greenough
- C Marilyn Albert
- D Robert Sapolsky
- E Linda Gilbert
- F Peter Davies
- G Paddy Spence

- 32 Alzheimer's was probably a kind of disease rather than a normal aging process.
- 33 Keeping neurons busy, people may be able to endure more harm to your brain
- 34 Regular exercises boost blood flow to the brain and increase anti-disease disability.
- 35 Significant increase in Sales of ginkgo has been shown.
- 36 More links between their neurons are found among stimulated animals.
- 37 Effectiveness of the use of brains supplements products can be of little scientific proof.
- 38 Heightened levels of stress may damage brain cells and cause part of the brain to deteriorate.
- 39 Products that upgrade and preserve mental competence are still a newly developing industry.

答案

- 32. D
- 33. F
- 34. C
- 35. G
- 36. B
- 37. A
- 38. C
- 39. E

Test 20

READING PASSAGE 1

You should spend about 20 minutes on Questions 1-13 which are based on Reading Passage 1 below.

Organic farming and chemical fertilisers

A

The world's population continues to climb. And despite the rise of high-tech agriculture, 800 million people don't get enough to eat. Clearly, it's time to rethink the food we eat and where it comes from. Feeding 9 billion people will take more than the same old farming practices, especially if we want to do it without felling rainforests and planting every last scrap of prairie. Finding food for all those people will tax farmers' – and researchers' – ingenuity to the limit. Yet already, precious aquifers that provide irrigation water for some of the world's most productive farmlands are drying up or filling with seawater, and arable land in China is eroding to create vast dust storms that redden sunsets as far away as North America.

“Agriculture must become the solution to environmental problems in 50 years. If we don't have systems that make the environment better – not just hold the fort-then we're in trouble,” says Kenneth Cassman, an agronomist at the University of Nebraska at Lincoln. That view was echoed in January by the Curry report, a government panel that surveyed the future of farming and food in Britain.

B

It's easy to say agriculture has to do better, but what should this friendly farming of the future look like? Concerned consumers come up short at this point, facing what appears to be an ever-widening ideological divide. In one corner are the techno-optimists who put their faith in genetically modified crops, improved agrochemicals and computer-enhanced machinery; in the other are advocates of organic farming, who reject artificial chemicals and embrace back-to-nature techniques such as composting. Both sides cite plausible science to back their claims to the moral high ground, and both bring enough passion to the debate for many people to come away thinking we're faced with a stark choice between two mutually incompatible options.

C

Not so. If you take off the ideological blinkers and simply ask how the world can produce the food it needs with the least environmental cost, a new middle way opens. The key is sustainability: whatever we do must not destroy the capital of soil and water we need to keep on producing. Like today's organic farming, the intelligent farming of the future should pay much more attention to the health of its soil and the ecosystem it's part of. But intelligent farming should also make shrewd and locally appropriate use of chemical fertilisers and pesticides. The most crucial ingredient in

this new style of agriculture is not chemicals but information about what's happening in each field and how to respond. Yet ironically, this key element may be the most neglected today.

D

Clearly, organic farming has all the warm, fuzzy sentiment on its side. An approach that eschews synthetic chemicals surely runs no risk of poisoning land and water. And its emphasis on building up natural ecosystems seems to be good for everyone. Perhaps these easy assumptions explain why sales of organic food across Europe are increasing by at least 50 per cent per year.

E

Going organic sounds idyllic – but it's native, too. Organic agriculture has its own suite of environmental costs, which can be worse than those of conventional farming, especially if it were to become the world norm. But more fundamentally, the organic versus-chemical debate focuses on the wrong question. The issue isn't what you put into a farm, but what you get out of it, both in terms of crop yields and pollutants, and what condition the farm is in when you're done.

F

Take chemical fertilisers, which deliver nitrogen, an essential plant nutrient, to crops along with some phosphorus and potassium. It is a mantra of organic farming that these fertilisers are unwholesome, and plant nutrients must come from natural sources. But in fact, the main environmental damage done by chemical fertilisers as opposed to any other kind is through greenhouse gases-carbon dioxide from the fossil fuels used in their synthesis and nitrogen oxides released by their degradation. Excess nitrogen from chemical fertilisers can pollute groundwater, but so can excess nitrogen from organic manures.

G

On the other hand, relying solely on chemical fertilisers to provide soil nutrients without doing other things to build healthy soil is damaging. Organic farmers don't use chemical fertilisers, so they are very good at building soil fertility by working crop residues and manure into the soil, rotating grain with legumes that fix atmospheric nitrogen, and other techniques.

H

This generates vital soil nutrients and also creates a soil that is richer in organic matter, so it retains better and is hospitable to the crop's roots and creatures such as earthworms that help maintain soil fertility. Such soil also holds water better and therefore make more efficient use of both rainfall and irrigation water. And organic matter ties up CO₂ in the soil, helping to offset emissions from burning fossil fuels and reduce global warming.

I

Advocates of organic farming like to point out that fields managed in this way can produce yields just as high as fields juiced up with synthetic fertilisers. For example, Bill Liebhardt, research manager at the Rodale Institute in Kutztown, Pennsylvania, recently compiled the results of such comparisons for corn, wheat, soybeans and tomatoes in the US and found that the organic fields averaged between 94 and 100 per

cent of the yields of nearby conventional crops.

J

But this optimistic picture tells only half the story. Farmers can't grow such crops every year if they want to maintain or build soil nutrients without synthetic fertilisers. They need to alternate with soil-building crops such as pasture grasses and legumes such as alfalfa. So in the long term, the yield of staple grains such as wheat, rice and corn must go down. This is the biggest cost of organic farming. Vaclav Smil of the University of Manitoba in Winnipeg, Canada, estimates that if farmers worldwide gave up the 80 million tonnes of synthetic fertiliser they now use each year, total grain production would fall by at least half. Either farmers would have to double the amount of land they cultivate – at catastrophic cost to natural habitats – or billions of people would starve.

K

That doesn't mean farmers couldn't get by with less fertiliser. Technologically advanced farmers in wealthy countries, for instance, can now monitor their yields hectare by hectare, or even more finely, throughout a huge field. They can then target their fertiliser to the parts of the field where it will do the most good, instead of responding to average conditions. This increases yield and decreases fertiliser use. Eventually, farmers may incorporate long-term weather forecasts into their planning as well, so that they can cut back on fertiliser use when the weather is likely to make harvests poor anyway, says Ron Olson, an agronomist with Cargill Fertilizer in Tampa, Florida.

L

Organic techniques certainly have their benefits, especially for poor farmers. But strict "organic agriculture", which prohibits certain technologies and allows others, isn't always better for the environment. Take herbicides, for example. These can leach into waterways and poison both wildlife and people. Just last month, researchers led by Tyrone Hayes at the University of California at Berkeley found that even low concentrations of atrazine, the most commonly used weedkiller in the US, can prevent frog tadpoles from developing properly.

Questions 1-4

Use the information in the passage to match the people (listed A-D) with opinions or deeds below.

Write the appropriate letters A-D in boxes 1-4 on your answer sheet.

- A Vaclav Smil
- B Bill Liebhardt
- C Kenneth Cassman
- D Ron Olson

- 1 Use of chemical fertilizer can be optimised by combining weather information.
- 2 Organic farming yield is nearly equal to traditional ones.
- 3 Better agricultural setting is a significant key to solve environmental tough nut.
- 4 Substantial production loss would happen in case all farmers shifted from using synthetic fertiliser.

答案

1. D
2. B
3. C
4. A

READING PASSAGE 2

You should spend about 20 minutes on Questions 14-27 which are based on Reading Passage 2 below.

Stealth Forces in Weight Loss

The field of weight loss is like the ancient fable about the blind men and the elephant. Each man investigates a different part of the animal and reports back, only to discover their findings are bafflingly incompatible.

A

The various findings by public-health experts, physicians, psychologists, geneticists, molecular biologists, and nutritionists are about as similar as an elephant's tusk is to its tail. Some say obesity is largely predetermined by our genes and biology; others attribute it to an overabundance of fries, soda, and screen-sucking; still, others think we're fat because of viral infection, insulin, or the metabolic conditions we encountered in the womb. "Everyone subscribes to their own little theory," says Robert Berkowitz, medical director of the Center for Weight and Eating Disorders at the University of Pennsylvania School of Medicine. We're programmed to hang onto the fat we have, and some people are predisposed to create and carry more fat than others. Diet and exercise help, but in the end, the solution will inevitably be more complicated than pushing away the plate and going for a walk. "It's not as simple as 'You're fat because you're lazy'," says Nikhil Dhurandhar, an associate professor at Pennington Biomedical Research Center in Baton Rouge. "Willpower is not a prerogative of thin people. It's distributed equally."

B

Science may still be years away from giving us a miracle formula for fat-loss. The hormone leptin is a crucial player in the brain's weight-management circuitry. Some people produce too little leptin; others become desensitized to it. And when obese people lose weight, their leptin levels plummet along with their metabolism. The body becomes more efficient at using fuel and conserving fat, which makes it tough to keep the weight off. Obese dieters' bodies go into a state of chronic hunger, a feeling Rudolph Leibel, an obesity researcher at Columbia University, compares to thirst. "Some people might be able to tolerate chronic thirst, but the majority couldn't stand it," says Leibel. "Is that a behavioral problem – a lack of willpower? I don't think so."

C

The government has long espoused moderate daily exercise – of the evening-walk or take-the-stairs variety – but that may not do much to budge the needle on the scale. A 150-pound person burns only 150 calories on a half-hour walk, the equivalent of two apples. It's good for the heart, less so for the gut. "Radical changes are necessary," says Deirdre Barrett, a psychologist at Harvard Medical School and author of *Waistland*. "People don't lose weight by choosing the small fries or taking a little walk every other day," Barrett suggests taking a cue from the members of the National Weight Control Registry (NWCR), a self-selected group of more than 5,000

successful weight-losers who have shed an average of 66 pounds and kept it off 5.5 years. Some registry members lost weight using low-carb diets; some went low-fat; others eliminated refined foods. Some did it on their own; others relied on counseling. That said, not everyone can lose 66 pounds and not everyone needs to. The goal shouldn't be getting thin, but getting healthy. It's enough to whittle your weight down to the low end of your set range, says Jeffrey Friedman, a geneticist at Rockefeller University. Losing even 10 pounds vastly decrease your risk of diabetes, heart disease, and high blood pressure. The point is to not give up just because you don't look like a swimsuit model.

D

The negotiation between your genes and the environment begins on day one. Your optimal weight, writ by genes, appears to get edited early on by conditions even before birth, inside the womb. If a woman has high blood-sugar levels while she's pregnant, her children are more likely to be overweight or obese, according to a study of almost 10,000 mother-child pairs. Maternal diabetes may influence a child's obesity risk through a process called metabolic imprinting, says Teresa Hillier, an endocrinologist with Kaiser Permanente's Center for Health Research and the study's lead author. The implication is clear: Weight may be established very early on, and obesity largely passed from mother to child. Numerous studies in both animals and humans have shown that a mother's obesity directly increases her child's risk for weight gain. The best advice for mons-to-be: Get fit before you get pregnant. You'll reduce your risk of complications during pregnancy and increase your chances of having a normal-weight child.

E

It's the \$64,000 question: Which diets work? It got people wondering. Isn't there a better way to diet? A study seemed to offer an answer. The paper compared two groups of adults: those who, after eating, secreted high levels of insulin, a hormone that sweeps blood sugar out of the bloodstream and promotes its storage as fat, and those who secreted less. Within each group, half were put on a low-fat diet and half on a low-glycemic-load diet. On average, the low-insulin-secreting group fared the same on both diets, losing nearly 10 pounds in the first six months – but they gained about half of it back by the end of the 18-month study. The high-insulin group didn't do as well on the low-fat plan, losing about 4.5 pounds, and gaining back more than half by the end. But the most successful was the high-insulin-secretors on the low-glycemic-load diet. They lost nearly 13 pounds and kept it off.

F

What if your fat is caused not by diet or genes, but by germs – says, a virus? It sounds like a sci-fi horror movie, but research suggests some dimension of the obesity epidemic may be attributable to infection by common viruses, says Dhurandhar. The idea of “infectobesity” came to him 20 years ago when he was a young doctor treating obesity in Bombay. He discovered that a local avian virus, SMAM-1, caused chickens to die, sickened with organ damage but also, strangely, with lots of abdominal fat. In experiments, Dhurandhar found that SMAM-1-infected chickens became obese on the same diet as uninfected ones, which stayed svelte.

G

He later moved to the U.S. and onto a bona fide human virus, adenovirus 36 (AD-36). In the lab, every species of animal Dhurandhar infected with the virus became obese – chickens got fat, mice got fat, even rhesus monkeys at the zoo that picked up the virus from the environment suddenly gained 15 percent of their body weight upon exposure. In his latest studies, Dhurandhar has isolated a gene that, when blocked from expressing itself, seems to turn off the virus' s fattening power. Stem cells extracted from fat cells and the exposed to AD-36 virus with the key gene inhibited, the stems cells don' t differentiate. The gene appears to be necessary and sufficient to trigger AD-36-related obesity, and the goal is to use the research to create a sort of obesity vaccine.

Researchers have discovered 10 microbes so far that trigger obesity – seven of their viruses. It may be a long shot, but for people struggling desperately to be thin, even the possibility of an alternative cause of obesity offers some solace. “They feel better knowing there may be something beyond them that could be responsible,” says Dhurandhar. “The thought that there could be something besides what they’ ve heard all their lives – that they are greedy and lazy – helps.”

Questions 14-18

The Reading Passage 2 has seven sections A-G

Which section contains the following information?

Write the correct letter A-G, in boxes 14-18 on your answer sheet.

NB You may use any letter more than once.

- 14 evaluation on the effect of weight loss on different kind of diets
- 15 an example of research which include relatives of participants
- 16 Example of a group of people who never regain weight immediately after.
- 17 long term hunger may appear to be acceptable to most of the participants while losing weight
- 18 a continuous experiment may lead to a practical application besides diet or hereditary resort.

Questions 19-23

Look at the following researchers and the list of findings below.

Match each researcher with the correct finding.

Write the correct letter in boxes 19-23 on your answer sheet.

List Of Researchers

- A Robert Berkowitz
- B Rudolph Leibel
- C Nikhil Dhurandhar
- D Deirdre Barrett
- E Jeffery Friedman
- F Teresa Hillier

-
- 19 A person's weight is predetermined to a set point by the DNA.
- 20 Pregnant mother who are overweight may risk their fetus
- 21 The aim of losing weight should be keeping healthy rather than attractiveness.
- 22 Small changes in lifestyle will not have great impact on reducing much weight
- 23 Researchers should be divided into different groups with their own point of view about weight loss.

答案

14. E
15. D
16. C
17. B
18. G
19. C
20. F
21. E
22. D
23. A

Test 21

READING PASSAGE 2

You should spend about 20 minutes on Questions 14-27 which are based on Reading Passage 2 below.

Is Graffiti Art or Crime?

A

The term graffiti derives from the Italian graffito meaning ‘scratching’ and can be defined as uninvited marking or writing scratched or applied to objects, built structures and natural features. It is not a new phenomenon: examples can be found on ancient structures around the world, in some cases predating the Greeks and Romans. In such circumstances it has acquired invaluable historical and archaeological significance, providing a social history of life and events at that time. Graffiti is now a problem that has become pervasive, as a result of the availability of cheap and quick means of mark-making. History

B

It is usually considered a priority to remove graffiti as quickly as possible after it appears. This is for several reasons. The first is to prevent ‘copy-cat’ emulation which can occur rapidly once a clean surface is defaced. It may also be of a racist or otherwise offensive nature and many companies and councils have a policy of removing this type of graffiti within an hour or two of it being reported. Also, as paints, glues and inks dry out over time they can become increasingly difficult to remove and are usually best dealt with as soon as possible after the incident. Graffiti can also lead to more serious forms of vandalism and, ultimately, the deterioration of an area, contributing to social decline.

C

Although graffiti may be regarded as an eyesore, any proposal to remove it from sensitive historic surfaces should be carefully considered: techniques designed for more robust or utilitarian surfaces may result in considerable damage. In the event of graffiti incidents, it is important that the owners of buildings or other structures and their consultants are aware of the approach they should take in dealing with the problem. The police should be informed as there may be other related attacks occurring locally. An incidence pattern can identify possible culprits, as can stylised signatures or nicknames, known as ‘tags’, which may already be familiar to local police. Photographs are useful to record graffiti incidents and may assist the police in bringing a prosecution. Such images are also required for insurance claims and can be helpful in cleaning operatives, allowing them to see the problem area before arriving on site.

D

There are a variety of methods that are used to remove graffiti. Broadly these divide

between chemical and mechanical systems. Chemical preparations are based on dissolving the media; these solvents can range from water to potentially hazardous chemical ‘cocktails’. Mechanical systems such as wire-brushing and grit-blasting attempt to abrade or chip the media from the surface. Care should be taken to comply with health and safety legislation with regard to the protection of both passers-by and any person carrying out the cleaning. Operatives should follow product guidelines in terms of application and removal, and wear the appropriate protective equipment. Measures must be taken to ensure that run-off, aerial mists, drips and splashes do not threaten unprotected members of the public. When examining a graffiti incident it is important to assess the ability of the substrate to withstand the prescribed treatment. If there is any doubt regarding this, then small trial areas should be undertaken to assess the impact of more extensive treatment.

E

A variety of preventive strategies can be adopted to combat a recurring problem of graffiti at a given site. As no two sites are the same, no one set of protection measures will be suitable for all situations. Each site must be looked at individually. Surveillance systems such as closed-circuit television may also help. In cities and towns around the country, prominently placed cameras have been shown to reduce anti-social behavior of all types including graffiti. Security patrols will also act as a deterrent to prevent recurring attacks. However, the cost of this may be too high for most situations. A physical barriers such as a wall, railings, doors or gates can be introduced to discourage unauthorized access to a vulnerable site. However, consideration has to be given to the impact measures have on the structure being protected. In the worst cases, they can be almost as damaging to the quality of the environment as the graffiti they prevent. In others, they might simply provide a new surface for graffiti.

F

One of the most significant problems associated with graffiti removal is the need to remove it from surfaces that are repeatedly attacked. Under these circumstances, the repeated removal of graffiti using even the most gentle methods will ultimately cause damage to the surface material. There may be situations where the preventive strategies mentioned above do not work or are not a viable proposition at a given site. Anti-graffiti coatings are usually applied by brush or spray leaving a thin veneer that essentially serves to isolate the graffiti from the surface.

G

Removal of graffiti from a surface that has been treated in this way is much easier, usually using low-pressure water which reduces the possibility of damage. Depending on the type of barrier selected it may be necessary to reapply the coating after each graffiti removal exercise.

Questions 14-19

Reading passage 2 has seven paragraphs, A-G.

Which paragraph contains the following information?

Write the correct letter, A-G, in boxes 14-19 on your answer sheet. Books & Literature

NB You may use any letter more than once.

- 14 why chemically cleaning graffiti may cause damage
- 15 the benefit of a precautionary strategy on the gentle removal
- 16 the damaging and accumulative impact of graffiti on the community
- 17 the need for different preventive measures being taken to cope with graffiti
- 18 a legal proposal made to the owner of building against graffiti
- 19 the reasons for removing graffiti as soon as possible.

答案

14.D

15.F

16.B

17.E

18.C

19.B

Test 24

READING PASSAGE 1

You should spend about 20 minutes on Questions 1-13 which are based on Reading Passage 1 below.

Bovids

A

The family of mammals called bovids belongs to the Artiodactyl class, which also includes giraffes. Bovids are a highly diverse group consisting of 137 species, some of which are man's most important domestic animals.

B

Bovids are well represented in most parts of Eurasia and Southeast Asian islands, but they are by far the most numerous and diverse in the latter. Some species of bovid are solitary, but others live in large groups with complex social structures. Although bovids have adapted to a wide range of habitats, from arctic tundra to deep tropical forest, the majority of species favour open grassland, scrub or desert. This diversity of habitat is also matched by great diversity in size and form: at one extreme is the royal antelope of West Africa, which stands a mere 25 cm at the shoulder; at the other, the massively built bison of North America and Europe, growing to a shoulder height of 2.2m.

C

Despite differences in size and appearance, bovids are united by the possession of certain common features. All species are ruminants, which means that they retain undigested food in their stomachs, and regurgitate it as necessary. Bovids are almost exclusively herbivorous: plant-eating "incisors: front teeth herbivorous".

D

Typically their teeth are highly modified for browsing and grazing: grass or foliage is cropped with the upper lip and lower incisors** (the upper incisors are usually absent), and then ground down by the cheek teeth. As well as having cloven, or split, hooves, the males of all bovid species and the females of most carry horns. Bovid horns have bony cores covered in a sheath of horny material that is constantly renewed from within; they are unbranched and never shed. They vary in shape and size: the relatively simple horns of a large Indian buffalo may measure around 4 m from tip to tip along the outer curve, while the various gazelles have horns with a variety of elegant curves.

E

Five groups, or sub-families, may be distinguished: Bovinae, Antelope, Caprinae, Cephalophinae and Antilocapridae. The sub-family Bovinae comprises most of the larger bovids, including the African bongo, and nilgai, eland, bison and cattle. Unlike most other bovids they are all non-territorial. The ancestors of the various species of

domestic cattle banteng, gaur, yak and water buffalo are generally rare and endangered in the wild, while the auroch (the ancestor of the domestic cattle of Europe) is extinct.

F

The term ‘antelope’ is not a very precise zoological name – it is used to loosely describe a number of bovids that have followed different lines of development. Antelopes are typically long-legged, fast-running species, often with long horns that may be laid along the back when the animal is in full flight. There are two main sub-groups of antelope: Hippotraginae, which includes the oryx and the addax, and Antilopinae, which generally contains slighter and more graceful animals such as gazelle and the springbok. Antelopes are mainly grassland species, but many have adapted to flooded grasslands: puku, waterbucks and lechwes are all good at swimming, usually feeding in deep water, while the sitatunga has long, splayed hooves that enable it to walk freely on swampy ground.

G

The sub-family Caprinae includes the sheep and the goat, together with various relatives such as the goral and the tahr. Most are woolly or have long hair. Several species, such as wild goats, chamois and ibex, are agile cliff – and mountain-dwellers. Tolerance of extreme conditions is most marked in this group: Barbary and bighorn sheep have adapted to arid deserts, while Rocky Mountain sheep survive high up in mountains and musk oxen in arctic tundra.

H

The duiker of Africa belongs to the Cephalophinae sub-family. It is generally small and solitary, often living in thick forest. Although mainly feeding on grass and leaves, some duikers – unlike most other bovids – are believed to eat insects and feed on dead animal carcasses, and even to kill small animals.

I

The pronghorn is the sole survivor of a New World sub-family of herbivorous ruminants, the Antilocapridae in North America. It is similar in appearance and habits to the Old World antelope. Although greatly reduced in numbers since the arrival of Europeans, and the subsequent enclosure of grasslands, the pronghorn is still found in considerable numbers throughout North America, from Washington State to Mexico. When alarmed by the approach of wolves or other predators, hairs on the pronghorn’s rump stand erect, so showing and emphasizing the white patch there. At this signal, the whole herd gallops off at speed of over 60 km per hour.

Questions 4-8

Look at the following characteristics (Questions 4-8) and the list of sub-families below.

Match each characteristic with the correct sub-family, A, B, C or D.

Write the correct letter, A, B, C or D, in boxes 4-8 on your answer sheet.

NB You may use any letter more than once

4 can endure very harsh environments

5 includes the ox and the cow

-
- 6 may supplement its diet with meat
7 can usually move a speed
8 does not defend a particular area of land

List of sub-families

- A Antelope
B Bovinae
C Caprinae
D Cephalophinae

答案

4. C
5. B
6. D
7. A
8. B

READING PASSAGE 2

You should spend about 20 minutes on Questions 14-27 which are based on Reading Passage 2 below.

Art in Iron and Steel

A

Works of engineering and technology are sometimes viewed as the antitheses of art and humanity. Think of the connotations of assembly lines, robots, and computers. Any positive values there might be in such creations of the mind and human industry can be overwhelmed by the associated negative images of repetitive, stressful, and threatened jobs. Such images fuel the arguments of critics of technology even as they may drive powerful cars and use the Internet to protest what they see as the artless and dehumanizing aspects of living in an industrialized and digitized society. At the same time, landmark megastructures such as the Brooklyn and Golden Gate bridges are almost universally hailed as majestic human achievements as well as great engineering monuments that have come to embody the spirits of their respective cities. The relationship between art and engineering has seldom been easy or consistent.

B

The human worker may have appeared to be but a cog in the wheel of industry, yet photographers could reveal the beauty of line and composition in a worker doing something as common as using a wrench to turn a bolt. When Henry Ford's enormous River Rouge plant opened in 1927 to produce the Model A, the painter/photographer Charles Sheeler was chosen to photograph it. The world's largest car factory captured the imagination of Sheeler, who described it as the most thrilling subject he ever had to work with. The artist also composed oil paintings of the plant, giving them titles such as American Landscape and Classic Landscape.

C

Long before Sheeler, other artists, too, had seen the beauty and humanity in works of engineering and technology. This is perhaps no more evident than in Coalbrookdale, England, where iron, which was so important to the industrial revolution, was worked for centuries. Here, in the late eighteenth century, Abraham Darby III cast on the banks of the Severn River the large ribs that formed the world's first iron bridge, a dramatic departure from the classic stone and timber bridges that dotted the countryside and were captured in numerous serene landscape paintings. The metal structure, simply but appropriately called Iron Bridge, still spans the river and still beckons engineers, artists, and tourists to gaze upon and walk across it, as if on a pilgrimage to a revered place.

D

At Coalbrookdale, the reflection of the ironwork in the water completes the semicircular structure to form a wide-open eye into the future that is now the past. One artist's bucolic depiction shows pedestrians and horsemen on the bridge, as if on a woodland trail. On one shore, a pair of well-dressed onlookers interrupts their stroll along the riverbank, perhaps to admire the bridge. On the other side of the gently

flowing river, a lone man leads two mules beneath an arch that lets the towpath pass through the bridge's abutment. A single boatman paddles across the river in a tiny tub boat. He is in no rush because there is no towline to carry from one side of the bridge to the other. This is how Michael Rooker was Iron Bridge in his 1792 painting. A colored engraving of the scene hangs in the nearby Coalbrookdale museum, along with countless other contemporary renderings of the bridge in its full glory and in its context, showing the iron structure not as a blight on the landscape but at the center of it. The surrounding area at the same time radiates out from the bridge and pales behind it.

E

In the nineteenth century, the railroads captured the imagination of artists, and the steam engine in the distance of a landscape became as much a part of it as the herd of cows in the foreground. The Impressionist Claude Monet painted man-made structures like railway stations and cathedrals as well as water lilies. Portrait painters such as Christian Schussele found subjects in engineers and inventors – and their inventions – as well as in the American founding fathers. By the twentieth century, engineering, technology, and industry were very well established as subjects for artists.

F

American-born Joseph Pennell illustrated many European travel articles and books. Pennell, who early in his career made drawings of buildings under construction and shrouded in scaffolding, returned to America late in life and recorded industrial activities during World War I. He is perhaps best known among engineers for his depiction of the Panama Canal as it neared completion and his etchings of the partially completed Hell Gate and Delaware River bridges.

G

Pennell has often been quoted as saying, “Great engineering is great art,” a sentiment that he expressed repeatedly. He wrote of his contemporaries, “I understand nothing of engineering, but I know that engineers are the greatest architects and the most pictorial builders since the Greeks.” Where some observers saw only utility, Pennell saw also beauty, if not in form then at least in scale. He felt he was not only rendering a concrete subject but also conveying through his drawings the impression that it made on him. Pennell called the sensation that he felt before a great construction project “The Wonder of Work”. He saw engineering as a process. That process is memorialized in every completed dam, skyscraper, bridge, or other great achievement of engineering.

H

If Pennell experienced the wonder of work in the aggregate, Lewis Hine focused on the individuals who engaged in the work. Hine was trained as a sociologist but became best known as a photographer who exposed the exploitation of children. His early work documented immigrants passing through Ellis Island, along with the conditions in the New York tenements where they lived and the sweatshops where they worked. Upon returning to New York, he was given the opportunity to record the construction of the Empire State Building, which resulted in the striking photographs

that have become such familiar images of daring and insouciance. He put his own life at risk to capture workers suspended on cables hundreds of feet in the air and sitting on a high girder eating lunch. To engineers today, one of the most striking features of these photos, published in 1932 in *Men at Work*, is the absence of safety lines and hard hats. However, perhaps more than anything, the photos evoke Pennell's "The Wonder of Work" and inspire admiration for the bravery and skill that bring a great engineering project to completion.

Questions 14-18

The Reading Passage has eight paragraphs A-H

Which paragraph contains the following information?

Write the correct letter A-H, in boxes 14-18 on your answer sheet.

- 14 Art connected with architecture for the first time.
- 15 small artistic object and constructions built are put together
- 16 the working condition were recorded by the artist as an exciting subject.
- 17 mention of one engineers' artistic work on an unfinished engineering project
- 18 Two examples of famous bridges which became the iconic symbols of those cities

Questions 19-23

Use the information in the passage to match the people (listed A-F) with opinions or deeds below.

Write the appropriate letters A-F in boxes 19-23 on your answer sheet.

List of people

- A Charles Sheeler
- B Michael Rooker
- C Claude Monet
- D Christian Schussele
- E Joseph Pennell
- F Lewis Hine

- 19 who made a comment that concrete constructions have a beauty just as artistic processes created by engineers the architects
- 20 who made a romantic depiction of an old bridge in one painting
- 21 who produced art pieces demonstrating the courage of workers in the site
- 22 who produced portraits involving subjects in engineers and inventions and historical human heroes.
- 23 who produced a painting of factories and named them ambitiously

答案

- 14. C
- 15. E
- 16. B
- 17. F

-
- 18. A
 - 19. E
 - 20. B
 - 21. F
 - 22. D
 - 23. A

READING PASSAGE 3

You should spend about 20 minutes on Questions 28-40 which are based on Reading Passage 3 below.

Psychology Of New Product Adoption

A

In today's hypercompetitive marketplace, companies that successfully introduce new products are more likely to flourish than those that don't. Businesses spend billions of dollars making better "mousetraps" only to find consumers roundly rejecting them. Studies show that new products fail at the stunning rate of between 40% and 90%, depending on the category, and the odds haven't changed much in the past 25 years. In the U.S. packaged goods industry, for instance, companies introduce 30,000 products every year, but 70% to 90% of them don't stay on store shelves for more than 12 months. Most innovative products – those that create new product categories or revolutionize old ones – are also unsuccessful. According to one study, 47% of first movers have failed, meaning that approximately half the companies that pioneered new product categories later pulled out of those businesses.

B

After the fact, experts and novices alike tend to dismiss unsuccessful innovations as bad ideas that were destined to fail. Why do consumers fail to buy innovative products even when they offer distinct improvements over existing ones? Why do companies invariably have more faith in new products than is warranted? Few would question the objective advantages of many innovations over existing alternatives, but that's often not enough for them to succeed. To understand why new products fail to live up to companies' expectations, we must delve into the psychology of behavior change.

C

New products often require consumers to change their behavior. As companies know, those behavior changes entail costs. Consumers' costs, such as the activation fees they have to pay when they switch from one cellular service provider to another. They also bear learning costs, such as when they shift from manual to automatic automobile transmissions. People sustain obsolescence costs, too. For example, when they switch from VCRs to DVD players, their videotape collections become useless. All of these are economic switching costs that most companies routinely anticipate.

D

What businesses don't take into account, however, are the psychological costs associated with behavior change. Many products fail because of a universal, but largely ignored, psychological bias: People irrationally overvalue benefits they currently possess relative to those they don't. The bias leads consumers to value the advantages of products they own more than the benefits of new ones. It also leads

executives to value the benefits of innovations they've developed over the advantages of incumbent products.

E

Companies have long assumed that people will adopt new products that deliver more value or utility than existing ones. Thus, businesses need only to develop innovations that are objectively superior to incumbent products, and consumers will have sufficient incentive to purchase them. In the 1960s, communications scholar Everett Rogers called the concept "relative advantage" and identified it as the most critical driver of new-product adoption. This argument assumes that companies make unbiased assessments of innovations and of consumers, likelihood of adopting them. Although compelling, the theory has one major flaw: It fails to capture the psychological biases that affect decision making.

F

In 2002, psychologist Daniel Kahneman won the Nobel Prize in economics for a body of work that explores why and when individuals deviate from rational economic behavior. One of the cornerstones of that research, developed with psychologist Amos Tversky, is how individuals value prospects, or choices, in the marketplace. Kahneman and Tversky showed, and others have confirmed, that human beings' responses to the alternatives before them have four distinct characteristics.

G

First, people evaluate the attractiveness of an alternative based not on its objective, or actual, value but on its subjective, or perceived value. Second, consumers evaluate new products or investments relative to a reference point, usually the products they already own or consume. Third, people view any improvements relative to this reference point as gains and treat all shortcomings as losses. Fourth, and most important, losses have a far greater impact on people than similarly sized gains, a phenomenon that Kahneman and Tversky called "loss aversion." For instance, studies show that most people will not accept a bet in which there is a 50% chance of winning \$100 and a 50% chance of losing \$100. The gains from the wager must outweigh the losses by a factor of between two and three before most people find such a bet attractive. Similarly, a survey of 1,500 customers of Pacific Gas and Electric revealed that consumers demand three to four times more compensation to endure a power outage – and suffer a loss – than they are willing to pay to avoid the problem, a potential gain. As Kahneman and Tversky wrote, "losses loom larger than gains."

H

Loss aversion leads people to value products that they already possess – those that are part of their endowment – more than those they don't have. According to behavioral economist Richard Thaler, consumers value what they own, but many have to give up, much more than they value what they don't own but could obtain. Thaler called that bias the "endowment effect."

I

In a 1990 paper, Thaler and his colleagues describe a series of experiments they conducted to measure the magnitude of the endowment effect. In one such experiment,

they gave coffee mugs to a group of people, the Sellers, and asked at what price point – from 25 cents to \$9.25 – the Sellers would be willing to part with those mugs. They asked another group – the Choosers – to whom they didn't give coffee mugs, to indicate whether they would choose the mug or the money at each price point. In objective terms, all the Sellers and Choosers were in the same situation: They were choosing between a mug and a sum of money. In one trial of this experiment, the Sellers priced the mug at \$7.12, on average, but the Choosers were willing to pay only \$3.12. In another trial, the Sellers and the Choosers valued the mug at \$7.00 and \$3.50, respectively. Overall, the Sellers always demanded at least twice as much to give up the mugs as the Choosers would pay to obtain them.

J

Kahneman and Tversky's research also explains why people tend to stick with what they have even if a better alternative exists. In a 1989 paper, economist Jack Knetsch provided a compelling demonstration of what economists William Samuelson and Richard Zeckhauser called the “status quo bias.” Knetsch asked one group of students to choose between an attractive coffee mug and a large bar of Swiss chocolate. He gave a second group of students the coffee mugs but a short time later allowed each student to exchange his or her mug for a chocolate bar. Finally, Knetsch gave chocolate bars to a third group of students but much later allowed each student to exchange his or her bar for a mug. Of the students given a choice at the outset, 56% chose the mug, and 44% chose the chocolate bar, indicating a near even split in preferences between the two products. Logically, therefore, about half of the students to whom Knetsch gave the coffee mug should have traded for the chocolate bar and vice versa. That didn't happen. Only 11% of the students who had been given the mugs and 10% of those who had been given the chocolate bars wanted to exchange their products. To approximately 90% of the students, giving up what they already had seemed like a painful loss and shrank their desire to trade.

K

Interestingly, most people seem oblivious to the existence of the behaviors implicit in the endowment effect and the status quo bias. In study after study, when researchers presented people with evidence that they had irrationally overvalued the status quo, they were shocked, skeptical, and more than a bit defensive. These behavioral tendencies are universal, but awareness of them is not.

Questions 28-31

Use the information in the passage to match the people (listed A-C) with opinions or deeds below.

Write the appropriate letters A-C in boxes 28-31 on your answer sheet.

A Richard Thaler

B Everett Rogers

C Kahneman and Tversky

28 stated a theory which bears potential fault in the application

29 decided the consumers' several behavior features when they face other options

30 generalised that customers value more of their possession they are going to

abandon for a purpose than alternative they are going to swap in

31 answered the reason why people don't replace existing products

答案

28. B

29. C

30. A

31. C

Test 22

READING PASSAGE 1

You should spend about 20 minutes on Questions 1-13 which are based on Reading Passage 1 below.

Plant Scents

A

Everyone is familiar with scented flowers, and many people have heard that floral odors help the plant attract pollinators. This common notion is mostly correct, but it is surprising how little scientific proof of it exists. Of course, not all flowers are pollinated by biological agents – for example, many grasses are wind-pollinated – but the flowers of the grasses may still emit volatiles. In fact, plants emit organic molecules all the time, although they may not be obvious to the human nose. As for flower scents that we can detect with our noses, bouquets that attract moths and butterflies generally smell “sweet,” and those that attract certain flies seem “rotten” to us.

B

The release of volatiles from vegetative parts of the plant is familiar, although until recently the physiological functions of these chemicals were less clear and had received much less attention from scientists. When the trunk of a pine tree is injured – for example, when a beetle tries to burrow into it – it exudes a very smelly resin. This resin consists mostly of terpenes – hydrocarbons with a backbone of 10, 15 or 20 carbons that may also contain atoms of oxygen. The heavier C₂₀ terpenes, called diterpenes, are glue-like and can cover and immobilize insects as they plug the hole. This defense mechanism is as ancient as it is effective: Many samples of fossilized resin, or amber, contain the remains of insects trapped inside. Many other plants emit volatiles when injured, and in some cases the emitted signal helps defend the plant. For example, (Z)-3-hexenyl acetate, which is known as a “green leaf volatile” because it is emitted by many plants upon injury, deters females of the moth *Heliothis virescens* from laying eggs on injured tobacco plants. Interestingly, the profile of emitted tobacco volatiles is different at night than during the day, and it is the nocturnal blend, rich in several (Z)-3-hexen-1-olesters, that is most effective in repelling the night-active *H. virescens* moths.

C

Herbivore induced volatiles often serve as indirect defenses. These bulwarks exist in a variety of plant species, including corn, beans, and the model plant species *Arabidopsis thaliana*. Plants not only emit volatiles acutely, at the site where caterpillars, mites, aphids or similar insects are eating them but also generally from non-damaged parts of the plant. These signals attract a variety of predatory insects that prey on the plant-eaters. For example, some parasitic wasps can detect the

volatile signature of a damaged plant and will lay their eggs inside the offending caterpillar; eventually, the wasp eggs hatch, and the emerging larvae feed on the caterpillar from the inside hatch, and the emerging larvae feed on the caterpillar from the inside out. The growth of infected caterpillars is retarded considerably, to the benefit of the plant. Similarly, volatiles released by plants in response to herbivore egg laying can attract parasites of the eggs, thereby preventing them from hatching and avoiding the onslaught of hungry herbivores that would have emerged. Plant volatiles can also be used as a kind of currency in some very indirect defensive schemes. In the rainforest understory tree *Leonardoxa Africana*, ants of the species *Petalomyrmex phylax* patrol young leaves and attack any herbivorous insects that they encounter. The young leaves emit high levels of the volatile compound methyl salicylate, a compound that the ants use either as a pheromone or as an antiseptic in their nests. It appears that methyl salicylate is both an attractant and a reward offered by the tree to get the ants to perform this valuable deterrent role.

D

Floral scent has a strong impact on the economic success of many agricultural crops that rely on insect pollinators, including fruit trees such as the bee-pollinated cherry, apple, apricot and peach, as well as vegetables and tropical plants such as papaya. Pollination not only affects crop yield, but also the quality and efficiency of crop production. Many crops require most, if not all, ovules to be fertilized for optimum fruit size and shape. A decrease in fragrance emission reduces the ability of flowers to attract pollinators and results in considerable losses for growers, particularly for introduced species that had a specialized pollinator in their place of origin. This problem has been exacerbated by recent disease epidemics that have killed many honeybees, the major insect pollinators in the United States.

E

One means by which plant breeders circumvent the pollination problem is by breeding self-compatible, or apomictic, varieties that do not require fertilization. Although this solution is adequate, its drawbacks include near genetic uniformity and consequent susceptibility to pathogens. Some growers have attempted to enhance honeybee foraging by spraying scent compounds on orchard trees, but this approach was costly, had to be repeated, had potentially toxic effects on the soil or local biota, and, in the end, proved to be inefficient. The poor effectiveness of this strategy probably reflects inherent limitations of the artificial, topically applied compounds, which clearly fail to convey the appropriate message to the bees. For example, general spraying of the volatile mixture cannot tell the insects where exactly the blossoms are. Clearly, a more refined strategy is needed. The ability to enhance existing floral scent, which could all be accomplished by genetic engineering, would allow us to manipulate the types of insect pollinators and the frequency of their visits. Moreover, the metabolic engineering of fragrance could increase crop protection against pathogens and pests.

F

Genetic manipulation of the scent will also benefit the floriculture industry. Ornamentals, including cut flowers, foliage and potted plants, play an important aesthetic role in human life. Unfortunately, traditional breeding has often produced

cultivars with improved vase life, shipping characteristics, color and shape while sacrificing desirable perfumes. The loss of scent among ornamentals, which have a worldwide value of more than \$30 billion, makes them important targets for the genetic manipulation of flower fragrance. Some work has already begun in this area, as several groups have created petunia and carnation plants that express the linalool synthase gene from *C. Breweri*. These experiments are still preliminary: For technical reasons, the gene was expressed everywhere in the plant, and although the transgenic plants did create small amounts of linalool, the level was below the threshold of detection for the human nose. Similar experiments in tobacco used genes for other monoterpene synthases, such as the one that produces limonene, but gave similar results.

G

The next generation of experiments, already in progress, includes sophisticated schemes that target the expression of scent genes specifically to flowers or other organs – such as special glands that can store antimicrobial or herbivore-repellent compounds.

Questions 1-4

The Reading Passage has seven paragraphs A-G

Which paragraph contains the following information?

Write the correct letter A-G, in boxes 1-4 on your answer sheet.

- 1 Substance released to help plants themselves.
- 2 Scent helps plant' s pollination.
- 3 Practice on genetic experiment of fragrance.
- 4 Plant' s scent attracts herbivore' s enemy for protection.

答案

1. B
2. A
3. F
4. C

READING PASSAGE 2

You should spend about 20 minutes on Questions 14-26 which are based on Reading Passage 2 below.

We have Star performers!

A

The difference between companies is people. With capital and technology in plentiful supply, the critical resource for companies in the knowledge era will be human talent. Companies full of achievers will, by definition, outperform organisations of plodders. Ergo, compete ferociously for the best people. Poach and pamper stars; ruthlessly weed out second-raters. This, in essence, has been the recruitment strategy of the ambitious company of the past decade. The ‘talent mindset’ was given definitive form in two reports by the consultancy McKinsey famously entitled *The War for Talent*. Although the intensity of the warfare subsequently subsided along with the air in the internet bubble, it has been warming up again as the economy tightens: labour shortages, for example, are the reason the government has laid out the welcome mat for immigrants from the new Europe.

B

Yet while the diagnosis – people are important – is evident to the point of platitude, the apparently logical prescription – hire the best – like so much in management is not only not obvious: it is in fact profoundly wrong. The first suspicions dawned with the crash to earth of the dotcom meteors, which showed that dumb is dumb whatever the IQ of those who perpetrate it. The point was illuminated in brilliant relief by Enron, whose leaders, as a New Yorker article called ‘The Talent Myth’ entertainingly related, were so convinced of their own cleverness that they never twigged that collective intelligence is not the sum of a lot of individual intelligence. In fact, in a profound sense, the two are opposites. Enron believed in stars, noted author Malcolm Gladwell, because they didn’t believe in systems. But companies don’t just create: ‘they execute and compete and coordinate the efforts of many people, and the organisations that are most successful at that task are the ones where the system is the star’. The truth is that you can’t win the talent wars by hiring stars – only lose it. New light on why this should be so is thrown by an analysis of star behaviour in this month’s Harvard Business Review. In a study of the careers of 1,000 star-stock analysts in the 1990s, the researchers found that when a company recruited a star performer, three things happened.

C

First, stardom doesn’t easily transfer from one organisation to another. In many cases, performance dropped sharply when high performers switched employers and in some instances never recovered. More of success than commonly supposed is due to the working environment – systems, processes, leadership, accumulated embedded learning that are absent in and can’t be transported to the new firm. Moreover, precisely because of their past stellar performance, stars were unwilling to learn new tricks and antagonised those (on whom they now unwittingly depended) who could

teach them. So they moved, upping their salary as they did – 36 per cent moved on within three years, fast even for Wall Street. Second, group performance suffered as a result of tensions and resentment by rivals within the team. One respondent likened hiring a star to an organ transplant. The new organ can damage others by hogging the blood supply, other organs can start aching or threaten to stop working or the body can reject the transplants altogether, he said. ‘You should think about it very carefully before you do a transplant to a healthy body.’ Third, investors punished the offender by selling its stock. This is ironic since the motive for importing stars was often a suffering share price in the first place. Shareholders evidently believe that the company is overpaying, the hiree is cashing in on a glorious past rather than preparing for a glowing present, and a spending spree is in the offing.

D

The result of mass star hirings as well as individual ones seems to confirm such doubts. Look at County NatWest and Barclays de Zoete Wedd, both of which hired teams of stars with loud fanfare to do great things in investment banking in the 1990s. Both failed dismally. Everyone accepts the cliché that people make the organisation – but much more does the organisation make the people. When researchers studied the performance of fund managers in the 1990s, they discovered that just 30 per cent of the variation in fund performance was due to the individual, compared to 70 per cent to the company-specific setting.

E

That will be no surprise to those familiar with systems thinking. W Edwards Deming used to say that there was no point in beating up on people when 90 per cent of performance variation was down to the system within which they worked. Consistent improvement, he said, is a matter not of raising the level of individual intelligence, but of the learning of the organisation as a whole. The star system is glamorous – for the few. But it rarely benefits the company that thinks it is working it. And the knock-on consequences indirectly affect everyone else too. As one internet response to Gladwell’s New Yorker article put it: after Enron, ‘the rest of corporate America is stuck with overpaid, arrogant, underachieving, and relatively useless talent.’

F

Football is another illustration of the star vs systems strategic choice. As with investment banks and stockbrokers, it seems obvious that success should ultimately be down to money. Great players are scarce and expensive. So the club that can afford more of them than anyone else will win. But the performance of Arsenal and Manchester United on one hand and Chelsea and Real Madrid on the other proves that it’s not as easy as that. While Chelsea and Real have the funds to be compulsive star collectors – as with Juan Sebastian Veron – they are less successful than Arsenal and United which, like Liverpool before them, have put much more emphasis on developing a setting within which stars-in-the-making can flourish. Significantly, Thierry Henry, Patrick Veira and Robert Pires are much bigger stars than when Arsenal bought them, their value (in all senses) enhanced by the Arsenal system. At Chelsea, by contrast, the only context is the stars themselves – managers with different outlooks come and go every couple of seasons. There is no settled system for

the stars to blend into. The Chelsea context has not only not added value, but it has also subtracted it. The side is less than the sum of its exorbitantly expensive parts. Even Real Madrid's galacticos, the most extravagantly gifted on the planet, are being outperformed by less talented but better-integrated Spanish sides. In football, too, stars are trumped by systems.

G

So if not by hiring stars, how do you compete in the war for talent? You grow your own. This worked for investment analysts, where some companies were not only better at creating stars but also at retaining them. Because they had a much more sophisticated view of the interdependent relationship between star and system, they kept them longer without resorting to the exorbitant salaries that were so destructive to rivals.

Questions 14-17

The Reading Passage has seven paragraphs A-G

Which paragraph contains the following information?

Write the correct letter A-G, in boxes 14-17 on your answer sheet.

NB You may use any letter more than once.

- 14 One example from non-commerce/business settings that better system win bigger stars
- 15 One failed company that believes stars rather than the system
- 16 One suggestion that the author made to acquire employees than to win the competition nowadays
- 17 One metaphor to human medical anatomy that illustrates the problems of hiring stars.

答案

- 14. F
- 15. B
- 16. G
- 17. C

READING PASSAGE 3

You should spend about 20 minutes on Questions 27-40 which are based on Reading Passage 3 below.

Memory Decoding

Try this memory test: Study each face and compose a vivid image for the person's first and last name. Rose Leo, for example, could be a rosebud and a lion. Fill in the blanks on the next page. The Examinations School at Oxford University is an austere building of oak-paneled rooms, large Gothic windows, and looming portraits of eminent dukes and earls. It is where generations of Oxford students have tested their memory on final exams, and it is where, last August, 34 contestants gathered at the World Memory Championships to be examined in an entirely different manner.

A

In timed trials, contestants were challenged to look at and then recite a two-page poem, memorize rows of 40-digit numbers, recall the names of 110 people after looking at their photographs, and perform seven other feats of extraordinary retention. Some tests took just a few minutes; others lasted hours. In the 14 years since the World Memory Championships was founded, no one has memorized the order of a shuffled deck of playing cards in less than 30 seconds. That nice round number has become the four-minute mile of competitive memory, a benchmark that the world's best "mental athletes," as some of them like to be called, is closing in on. Most contestants claim to have just average memories, and scientific testing confirms that they're not just being modest. Their feats are based on tricks that capitalize on how the human brain encodes information. Anyone can learn them.

B

Psychologists Elizabeth Valentine and John Wilding, authors of the monograph *Superior Memory*, recently teamed up with Eleanor Maguire, a neuroscientist at University College London to study eight people, including Karsten, who had finished near the top of the World Memory Championships. They wondered if the contestants' brains were different in some way. The researchers put the competitors and a group of control subjects into an MRI machine and asked them to perform several different memory tests while their brains were being scanned. When it came to memorizing sequences of three-digit numbers, the difference between the memory contestant and the control subjects was, as expected, immense. However, when they were shown photographs of magnified snowflakes, images that the competitors had never tried to memorize before, the champions did no better than the control group. When the researchers analyzed the brain scans, they found that the memory champs were activating some brain regions that were different from those the control subjects were using. These regions, which included the right posterior hippocampus, are known to be involved in visual memory and spatial navigation.

C

It might seem odd that the memory contestants would use visual imagery and spatial navigation to remember numbers, but the activity makes sense when their techniques

are revealed. Cooke, a 23-year-old cognitive-science graduate student with a shoulder-length mop of curly hair, is a grand master of brain storage. He can memorize the order of 10 decks of playing cards in less than an hour or one deck of cards in less than a minute. He is closing in on the 30-second deck. In the Lamb and Flag, Cooke pulled out a deck of cards and shuffled it. He held up three cards – the 7 of spades, the queen of clubs, and the 10 of spades. He pointed at a fireplace and said, “Destiny’s Child is whacking Franz Schubert with handbags.” The next three cards were the king of hearts, the king of spades, and the jack of clubs.

D

How did he do it? Cooke has already memorized a specific person, verb, and object that he associates with each card in the deck. For example, for the 7 of spades, the person (or, in this case, persons) is always the singing group Destiny’s Child, the action is surviving a storm, and the image is a dinghy. The queen of clubs is always his friend Henrietta, the action is thwacking with a handbag, and the image is of wardrobes filled with designer clothes. When Cooke commits a deck to memory, he does it three cards at a time. Every three-card group forms a single image of a person doing something to an object. The first card in the triplet becomes the person, the second the verb, the third the object. He then places those images along a specific familiar route, such as the one he took through the Lamb and Flag. In competitions, he uses an imaginary route that he has designed to be as smooth and downhill as possible. When it comes time to recall, Cooke takes a mental walk along his route and translates the images into cards. That’s why the MRIs of the memory contestants showed activation in the brain areas associated with visual imagery and spatial navigation.

E

The more resonant the images are, the more difficult they are to forget. But even meaningful information is hard to remember when there’s a lot of it. That’s why competitive memorizers place their images along an imaginary route. That technique, known as the loci method, reportedly originated in 477 B.C. with the Greek poet Simonides of Ceos. Simonides was the sole survivor of a roof collapse that killed all the other guests at a royal banquet. The bodies were mangled beyond recognition, but Simonides was able to reconstruct the guest list by closing his eyes and recalling each individual around the dinner table. What he had discovered was that our brains are exceptionally good at remembering images and spatial information. Evolutionary psychologists have offered an explanation: Presumably, our ancestors found it important to recall where they found their last meal or the way back to the cave. After Simonides’ discovery, the loci method became popular across ancient Greece as a trick for memorizing speeches and texts. Aristotle wrote about it, and later a number of treatises on the art of memory were published in Rome. Before printed books, the art of memory was considered a staple of classical education, on a par with grammar, logic, and rhetoric.

F

The most famous of the naturals was the Russian journalist S.V. Shereshevski, who could recall long lists of numbers memorized decades earlier, as well as poems,

strings of nonsense syllables, and just about anything else he was asked to remember.

“The capacity of his memory had no distinct limits,” wrote Alexander Luria, the Russian psychologist who studies Shereshevski also had synesthesia, a rare condition in which the senses become intertwined. For example, every number may be associated with a color or every word with a taste. Synesthetic reactions evoke a response in more areas of the brain, making memory easier.

G

K. Anders Ericsson, a Swedish-born psychologist at Florida State University, thinks anyone can acquire Shereshevski's skills. He cites an experiment with S. F., an undergraduate who was paid to take a standard test of memory called the digit span for one hour a day, two or three days a week. When he started, he could hold, like most people, only about seven digits in his head at any given time (conveniently, the length of a phone number). Over two years, S. F. completed 250 hours of testing. By then, he had stretched his digit span from 7 to more than 80. The study of S. F. led Ericsson to believe that innately superior memory doesn't exist at all. When he reviewed original case studies of naturals, he found that exceptional memorizers were using techniques – sometimes without realizing it – and lots of practice. Often, exceptional memory was only for a single type of material, like digits. “If we look at some of these memory tasks, they're the kind of thing most people don't even waste one hour practicing, but if they wasted 50 hours, they'd be exceptional at it,” Ericsson says. It would be remarkable, he adds, to find a “person who is exceptional across a number of tasks. I don't think that there's any compelling evidence that there are such people.”

Questions 27-31

The Reading Passage has seven paragraphs A-G.

Which paragraph contains the following information?

Write the correct letter A-G, in boxes 27-31 on your answer sheet.

- 27 The reason why the competence of super memory is significant in academic settings
- 28 Mention of a contest for extraordinary memory held in consecutive years
- 29 A demonstrative example of extraordinary person did an unusual recalling game
- 30 A belief that extraordinary memory can be gained through enough practice
- 31 A depiction of the rare ability which assists the extraordinary memory reactions

答案

27. E

28. A

29. C

30. G

31. F

Test 23

READING PASSAGE 1

You should spend about 20 minutes on Questions 1-13 which are based on Reading Passage 1 below.

Ants Could Teach Ants

A

The ants are tiny and usually nest between rocks on the south coast of England. Transformed into research subjects at the University of Bristol, they raced along a tabletop foraging for food – and then, remarkably, returned to guide others. Time and again, followers trailed behind leaders, darting this way and that along the route, presumably to memorise landmarks. Once a follower got its bearings, it tapped the leader with its antennae, prompting the lesson to literally proceed to the next step. The ants were only looking for food, but the researchers said the careful way the leaders led followers – thereby turning them into leaders in their own right – marked the *Temnothorax Albipennis* ant as the very first example of a non-human animal exhibiting teaching behaviour.

B

“Tandem running is an example of teaching, to our knowledge the first in a non-human animal, that involves bidirectional feedback between teacher and pupil,” remarks Nigel Franks, professor of animal behaviour and ecology, whose paper on the ant educators was published last week in the journal *Nature*.

C

No sooner was the paper published, of course, than another educator questioned it. Marc Hauser, a psychologist and biologist and one of the scientists who came up with the definition of teaching, said it was unclear whether the ants had learned a new skill or merely acquired new information.

D

Later, Franks took a further study and found that there were even races between leaders. With the guidance of leaders, ants could find food faster. But the help comes at a cost for the leader, who normally would have reached the food about four times faster if not hampered by a follower. This means the hypothesis that the leaders deliberately slowed down in order to pass the skills on to the followers seems potentially valid. His ideas were advocated by the students who carried out the video project with him.

E

Opposing views still arose, however, Hauser noted that mere communication of information is commonplace in the animal world. Consider a species, for example, that uses alarm calls to warn fellow members about the presence of a predator. Sounding the alarm can be costly, because the animal may draw the attention of the

predator to itself. But it allows others to flee to safety. “Would you call this teaching?” wrote Hauser. “The caller incurs a cost. The naïve animals gain a benefit and new knowledge that better enables them to learn about the predator’s location than if the caller had not called. This happens throughout the animal kingdom, but we don’t call it teaching, even though it is clearly a transfer of information.”

F

Tim Caro, a zoologist, presented two cases of animal communication. He found that cheetah mothers that take their cubs along on hunts gradually allow their cubs to do more of the hunting – going, for example, from killing a gazelle and allowing young cubs to eat to merely tripping the gazelle and letting the cubs finish it off. At one level, such behaviour might be called teaching – except the mother was not really teaching the cubs to hunt but merely facilitating various stages of learning. In another instance, birds watching other birds using a stick to locate food such as insects and so on, are observed to do the same thing themselves while finding food later.

G

Psychologists study animal behaviour in part to understand the evolutionary roots of human behaviour, Hauser said. The challenge in understanding whether other animals truly teach one another, he added, is that human teaching involves a “theory of mind” teachers are aware that students don’t know something. He questioned whether Franks’ leader ants really knew that the follower ants were ignorant. Could they simply have been following an instinctive rule to proceed when the followers tapped them on the legs or abdomen? And did leaders that led the way to food – only to find that it had been removed by the experimenter – incur the wrath of followers? That, Hauser said, would suggest that the follower ant actually knew the leader was more knowledgeable and not merely following an instinctive routine itself.

H

The controversy went on, and for a good reason. The occurrence of teaching in ants, if proven to be true, indicates that teaching can evolve in animals with tiny brains. It is probably the value of information in social animals that determines when teaching will evolve, rather than the constraints of brain size.

I

Bennett Galef Jr., a psychologist who studies animal behaviour and social learning at McMaster University in Canada, maintained that ants were unlikely to have a “theory of mind” – meaning that leaders and followers may well have been following instinctive routines that were not based on an understanding of what was happening in another ant’s brain. He warned that scientists may be barking up the wrong tree when they look not only for examples of humanlike behaviour among other animals but humanlike thinking that underlies such behaviour among other animals but humanlike thinking that underlies such behaviour. Animals may behave in ways similar to humans without a similar cognitive system, he said, so the behaviour is not necessarily a good guide into how humans came to think the way they do.

Questions 1-5

Look at the following statements (Questions 1-5) and the list of people in the box below.

Match each statement with the correct person, A, B, C or D.

Write the correct letter, A, B, C or D, in boxes 1-5 on your answer sheet.

NB You may use any letter more than once.

- 1 Animals could use objects to locate food.
- 2 Ants show two-way, interactive teaching behaviours.
- 3 It is risky to say ants can teach other ants like human beings do.
- 4 Ant leadership makes finding food faster.
- 5 Communication between ants is not entirely teaching.

List of People

- | | |
|---|-------------------|
| A | Nigel Franks |
| B | Marc Hauser |
| C | Tim Caro |
| D | Bennett Galef Jr. |

答案

1. C
2. A
3. D
4. A
5. B

READING PASSAGE 2

You should spend about 20 minutes on Questions 14-27 which are based on Reading Passage 2 below.

Father of modern management 2

A

Peter Drucker was one of the most important management thinkers of the past hundred years. He wrote about 40 books and thousands of articles and he never rested in his mission to persuade the world that management matters. “Management is an organ of institutions ... the organ that converts a mob into an organisation, and human efforts into performance.” Did he succeed? The range of his influence was extraordinary. Wherever people grapple with tricky management problems, from big organizations to small ones, from the public sector to the private, and increasingly in the voluntary sector, you can find Drucker’s fingerprints.

B

His first two books – The End of Economic Man (1939) and The Future of Industrial Man (1942) – had their admirers, including Winston Churchill, but they annoyed academic critics by ranging so widely over so many different subjects. Still, the second of these books attracted attention with its passionate insistence that companies had a social dimension as well as an economic purpose. His third book, The Concept of the Corporation, became an instant bestseller and has remained in print ever since.

C

The two most interesting arguments in The Concept of the Corporation actually had little to do with the decentralization fad. They were to dominate his work. The first had to do with “empowering” workers. Drucker believed in treating workers as resources rather than just as costs. He was a harsh critic of the assembly-line system of production that then dominated the manufacturing sector – partly because assembly lines moved at the speed of the slowest and partly because they failed to engage the creativity of individual workers. The second argument had to do with the rise of knowledge workers. Drucker argued that the world is moving from an “economy of goods” to an economy of “knowledge” – and from a society dominated by an industrial proletariat to one dominated by brain workers. He insisted that this had profound implications for both managers and politicians. Managers had to stop treating workers like cogs in a huge inhuman machine and start treating them as brain workers. In turn, politicians had to realise that knowledge, and hence education, was the single most important resource for any advanced society. Yet Drucker also thought that this economy had implications for knowledge workers themselves. They had to come to terms with the fact that they were neither “bosses” nor “workers”, but something in between: entrepreneurs who had responsibility for developing their most important resource, brainpower, and who also needed to take more control of their own careers, including their pension plans.

D

However, there was also a hard side to his work. Drucker was responsible for inventing one of the rational school of management's most successful products – “management by objectives”. In one of his most substantial works, *The Practice of Management* (1954), he emphasised the importance of managers and corporations setting clear long-term objectives and then translating those long-term objectives into more immediate goals. He argued that firms should have an elite corps of general managers, who set these long-term objectives, and then a group of more specialised managers. For his critics, this was a retreat from his earlier emphasis on the soft side of management. For Drucker it was all perfectly consistent: if you rely too much on empowerment you risk anarchy, whereas if you rely too much on command-and-control you sacrifice creativity. The trick is for managers to set long-term goals, but then allow their employees to work out ways of achieving those goals. If Drucker helped make management a global industry, he also helped push it beyond its business base. He was emphatically a management thinker, not just a business one. He believed that management is “the defining organ of all modern institutions”, not just corporations.

E

There are three persistent criticisms of Drucker's work. The first is that he focused on big organisations rather than small ones. The Concept of the Corporation was in many ways a fanfare to big organisations. As Drucker said, “We know today that in modern industrial production, particularly in modern mass production, the small unit is not only inefficient, it cannot produce at all.” The book helped to launch the “big organisation boom” that dominated business thinking for the next 20 years. The second criticism is that Drucker's enthusiasm for management by objectives helped to lead the business down a dead end. They prefer to allow ideas, including ideas for long-term strategies, to bubble up from the bottom and middle of the organisations rather than being imposed from on high. Thirdly, Drucker is criticised for being a maverick who has increasingly been left behind by the increasing rigour of his chosen field. There is no single area of academic management theory that he made his own.

F

There is some truth in the first two arguments. Drucker never wrote anything as good as *The Concept of the Corporation* on entrepreneurial start-ups. Drucker's work on management by objectives sits uneasily with his earlier and later writings on the importance of knowledge workers and self-directed teams. But the third argument is short-sighted and unfair because it ignores Drucker's pioneering role in creating the modern profession of management. He produced one of the first systematic studies of a big company. He pioneered the idea that ideas can help galvanise companies. The biggest problem with evaluating Drucker's influence is that so many of his ideas have passed into conventional wisdom. In other words, he is the victim of his own success. His writings on the importance of knowledge workers and empowerment may sound a little banal today. But they certainly weren't banal when he first dreamed them up in the 1940s, or when they were first put in to practice in the Anglo-Saxon world in the 1980s. Moreover, Drucker continued to produce new ideas up until his 90s. His work on the management of voluntary organisations remained at the cutting edge.

Questions 14-19

Reading Passage 2 has six paragraphs, A-F

Choose the correct heading for each paragraph from the list below.

Write the correct number, i-ix, in boxes 14-19 on your answer sheet.

List of Headings

- i The popularity and impact of Drucker's work
 - ii Finding fault with Drucker
 - iii The impact of economic globalisation
 - iv Government regulation of business
 - v Early publications of Drucker's
 - vi Drucker's view of balanced management
 - vii Drucker's rejection of big business
 - viii An appreciation of the pros and cons of Drucker's work
 - ix The changing role of the employee
- 14 Paragraph A
 - 15 Paragraph B
 - 16 Paragraph C
 - 17 Paragraph D
 - 18 Paragraph E
 - 19 Paragraph F

答案

- 14. i
- 15. v
- 16. ix
- 17. vi
- 18. ii
- 19. viii

READING PASSAGE 3

You should spend about 20 minutes on Questions 28-40 which are based on Reading Passage 3 below.

The secret of the Yawn

A

When a scientist began to study yawning in the 1980s, it was difficult to convince some of his research students of the merits of “yawning science.” Although it may appear quirky, his decision to study yawning was a logical extension to human beings of my research in developmental neuroscience, reported in such papers as “Wing-flapping during Development and Evolution.” As a neurobehavioral problem, there is not much difference between the wing-flapping of birds and the face – and body-flapping of human yawners.

B

Yawning is an ancient, primitive act. Humans do it even before they are born, opening wide in the womb. Some snakes unhinge their jaws to do it. One species of penguins yawns as part of mating. Only now are researchers beginning to understand why we yawn, when we yawn and why we yawn back. A professor of cognitive neuroscience at Drexel University in Philadelphia, Steven Platek, studies the act of contagious yawning, something done only by people and other primates.

C

In his first experiment, he used a psychological test to rank people on their empathic feelings. He found that participants who did not score high on compassion did not yawn back. “We literally had people saying, ‘Why am I looking at people yawning?’” Professor Platek said. “It just had no effect.”

D

For his second experiment, he put 10 students in a magnetic resonance imaging machine as they watched video tapes of people yawning. When the students watched the videos, the part of the brain which reacted was the part scientists believe controls empathy – the posterior cingulate, in the brain’s middle rear.” I don’t know if it’s necessarily that nice people yawn more, but I think it’s a good indicator of a state of mind,” said Professor Platek. “It’s also a good indicator if you’re empathizing with me and paying attention.”

E

His third experiment is studying yawning in those with brain disorders, such as autism and schizophrenia, in which victims have difficulty connecting emotionally with others. A psychology professor at the University of Maryland, Robert Provine, is one of the few other researchers into yawning. He found the basic yawn lasts about six seconds and they come in bouts with an interval of about 68 seconds. Men and women yawn or half-yawn equally often, but men are significantly less likely to cover

their mouths which may indicate complex distinction in genders.” A watched yawner never yawns,” Professor Provine said. However, the physical root of yawning remains a mystery. Some researchers say it’s coordinated within the hypothalamus of the brain, the area that also controls breathing.

F

Yawning and stretching also share properties and may be performed together as parts of a global motor complex. But they do not always co-occur – people usually yawn when we stretch, but we don’t always stretch when we yawn, especially before bedtime. Studies by J.I.P, G.H.A. Visser and H.F. Prechtl in the early 1980s, charting movement in the developing fetus using ultrasound, observed not just yawning but a link between yawning and stretching as early as the end of the first prenatal trimester.

G

The most extraordinary demonstration of the yawn-stretch linkage occurs in many people paralyzed on one side of their body because of brain damage caused by a stroke. The prominent British neurologist Sir Francis Walshe noted in 1923 what when these hemiplegics yawn, they are startled and mystified to observe that their otherwise paralyzed arm rises and flexes automatically in what neurologists term an “associated response.” Yawning apparently activates undamaged, unconsciously controlled connections between the brain and the cord motor system innervating the paralyzed limb. It is not known whether the associated response is a positive prognosis for recovery, nor whether yawning is therapeutic for reinnervation or prevention of muscular atrophy.

H

Clinical neurology offers other surprises. Some patients with “locked-in” syndrome, who are almost totally deprived of the ability to move voluntarily, can yawn normally. The neural circuits for spontaneous yawning must exist in the brain stem near other respiratory and vasomotor centers, because yawning is performed by anencephalic who possess only the medulla oblongata. The multiplicity of stimuli of contagious yawning, by contrast, implicates many higher brain regions.

Questions 33-37

Read paragraph A-H.

Which paragraph contains the following information?

Write the correct letter A-H for questions 33-37

NB You may use any letter more than once.

- 33 The rate for yawning shows some regular pattern.
- 34 Yawning is an inherent ability that appears in both animals and humans.
- 35 Stretching and yawning is not always going together.
- 36 Yawning may suggest people are having positive notice or response in communicating.
- 37 Some superior areas in the brain may deal with the infectious feature of yawning.

答案

33. E

-
34. B
35. F
36. D
37. H

Test 25

READING PASSAGE 2

You should spend about 20 minutes on Questions 14-27 which are based on Reading Passage 2 below.

Going nowhere fast New transport mode PRT RUF

A

This is ludicrous! We can talk to people anywhere in the world or fly to meet them in a few hours. We can even send probes to other planets. But when it comes to getting around our cities, we depend on systems that have scarcely changed since the days of Gottlieb Daimler.

B

In recent years, the pollution belched out by millions of vehicles has dominated the debate about transport. The problem has even persuaded California that home of car culture to curb traffic growth. But no matter how green they become, cars are unlikely to get us around crowded cities any faster. And persuading people to use trains and buses will always be an uphill struggle. Cars, after all, are popular for very good reasons, as anyone with small children or heavy shopping knows.

C

A professor of mechanical engineering sits typing at a computer keyboard, conjuring up a scene on his monitor that looks something like the classic computer game PacMan. White dots stream in from the right of the screen, switch to red and merge with green boxes, which swiftly change colour to yellow and then red, while moving through a bewildering maze. But this is not a video game. J. Edward Anderson of Boston University is testing an urban transit system that he believes could revolutionise public transport worldwide.

D

For the past quarter of a century, Anderson has been promoting his version of personal rapid transit (PRT). Other versions came and went in the 1970s, from Europe, Japan and elsewhere in the US, but he was so convinced of the idea's potential that he stuck with it and, in 1983, founded the Taxi 2000 Corporation to 'commercialise' the initiative. Although the University of Minnesota, Anderson's employer until 1986, holds the patents to the technology, he is licensed to develop it and to sub-license other developers. So politicians should be trying to lure people out of their cars, not forcing them out. There's certainly no shortage of alternatives. Perhaps the most attractive is the concept known as personal rapid transit (PRT), independently invented in the US and Europe in the 1950s.

E

The idea is to go to one of many stations and hop into a computer-controlled car which can whisk you to your destination along with a network of guideways. You

wouldn't have to share your space with strangers, and with no traffic lights, pedestrians or parked cars to slow things down, PRT guideways can carry far more traffic, nonstop, than any inner-city road. It's a wonderful vision, but the odds are stacked against PRT for a number of reasons. The first cars ran on existing roads, and it was only after they became popular – and after governments started earning revenue from them – that a road network designed specifically for motor vehicles was built. With PRT, the infrastructure would have to come first – and that would cost megabucks.

F

What's more, any transport system that threatened the car's dominance would be up against all those with a stake in maintaining the status quo, from private car owners to manufacturers and oil multinationals. Even if PRTs were spectacularly successful in trials, it might not make much difference. Superior technology doesn't always triumph, as the VHS versus Betamax and Windows versus Apple Mac battles showed.

G

But “dual-mode” systems might just succeed where PRT seems doomed to fail. The Danish RUF system envisaged by Palle Jensen, for example, resembles PRT but with one key difference: vehicles have wheels as well as a lot allowing them to travel on a monorail, so they can drive off the rail onto a normal road. Once on a road, the occupant would take over from the computer, and the RUF vehicle – the term comes from a Danish saying meaning to “go fast” – would become an electric car.

H

Build a fast network of guideways in a busy city centre and people would have a strong incentive not just to use public RUF vehicles, but also to buy their own dual-mode vehicle. Commuters could drive onto the guideway, sit back and read as they are chauffeured into the city. At work, they would jump out, leaving their vehicles to park themselves. Unlike PRT, such a system could grow organically, as each network would serve a large area around it and people nearby could buy into it. And a dual-mode system might even win the support of car manufacturers, who could easily switch to producing dual-mode vehicles.

I

The RUF system can reduce the energy consumption from individual traffic. The main factor is the reduction of air resistance due to close coupling of vehicles. The energy consumption per ruf can be reduced to less than 1/3 at 100 km/h. Since RUF is an electric system, renewable sources can be used without problems. A combination of windmills and a RUF rail could be used over water. Solar cells can also be integrated into the system and ensure completely sustainable transportation.

J

Of course, creating a new transport system will not be cheap or easy. But unlike adding a dedicated bus lane here or extending the underground railway there, an innovative system such as Jensen's could transform cities. The vehicles in a RUF system “rides” very safely on top of a triangular monorail. This means that derailments are impossible and that the users will feel safe because it is easy to understand that when the rail is actually inside the vehicle it is absolutely stable. The

special rail brake ensures that braking power is always available even during bad weather. The brake can squeeze as hard against the rail as required in order to bring the vehicle to a safe stop. If a vehicle has to be evacuated, a walkway between the two rails can be used.

K

And it's not just a matter of saving a few minutes a day. According to the Red Cross, more than 30 million people have died in road accidents in the past century – three times the number killed in the First World War – and the annual death toll is rising. And what's more, the Red Cross believes road accidents will become the third biggest cause of death and disability by 2020, ahead of diseases such as AIDS and tuberculosis. Surely we can find a better way to get around.

Questions 18-24

Use the information in the passage to match the category (listed A-C) with the description below.

Write the appropriate letters A-C in boxes 18-24 on your answer sheet.

A ONLY PRT

B ONLY RUF

C BOTH OF THEM

- 18 Totally apply computer system
- 19 Opposition to the system from companies
- 20 Reach destination fast
- 21 Not necessary to share with the public
- 22 Work on existing road
- 23 Individuals can buy cars after all
- 24 Controlled both by computer and manual

答案

- 18. A
- 19. A
- 20. C
- 21. C
- 22. B
- 23. B
- 24. B

READING PASSAGE 3

You should spend about 20 minutes on Questions 28-40 which are based on Reading Passage 3 below.

Internal and External Marketing

A

Employees need to hear the same messages that you send out to the marketplace. At most companies, however, internal and external communications are often mismatched. This can be very confusing, and it threatens employees' perceptions of the company's integrity: They are told one thing by management but observe that a different message is being sent to the public. One health insurance company, for instance, advertised that the welfare of patients was the company's number one priority, while employees were told that their main goal was to increase the value of their stock options through cost reductions. And one major financial services institution told customers that it was making a major shift in focus from being a financial retailer to a financial adviser, but, a year later, research showed that the customer experience with the company had not changed. It turned out that company leaders had not made an effort to sell the change internally, so employees were still churning out transactions and hadn't changed their behavior to match their new adviser role.

B

Enabling employees to deliver on customer expectations is important, of course, but it's not the only reason a company needs to match internal and external messages. Another reason is to help push the company to achieve goals that might otherwise be out of reach. In 1997, when IBM launched its e-business campaign (which is widely credited for turning around the company's image), it chose to ignore research that suggested consumers were unprepared to embrace IBM as a leader in e-business. Although to the outside world this looked like an external marketing effort, IBM was also using the campaign to align employees around the idea of the Internet as the future of technology. The internal campaign changed the way employees thought about everything they did, from how they named products to how they organized staff to how they approached selling. The campaign was successful largely because it gave employees a sense of direction and purpose, which in turn restored their confidence in IBM's ability to predict the future and lead the technology industry. Today, research shows that people are four times more likely to associate the term "e-business" with IBM than with its nearest competitor, Microsoft.

C

The type of "two-way branding" that IBM did so successfully strengthens both sides of the equation. Internal marketing becomes stronger because it can draw on the same "big idea" as advertising. Consumer marketing becomes stronger because the messages are developed based on employees' behavior and attitudes, as well as on the company's strengths and capabilities – indeed, the themes are drawn from the

company's very soul. This process can result in a more distinct advertising idea because marketers are more likely to create a message that's unique to the company.

D

Perhaps even more important, by taking employees into account, a company can avoid creating a message that doesn't resonate with staff or, worse, one that builds resentment. In 1996, United Airlines shelved its "Come Fly the Friendly Skies" slogan when presented with a survey that revealed the depth of customer resentment toward the airline industry. In an effort to own up to the industry's shortcomings, United launched a new campaign, "Rising," in which it sought to differentiate itself by acknowledging poor service and promising incremental improvements such as better meals. While this was a logical premise for the campaign given the tenor of the times, a campaign focusing on customers' distaste for flying was deeply discouraging to the staff. Employee resentment ultimately made it impossible for United to deliver the improvements it was promising, which in turn undermined the "Rising" pledge. Three years later, United decided employee opposition was undermining its success and pulled the campaign. It has since moved to a more inclusive brand message with the line "United," which both audiences can embrace. Here, a fundamental principle of advertising – find and address a customer concern – failed United because it did not consider the internal market.

E

When it comes to execution, the most common and effective way to link internal and external marketing campaigns is to create external advertising that targets both audiences. IBM used this tactic very effectively when it launched its e-business campaign. It took out an eight-page ad in the Wall Street Journal declaring its new vision, a message directed at both customers and internal stakeholders. This is an expensive way to capture attention, but if used sparingly, it is the most powerful form of communication; in fact, you need do it only once for everyone in the company to read it. There's a symbolic advantage as well. Such a tactic signals that the company is taking its pledge very seriously; it also signals transparency – the same message going out to both audiences.

F

Advertising isn't the only way to link internal and external marketing. At Nike, a number of senior executives now hold the additional title of "Corporate Storyteller." They deliberately avoid stories of financial successes and concentrate on parables of "just doing it," reflecting and reinforcing the company's ad campaigns. One tale, for example, recalls how legendary coach and Nike co-founder Bill Bowerman, in an effort to build a better shoe for his team, poured rubber into the family waffle iron, giving birth to the prototype of Nike's famous Waffle Sole. By talking about such inventive moves, the company hopes to keep the spirit of innovation that characterizes its ad campaigns alive and well within the company.

G

But while their messages must be aligned, companies must also keep external promises a little ahead of internal realities. Such promises provide incentives for employees and give them something to live up to. In the 1980s, Ford turned "Quality

is Job!” from an internal rallying cry into a consumer slogan in response to the threat from cheaper, more reliable Japanese cars. It did so before the claim was fully justified, but by placing it in the public arena, it gave employees an incentive to match the Japanese. If the promise is pushed too far ahead, however, it loses credibility. When a beleaguered British Rail launched a campaign announcing service improvement under the banner “We’ re Getting There,” it did so prematurely. By drawing attention to the gap between the promise and the reality, it prompted destructive press coverage. This, in turn, demoralized staff, who had been legitimately proud of the service advances they had made.

Questions 28-34

Use the information in the passage to match the company (listed A-F) with correct category or deeds below.

Write the appropriate letters A-F in boxes 28-34 on your answer sheet.

NB You may use any letter more than once

- A legendary anecdote inspire employee successfully
 - B advertisement campaign inspire employees and ensure a leading role in business
 - C improper ads campaign brings negative effect
 - D internal and external announcement are different
 - E campaign brings positive and realistic expectation internally
 - F a bad slogan that failed both to win support internally and raise standard to its poor service
- 28 One health insurance Company
 - 29 British Rail
 - 30 IBM
 - 31 United Airline
 - 32 A financial service company
 - 33 A Shoemaking company (Nike)
 - 34 The Company of (Ford)

答案

- 28. D
- 29. C
- 30. B
- 31. F
- 32. C
- 33. A
- 34. E

Test 26

READING PASSAGE 2

You should spend about 20 minutes on Questions 14-27 which are based on Reading Passage 2 below.

Saving the British Bitterns

A

Breeding bitterns became extinct in the UK by 1886 but, following re-colonisation early last century, numbers rose to a peak of about 70 booming (singing) males in the 1950s, falling to fewer than 20 by the 1990s. In the late 1980s, it was clear that the bittern was in trouble, but there was little information on which to base recovery actions.

B

Bitterns have cryptic plumage and shy nature, usually remaining hidden within the cover of reedbed vegetation. Our first challenge was to develop standard methods to monitor their numbers. The boom of the male bittern is its most distinctive feature during the breeding season, and we developed a method to count them using the sound patterns unique to each individual. This not only allows us to be much more certain of the number of booming males in the UK but also enables us to estimate the local survival of males from one year to the next.

C

Our first direct understanding of the habitat needs of breeding bitterns came from comparisons of reedbed sites that had lost their booming birds with those that retained them. This research showed that bitterns had been retained in reedbeds where the natural process of succession, or drying out, had been slowed through management. Based on this work, broad recommendations on how to manage and rehabilitate reedbeds for bitterns were made, and funding was provided through the EU LIFE Fund to manage 13 sites within the core breeding range. This project, though led by the RSPB, involved many other organisations.

D

To refine these recommendations and provide fine-scale, quantitative habitat prescriptions on the bitterns' preferred feeding habitat, we radio-tracked male bitterns on the RSPB's Minsmere and Leighton Moss reserves. This showed clear preferences for feeding in the wetter reedbed margins, particularly within the reedbed next to larger open pools. The average home range sizes of the male bitterns we followed (about 20 hectares) provided a good indication of the area of reedbed needed when managing or creating habitat for this species. Female bitterns undertake all the incubation and care of the young, so it was important to understand their needs as well. Over the course of our research, we located 87 bittern nests and found that female bitterns preferred to nest in areas of continuous vegetation, well into the reedbed, but

where water was still present during the driest part of the breeding season.

E

The success of the habitat prescriptions developed from this research has been spectacular. For instance, at Minsmere, booming bittern numbers gradually increased from one to 10 following reedbed lowering, a management technique designed to halt the drying out process. After a low point of 11 booming males in 1997, bittern numbers in Britain responded to all the habitat management work and started to increase for the first time since the 1950s.

F

The final phase of the research involved understanding the diet, survival and dispersal of bittern chicks. To do this we fitted small radio tags to young bittern chicks in the nest, to determine their fate through to fledge and beyond. Many chicks did not survive to fledging and starvation was found to be the most likely reason for their demise. The fish prey fed to chicks was dominated by those species penetrating into the reed edge. So, an important element of recent studies (including a PhD with the University of Hull) has been the development of recommendations on habitat and water conditions to promote healthy native fish populations.

G

Once independent, radio-tagged young bitterns were found to seek out new sites during their first winter; a proportion of these would remain on new sites to breed if the conditions were suitable. A second EU LIFE funded project aims to provide these suitable sites in new areas. A network of 19 sites developed through this partnership project will secure a more sustainable UK bittern population with successful breeding outside of the core area, less vulnerable to chance events and sea-level rise.

H

By 2004, the number of booming male bitterns in the UK had increased to 55, with almost all of the increase being on those sites undertaking management based on advice derived from our research. Although science has been at the core of the bittern story, success has only been achieved through the trust, hard work and dedication of all the managers, owners and wardens of sites that have implemented, in some cases very drastic, management to secure the future of this wetland species in the UK. The constructed bunds and five major sluices now control the water level over 82 ha, with a further 50 ha coming under control in the winter of 2005/06. Reed establishment has principally used natural regeneration or planted seedlings to provide small core areas that will in time expand to create a bigger reed area. To date, nearly 275,000 seedlings have been planted and reed cover is extensive. Over 3 km of new ditches have been formed, 3.7 km of the existing ditch have been re-profiled and 2.2 km of old meander (former estuarine features) have been cleaned out.

I

Bitterns now regularly winter on the site with some indication that they are staying longer into the spring. No breeding has yet occurred but a booming male was present in the spring of 2004. A range of wildfowl breed, as well as a good number of reedbed passerines including reed bunting, reed, sedge and grasshopper warblers. Numbers of wintering shoveler have increased so that the site now holds a UK important

wintering population. Malltraeth Reserve now forms part of the UK network of key sites for water vole (a UK priority species) and 12 monitoring transects have been established. Otter and brown-hare occur on the sites as does the rare plant, pillwort.

Questions 14-20

The Reading Passage has eight paragraphs A-H

Choose the correct heading for paragraphs A-H from the list below.

Write the correct number, i-viii, in boxes 14-20 on your answer sheet.

List of Headings

- i research findings into habitats and decisions made
- ii fluctuation in bittern number
- iii protect the young bittern
- iv international cooperation works
- v Began in calculation of the number
- vi importance of food
- vii Research has been successful
- viii research into the reedbed
- ix reserve established holding bittern in winter

14 Paragraph A

15 Paragraph B

16 Paragraph C

17 Paragraph D

Example

Paragraph A vii

18 Paragraph F

19 Paragraph G

20 Paragraph H

答案

14. ii

15. v

16. i

17. viii

18. vi

19. iii

20. iv

READING PASSAGE 3

You should spend about 20 minutes on Questions 28-40 which are based on Reading Passage 3 below.

Extinct: the Giant Deer

Toothed cats, mastodons, giant sloths, woolly rhinos, and many other big, shaggy mammals are widely thought to have died out around the end of the last ice age, some 10,500 years ago.

A

The Irish elk is also known as the giant deer (*Megaloceros giganteus*). Analysis of ancient bones and teeth by scientists based in Britain and Russia show the huge herbivore survived until about 5,000 B.C. – more than three millennia later than previously believed. The research team says this suggests additional factors, besides climate change, probably hastened the giant deer's eventual extinction. The factors could include hunting or habitat destruction by humans.

B

The Irish elk, so-called because its well-preserved remains are often found in lake sediments under peat bogs in Ireland, first appeared about 400,000 years ago in Europe and Central Asia. Through a combination of radiocarbon dating of skeletal remains and the mapping of locations where the remains were unearthed, the team shows the Irish elk was widespread across Europe before the last “big freeze.” The deer's range later contracted to the Ural Mountains, in modern-day Russia, which separate Europe from Asia.

C

The giant deer made its last stand in western Siberia, some 3,000 years after the ice sheets receded, said the study's co-author, Adrian Lister, professor of palaeobiology at University College London, England. “The eastern foothills of the Urals became very densely forested about 8,000 years ago, which could have pushed them on to the plain,” he said. He added that pollen analysis indicates the region then became very dry in response to further climatic change, leading to the loss of important food plants.

“In combination with human pressures, this could have finally snuffed them out,” Lister said.

D

Hunting by humans has often been put forward as a contributory cause of extinction of the Pleistocene megafauna. The team, though, said their new date for the Irish elk's extinction hints at an additional human-made problem – habitat destruction. Lister said, “We haven't got just hunting 7,000 years ago – this was also about the time the first Neolithic people settled in the region. They were farmers who would have cleared the land.” The presence of humans may help explain why the Irish elk was unable to tough out the latest of many climatic fluctuations – periods it had survived in the past.

E

Meanwhile, Lister cast doubt on another possible explanation for the deer's demise – the male's huge antlers. Some scientists have suggested this exaggerated feature – the result of females preferring stags with the largest antlers, possibly because they advertised a male's fitness – contributed to the mammal's downfall. They say such antlers would have been a serious inconvenience in the dense forests that spread northward after the last ice age. But, Lister said, "That's a hard argument to make because the deer previously survived perfectly well through wooded interglacials [warmer periods between ice ages]." Some research has suggested that a lack of sufficient high-quality forage caused the extinction of the elk. High amounts of calcium and phosphate compounds are required to form antlers, and therefore large quantities of these minerals are required for the massive structures of the Irish Elk. The males (and male deer in general) met this requirement partly from their bones, replenishing them from food plants after the antlers were grown or reclaiming the nutrients from discarded antlers (as has been observed in extant deer). Thus, in the antler growth phase, Giant Deer was suffering from a condition similar to osteoporosis. When the climate changed at the end of the last glacial period, the vegetation in the animal's habitat also changed towards species that presumably could not deliver sufficient amounts of the required minerals, at least in the western part of its range.

F

The extinction of megafauna around the world was almost completed by the end of the last ice age. It is believed that megafauna initially came into existence in response to glacial conditions and became extinct with the onset of warmer climates. Tropical and subtropical areas have experienced less radical climatic change. The most dramatic of these changes was the transformation of a vast area of North Africa into the world's largest desert. Significantly, Africa escaped major faunal extinction as did tropical and sub-tropical Asia. The human exodus from Africa and our entrance into the Americas and Australia were also accompanied by climate change. Australia's climate changed from cold-dry to warm-dry. As a result, surface water became scarce. Most inland lakes became completely dry or dry in the warmer seasons. Most large, predominantly browsing animals lost their habitat and retreated to a narrow band in eastern Australia, where there were permanent water and better vegetation. Some animals may have survived until about 7000 years ago. If people have been in Australia for up to 60 000 years, then megafauna must have co-existed with humans for at least 30 000 years. Regularly hunted modern kangaroos survived not only 10 000 years of Aboriginal hunting, but also an onslaught of commercial shooters.

G

The group of scientists led by A.J. Stuart focused on northern Eurasia, which he was taking as Europe, plus Siberia, essentially, where they've got the best data that animals became extinct in Europe during the Late Pleistocene. Some cold-adapted animals, go through into the last part of the cold stage and then become extinct up there. So you've actually got two phases of extinction. Now, neither of these coincide – these are Neanderthals here being replaced by modern humans. There's no obvious coincidence between the arrival of humans or climatic change alone and

these extinctions. There's a climatic change here, so there's a double effect here. Again, as animals come through to the last part of the cold stage, here there's a fundamental change in the climate, reorganization of vegetation, and the combination of the climatic change and the presence of humans – of advanced Paleolithic humans – causes this wave of extinction. There's a profound difference between the North American data and that of Europe, which summarize that the extinctions in northern Eurasia, in Europe, are moderate and staggered, and in North America severe and sudden. And these things relate to the differences in the timing of human arrival. The extinction follows from human predation, but only at times of fundamental changes in the environment.

Questions 36-39

Choose the letter A-D and write your answers in boxes 36-39 on your answer sheet.

- A Eurasia
- B Australia
- C Asia
- D Africa

- 36 the continents where humans imposed a little impact on large mammals extinction
- 37 the continents where the climatic change was mild and fauna remains
- 38 the continents where both humans and climatic change are the causes
- 39 the continents where the climatic change alone caused a massive extinction

答案

- 36. B
- 37. D
- 38. A
- 39. C

Test 27

READING PASSAGE 2

You should spend about 20 minutes on Questions 14-27 which are based on Reading Passage 2 below.

Malaria Combat in Italy

A

Mal-aria. Bad air. Even the word is Italian, and this horrible disease marked the life of those in the peninsula for thousands of years. Giuseppe Garibaldi's wife died of the disease, as did the country's first prime minister, Cavour, in 1861. Yet by 1962, Italy was officially declared malaria-free, and it has remained so ever since. Frank Snowden's study of this success story is a remarkable piece of historical work. Original, crystal-clear, analytical and passionate, Snowden (who has previously written about cholera) takes us to areas historians have rarely visited before.

B

Everybody now knows that malaria is carried by mosquitoes. Malaria has always been the subject of research for medical practitioners from time immemorial. However, many ancient texts, especially medical literature, mention of various aspects of malaria and even of its possible link with mosquitoes and insects. Early man, confronting the manifestations of malaria, attributed the fevers to supernatural influences: evil spirits, angered deities, or the black magic of sorcerers. But in the 19th century, most experts believed that the disease was not produced by unclean air ("miasma" or "poisoning of the air"). Two Americans, Josiah Clark Nott and Lewis Daniel Beauperthy, echoed Crawford's ideas. Nott in his essay "Yellow Fever Contrasted with Bilious Fever," published in 1850, dismissed the miasma theory as worthless, arguing that microscopic insects somehow transmitted by mosquitoes caused both malaria and yellow fever. Others made a link between swamps, water and malaria, but did not make the further leap towards insects. The consequences of these theories were that little was done to combat the disease before the end of the century. Things became so bad that 11m Italians (from a total population of 25m) were "permanently at risk". In malarial zones, the life expectancy of land workers was a terrifying 22.5 years. Those who escaped death were weakened or suffered from splenomegaly – a "painful enlargement of the spleen" and "a lifeless stare". The economic impact of the disease was immense. Epidemics were blamed on southern Italians, given the widespread belief that malaria was hereditary. In the 1880s, such theories began to collapse as the dreaded mosquito was identified as the real culprit.

C

Italian scientists, drawing on the pioneering work of French doctor Alphonse Laveran, were able to predict the cycles of fever but it was in Rome that further key discoveries were made. Giovanni Battista Grassi, a naturalist, found that a particular type of

mosquito was the carrier of malaria. By experimenting on healthy volunteers (mosquitoes were released into rooms where they drank the blood of the human guinea pigs), Grassi was able to make the direct link between the insects (all females of a certain kind) and the disease. Soon, doctors and scientists made another startling discovery: the mosquitoes themselves were also infected and not mere carriers. Every year, during the mosquito season, malarial blood was moved around the population by the insects. Definitive proof of these new theories was obtained after an extraordinary series of experiments in Italy, where healthy people were introduced into malarial zones but kept free of mosquito bites – and remained well. The new Italian state had the necessary information to tackle the disease.

D

A complicated approach was adopted, which made use of quinine – a drug obtained from tree bark which had long been used to combat fever but was now seen as a crucial part of the war on malaria. Italy introduced a quinine law and a quinine tax in 1904, and the drug was administered to large numbers of rural workers. Despite its often terrible side-effects (the headaches produced were known as the “quinine-buzz”), the drug was successful in limiting the spread of the disease, and in breaking cycles of infection. In addition, Italy set up rural health centres and invested heavily in education programmes. Malaria, as Snowden shows, was not just a medical problem, but a social and regional issue, and could only be defeated through multi-layered strategies. Politics was itself transformed by the anti-malarial campaigns.

E

It was originally decided to give quinine to all those in certain regions – even healthy people; peasants were often suspicious of the medicine being forced upon them. Doctors were sometimes met with hostility and refusal, and many were dubbed “poisoners”. Despite these problems, the strategy was hugely successful. Deaths from malaria fell by some 80% in the first decade of the 20th century and some areas escaped altogether from the scourge of the disease.

F

Shamefully, the Italian malaria expert Alberto Missiroli had a role to play in the disaster: he did not distribute quinine, despite being well aware of the epidemic to come. Snowden claims that Missiroli was already preparing a new strategy – with the support of the US Rockefeller Foundation – using a new pesticide, DDT. Missiroli allowed the epidemic to spread, in order to create the ideal conditions for a massive, and lucrative, human experiment. Fifty-five thousand cases of malaria were recorded in the province of Littoria alone in 1944. It is estimated that more than a bird of those in the affected area contracted the disease. Thousands, nobody knows how many, died.

G

With the war over, the US government and the Rockefeller Foundation were free to experiment. DDT was sprayed from the air and 3m Italians had their bodies covered with the chemical. The effects were dramatic, and nobody really cared about the toxic effects of the chemical. By 1962, malaria was more or less gone from the whole

peninsula. The last cases were noted in a poor region of Sicily. One of the final victims to die of the disease in Italy was the popular cyclist, Fausto Coppi. He had contracted malaria in Africa in 1960, and the failure of doctors in the north of Italy to spot the disease was a sign of the times. A few decades earlier, they would have immediately noticed the tell-tale signs; it was later claimed that a small dose of quinine would have saved his life.

H

As there are still more than 1m deaths every year from malaria worldwide, Snowden's book also has contemporary relevance. This is a disease that affects every level of the societies where it is rampant. As Snowden writes: "In Italy, malaria undermined agricultural productivity, decimated the army, destroyed communities and left families impoverished." The economic miracle of the 50s and 60s which made Italy into a modern industrial nation would not have been possible without the eradication of malaria. Moreover, this book convincingly argues that the disease was "an integral part of the big picture of modern Italian history". This magnificent study, beautifully written and impeccably documented, deserves an audience beyond specialists in history, or in Italy. It also provides us with "a message of hope for a world struggling with the great present-day medical emergency".

Questions 22-27

Reading Passage 2 has 8 paragraphs, A-H.

Which paragraph contains the following information?

Write the correct letter A-H in boxes 22-27 on your answer sheet.

- 22 A breakthrough was found that mosquito was the carrier of malaria
- 23 A scientist intentionally failed to restrict the epidemic area.
- 24 This successful story still holds true for today's readers worldwide.
- 25 One of the final cases reported dying of malaria in Italy
- 26 the negative symptoms of the highly effective drug
- 27 A list of the speculative hypothesis was cited.

答案

- 22. C
- 23. F
- 24. H
- 25. G
- 26. D
- 27. B

READING PASSAGE 3

You should spend about 20 minutes on Questions 28-40 which are based on Reading Passage 3 below.

Video Games'**Unexpected Benefits to Human Brain****A**

James Paul Gee, professor of education at the University of Wisconsin-Madison, played his first video game years ago when his six-year-old son Sam was playing Pajama Sam: No Need to Hide When It's Dark Outside. He wanted to play the game so he could support Sam's problem-solving. Though Pajama Sam is not an "educational game", it is replete with the types of problems psychologists study when they study thinking and learning. When he saw how well the game held Sam's attention, he wondered what sort of beast a more mature video game might be.

B

Video and computer games, like many other popular, entertaining and addicting kid's activities, are looked down upon by many parents as time-wasters, and worse, parents think that these games rot the brain. Violent video games are readily blamed by the media and some experts as the reason why some youth become violent or commit extreme anti-social behavior. Recent content analyses of video games show that as many as 89% of games contain some violent content, but there is no form of aggressive content for 70% of popular games. Many scientists and psychologists, like James Paul Gee, find that video games actually have many benefits – the main one being making kids smart. Video games may actually teach kids high-level thinking skills that they will need in the future.

C

"Video games change your brain," according to University of Wisconsin psychologist Shawn Green. Video games change the brain's physical structure the same way as do learning to read, playing the piano, or navigating using a map. Much like exercise can build muscle, the powerful combination of concentration and rewarding surges of neurotransmitters like dopamine, which strengthens neural circuits, can build the player's brain.

D

Video games give your child's brain a real workout. In many video games, the skills required to win involve abstract and high-level thinking. These skills are not even taught at school. Some of the mental skills trained by video games include: following instructions, problem-solving, logic, hand-eye coordination, fine motor and spatial skills. Research also suggests that people can learn iconic, spatial, and visual attention skills from video games. There have been even studies with adults showing that experience with video games is related to better surgical skills. Jacob Benjamin, a doctor from Beth Israel Medical Center NY, found a direct link between skill at video gaming and skill at keyhole or laparoscopic surgery. Also, a reason given by experts

as to why fighter pilots of today are more skillful is that this generation's pilots are being weaned on video games.

E

The players learn to manage resources that are limited and decide the best use of resources, the same way as in real life. In strategy games, for instance, while developing a city, an unexpected surprise like an enemy might emerge. This forces the player to be flexible and quickly change tactics. Sometimes the player does this almost every second of the game giving the brain a real workout. According to researchers at the University of Rochester, led by Daphne Bavelier, a cognitive scientist, games simulating stressful events such as those found in battle or action games could be a training tool for real-world situations. The study suggests that playing action video games primes the brain to make quick decisions. Video games can be used to train soldiers and surgeons, according to the study. Steven Johnson, author of *Everything Bad is Good For You: How Today's Popular Culture*, says gamers must deal with immediate problems while keeping their long-term goals on their horizon. Young gamers force themselves to read to get instructions, follow storylines of games, and get information from the game texts.

F

James Paul Gee, professor of education at the University of Wisconsin-Madison, says that playing a video game is similar to working through a science problem. Like students in a laboratory, gamers must come up with a hypothesis. For example, players in some games constantly try out combinations of weapons and powers to use to defeat an enemy. If one does not work, they change the hypothesis and try the next one. Video games are goal-driven experiences, says Gee, which is fundamental to learning. Also, using math skills is important to win in many games that involve quantitative analysis like managing resources. In higher levels of a game, players usually fail the first time around, but they keep on trying until they succeed and move on to the next level.

G

Many games are played online and involve cooperation with other online players in order to win. Video and computer games also help children gain self-confidence and many games are based on history, city building, and governance and so on. Such games indirectly teach children about aspects of life on earth.

H

In an upcoming study in the journal *Current Biology*, authors Daphne Bavelier, Alexandre Pouget, and C. Shawn Green report that video games could provide a potent training regimen for speeding up reactions in many types of real-life situations. The researchers tested dozens of 18 to 25-year-olds who were not ordinarily video game players. They split the subjects into two groups. One group played 50 hours of the fast-paced action video games "Call of Duty 2" and "Unreal Tournament," and the other group played 50 hours of the slow-moving strategy game "The Sims 2." After this training period, all of the subjects were asked to make quick decisions in several tasks designed by the researchers. The action game players were up to 25 percent faster at coming to a conclusion and answered just as many questions

correctly as their strategy game playing peers.

Questions 36-40

Use the information in the passage to match the people (listed A-F) with opinions or deeds below.

Write the appropriate letters A-F in boxes 36-40 on your answer sheet.

- A The writer's opinion
- B James Paul Gee
- C Shawn Green
- D Daphne Bavelier
- E Steven Johnson
- F Jacob Benjamin

- 36 Video games as other daily life skills alter the brain's physical structure.
- 37 The brain is ready to make decisions without hesitation when players are immersed in playing stressful games.
- 38 The purpose-motivated experience that video games offer plays an essential role in studying.
- 39 Players are good at tackling prompt issues with future intentions.
- 40 It helps children broaden their horizon in many aspects and gain self-confidence.

答案

- 36. C
- 37. D
- 38. B
- 39. E
- 40. A

Test 28

READING PASSAGE 1

You should spend about 20 minutes on Questions 1-13 which are based on Reading Passage 1 below.

Man or Machine

A

During July 2003, the Museum of Science in Cambridge, Massachusetts exhibited what Honda calls ‘the world’ s most advanced humanoid robot’ , ASIMO (the Advanced Step in Innovative Mobility). Honda’ s brainchild is on tour in North America and delighting audiences wherever it goes. After 17 years in the making, ASIMO stands at four feet tall, weighs around 115 pounds and looks like a child in an astronaut’ s suit. Though it is difficult to see ASIMO’ s face at a distance, on closer inspection it has a smile and two large ‘eyes’ that conceal cameras. The robot cannot work autonomously – its actions are ‘remote-controlled’ by scientists through the computer in its backpack. Yet watching ASIMO perform at a show in Massachusetts it seemed uncannily human. The audience cheered as ASIMO walked forwards and backwards, side to side and up and downstairs. After the show, a number of people told me that they would like robots to play more of a role in daily life – one even said that the robot would be like ‘another person’ .

B

While the Japanese have made huge strides in solving some of the engineering problems of human kinetics and bipedal movements, for the past 10 years scientists at MIT’ s former Artificial Intelligence (AI) lab (recently renamed the Computer Science and Artificial Intelligence Laboratory, CSAIL) have been making robots that can behave like humans and interact with humans. One of MIT’ s robots, Kismet, is an anthropomorphic head and has two eyes (complete with eyelids), ears, a mouth, and eyebrows. It has several facial expressions, including happy, sad, frightened and disgusted. Human interlocutors are able to read some of the robot’ s facial expressions, and often change their behavior towards the machine as a result – for example, playing with it when it appears ‘sad’ . Kismet is now in MIT’ s museum, but the ideas developed here continue to be explored in new robots.

C

Cog (short for Cognition) is another pioneering project from MIT’ s former AI lab. Cog has a head, eyes, two arms, hands and a torso – and its proportions were originally measured from the body of a researcher in the lab. The work on Cog has been used to test theories of embodiment and developmental robotics, particularly getting a robot to develop intelligence by responding to its environment via sensors, and to learn through these types of interactions.

D

MIT is getting furthest down the road to creating human-like and interactive robots. Some scientists argue that ASIMO is a great engineering feat but not an intelligent machine – because it is unable to interact autonomously with unpredictabilities in its environment in meaningful ways, and learn from experience. Robots like Cog and Kismet and new robots at MIT’s CSAIL and media lab, however, are beginning to do this.

E

These are exciting developments. Creating a machine that can walk, make gestures and learn from its environment is an amazing achievement. And watch this space: these achievements are likely rapidly to be improved upon. Humanoid robots could have a plethora of uses in society, helping to free people from everyday tasks. In Japan, for example, there is an aim to create robots that can do the tasks similar to an average human and also act in more sophisticated situations as firefighters, astronauts or medical assistants to the elderly in the workplace and in homes – partly in order to counterbalance the effects of an ageing population.

F

Such robots say much about the way in which we view humanity, and they bring out the best and worst of us. On one hand, these developments express human creativity – our ability to invent, experiment, and to extend our control over the world. On the other hand, the aim to create a robot like a human being is spurred on by dehumanized ideas – by the sense that human companionship can be substituted by machines; that humans lose their humanity when they interact with technology; or that we are a little more than surface and ritual behaviors, that can be simulated with metal and electrical circuits.

Questions 1-6

Reading passage 1 has six paragraphs, A-F.

Which paragraph contains the following information?

Write the correct letter, A-F, in boxes 1-6 on your answer sheet.

NB You may use any letter more than once

- 1 different ways of using robots
- 2 a robot whose body has the same proportion as that of an adult
- 3 the fact that human can be copied and replaced by robots
- 4 a comparison between ASIMO for Honda and other robots
- 5 the pros and cons of creating robots
- 6 a robot that has eyebrows

答案

1. E
2. C
3. F
4. D
5. F
6. B

READING PASSAGE 2

You should spend about 20 minutes on Questions 14-26 which are based on Reading Passage 2 below.

Photovoltaics on the rooftop

A natural choice for powering the family home

A

In the past, urban homeowners have not always had much choice in the way electricity is supplied to their homes. Now, however, there is a choice, and a rapidly increasing number of households worldwide are choosing the solar energy option. Solar energy, the conversion of sunlight into energy, is made possible through the use of ‘photovoltaics’, which are simple appliances that fit onto the roof of a house.

B

The photovoltaics-powered home remains connected to the power lines, but no storage is required on-site, only a box of electronics (the inverter) to the interface between the photovoltaics and the grid network. Figure 1 illustrates the system. During the day, when the home may not be using much electricity, excess power from the solar array is fed back to the grid, to factories and offices that need daytime power. At night, power flows the opposite way. The grid network effectively provides storage. If the demand for electricity is well matched to when the sun shines, solar energy is especially valuable. This occurs in places like California in the US and Japan, where air-conditioning loads for offices and factories are large but heating loads for homes are small.

C

The first systematic exploration of the use of photovoltaics on homes began in the US during the 1970s. A well-conceived program started with the sitting of a number of residential experiment stations’ at selected locations around the country, representing different climatic zones. These stations contained a number of ‘dummy’ houses, each with different solar-energy system design. Homes within the communities close to these stations were monitored to see how well their energy use matched the energy generated by the stations’ dummy roofs. A change in US government priorities in the early 1980s halted this program.

D

With the US effort dropping away, the Japanese Sunshine Project came to the fore. A large residential test station was installed on Rokko Island beginning in 1986. This installation consists of 18 ‘dummy’ homes. Each equipped with its own 2-5 kilowatt photovoltaic system (about 20 – 50 square meters for each system). Some of these simulated homes have their own electrical appliances inside, such as TV sets, refrigerators and air conditioning units, which switch on and off under computer control providing a lavish lifestyle for the non-existent occupants. For the other systems, electronics simulate these household loads. This test station has allowed being explored in a systematic way, under well-controlled test conditions. With no insurmountable problems identified, the Japanese have used the experience gained

from this station to begin their own massive residential photovoltaics campaign.

E

Meanwhile, Germany began a very important ‘1,000 roof program’ in 1990, aimed at installing photovoltaics on the roofs of 1,000 private homes. Large federal and regional government subsidies were involved, accounting in most cases for 70% of the total system costs. The program proved immensely popular, forcing its extension to over 2,000 homes scattered across Germany. The success of this program stimulated other European countries to launch a similar program.

F

Japan’s ‘one million roof program’ was prompted by the experience gained in the Rokko Island test site and the success of the German 1,000 roof program. The initially quoted aims of the Japanese New Energy Development Organization were to have 70,000 homes equipped with the photovoltaics by the year 2000, on the way to 1 million by 2010. The program made a modest start in 1994 when 539 systems were installed with a government subsidy of 50 percent. Under this program, entire new suburban developments are using photovoltaics.

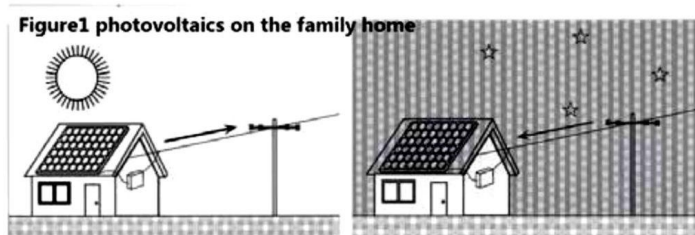
G

This is good news, not only for the photovoltaic industry but for everyone concerned with the environment. The use of fossil fuels to generate electricity is not only costly in financial terms, but also in terms of environmental damage. Gases produced by the burning of fossil fuels in the production of electricity are a major contributor to the greenhouse effect. To deal with this problem, many governments are now proposing stringent targets on the amount of greenhouse gas emissions permitted. These targets mean that all sources of greenhouse gas emissions including residential electricity use will receive closer attention in the future.

H

It is likely that in the future, governments will develop building codes that attempt to constrain the energy demands of new housing. For example, the use of photovoltaics or the equivalent may be stipulated to lessen demands on the grid network and hence reduce fossil fuel emissions. Approvals for building renovations may also be conditional upon taking such energy-saving measures. If this were to happen, everyone would benefit. Although there is an initial cost in attaching the system to the rooftop, the householder’s outlay is soon compensated with the savings on energy bills. In addition, everyone living on the planet stands to gain from the more benign environmental impact.

Photovoltaics on the family home



Residential use of photovoltaics – by day excess power is sent to the grid, and by night power is supplied to the home.

Questions 14-19

The Reading Passage 2 has nine paragraphs A-H

Which paragraph contains the following information?

Write the correct letter A-H, in boxes 14-19 on your answer sheet.

NB You may use any letter more than once.

- 14 examples of countries where electricity use is greater during the day than at night
- 15 a detailed description of an experiment that led to photovoltaics being promoted throughout the country
- 16 the negative effects of using conventional means of generating electricity
- 17 an explanation of the photovoltaic system.
- 18 the long-term benefits of using photovoltaics
- 19 a reference to wealthy countries being prepared to help less wealthy countries have access to photovoltaics

答案

- 14. B
- 15. D
- 16. H
- 17. B
- 18. I
- 19. E

READING PASSAGE 3

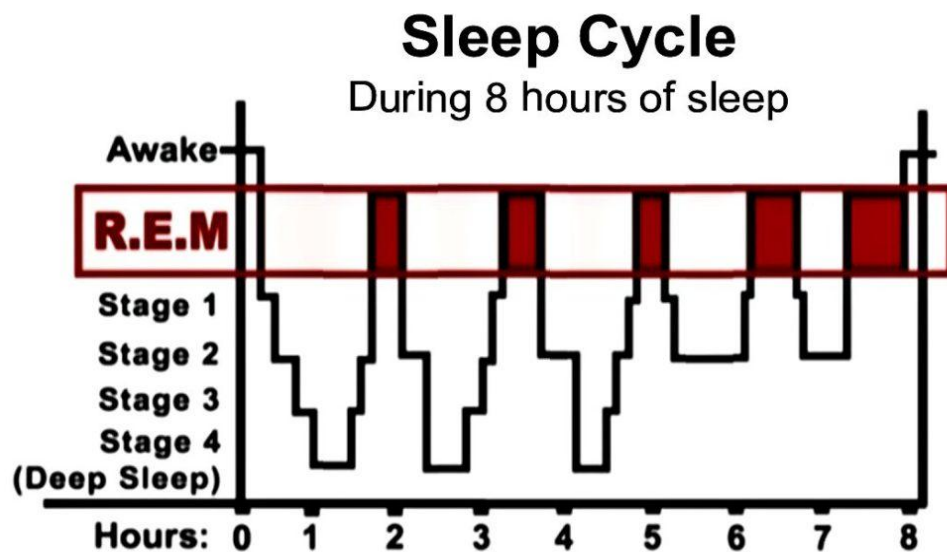
You should spend about 20 minutes on Questions 27-40 which are based on Reading Passage 3 below.

What Are Dreams?

A

Thousands of years ago, dreams were seen as messages from the gods, and in many cultures, they are still considered prophetic. In ancient Greece, sick people slept at the temples of Asclepius, the god of medicine, in order to receive dreams that would heal them. Modern dream science really begins at the end of the 19th century with Sigmund Freud, who theorized that dreams were the expression of unconscious desires often stemming from childhood. He believed that exploring these hidden emotions through analysis could help cure mental illness. The Freudian model of psychoanalysis dominated until the 1970s when new research into the chemistry of the brain showed that emotional problems could have biological or chemical roots, as well as environmental ones. In other words, we weren't sick just because of something our mothers did (or didn't do), but because of some imbalance that might be cured with medication.

B



After Freud, the most important event in dream science was the discovery in the early 1950s of a phase of sleep characterized by intense brain activity and rapid eye movement (REM). People awakened in the midst of REM sleep reported vivid dreams, which led researchers to conclude that most dreaming took place during REM. Using the electroencephalograph (EEG), researchers could see that brain activity during REM resembled that of the waking brain. That told them that a lot more was going on at night than anyone had suspected. But what, exactly?

C

Scientists still don't know for sure, although they have lots of theories. On one side

are scientists like Harvard's Allan Hobson, who believes that dreams are essentially random. In the 1970s, Hobson and his colleague Robert McCarley proposed what they called the "activation-synthesis hypothesis" which describes how dreams are formed by nerve signals sent out during REM sleep from a small area at the base of the brain called the pons. These signals, the researchers said, activate the images that we call dreams. That put a crimp in dream research; if dreams were meaningless nocturnal firings, what was the point of studying them?

D

Adult humans spend about a quarter of their sleep time in REM, much of it dreaming. During that time, the body is essentially paralyzed but the brain is buzzing. Scientists using PET and fMRI technology to watch the dreaming brain have found that one of the most active areas during REM is the limbic system, which controls our emotions. Much less active is the prefrontal cortex, which is associated with logical thinking. That could explain why dreams in REM sleep often lack a coherent storyline (some researchers have also found that people dream in non-REM sleep as well, although those dreams generally are less vivid.) Another active part of the brain in REM sleep is the anterior cingulate cortex, which detects discrepancies. Eric Nofzinger, director of the Sleep Neuroimaging Program at the University of Pittsburgh Medical Center, thinks that could be why people often figure out thorny problems in their dreams.

"As if the brain surveys the internal milieu and tries to figure out what it should be doing, and whether our actions conflict with who we are," he says.

E

These may seem like vital mental functions, but no one has yet been able to say that REM sleep or dreaming is essential to life or even sanity. MAO inhibitors, an older class of antidepressants, essentially block REM sleep without any detectable effects, although people do get a "REM rebound" – extra REM – if they stop the medication. That's also true of selective serotonin reuptake inhibitors (SSRIs) like Prozac, which reduce dreaming by a third to a half. Even permanently losing the ability to dream doesn't have to be disabling. Israeli researcher Peretz Lavie has been observing a patient named Yuval Chamtzani, who was injured by a fragment of shrapnel that penetrated his brain when he was 19. As a result, he gets no REM sleep and doesn't remember any dreams. But Lavie says that Chamtzani, now 55, "is probably the most normal person I know and one of the most successful ones." He's a lawyer, a painter and the editor of a puzzle column in a popular Israeli newspaper.

F

The mystery of REM sleep is that even though it may not be essential, it is ubiquitous – at least in mammals and birds. But that doesn't mean all mammals and birds dream (or if they do, they're certainly not – talking about it). Some researchers think REM may have evolved for physiological reasons. "One thing that's unique about mammals and birds is that they regulate body temperature", says neuroscientist Jerry Siegel, director of UCLA's Center for Sleep Research. "There's no good evidence that any coldblooded animal has REM sleep." REM sleep heats up the brain and non-REM cools it off, Siegel says, and that could mean that the changing sleep cycles allow the brain to repair itself. "It seems likely that REM sleep

is filling a basic physiological function and that dreams are a kind of epiphenomenon,” Siegel says – an extraneous byproduct; like foam on beer.

G

Whatever the function of dreams at night, they clearly can play a role in therapy during the day. The University of Maryland’s Clara Hill, who has studied the use of dreams in therapy, says that dreams are a ‘backdoor’, into a patient’s thinking.

“Dreams reveal stuff about you that you didn’t know was there,” she says. The therapists she trains to work with patients’ dreams are, in essence, heirs to Freud, using dream imagery to uncover hidden emotions and feelings. Dreams provide clues to the nature of the more serious mental illness. Schizophrenics, for example, have poor-quality dreams, usually about objects rather than people. “If you’re going to understand human behavior,” says Rosalind Cartwright, a chairman of psychology at Rush University Medical Center in Chicago, “here’s a big piece of it. Dreaming is our own storytelling time – to help us know who we are, where we’re going and how we’re going to get there.” Cartwright has been studying depression in divorced men and women, and she is finding that “good dreamers,” people who have vivid dreams with strong storylines, are less likely to remain depressed. She thinks that dreaming helps diffuse strong emotions. “Dreaming is a mental-health activity,” she says.

Questions 27-31

Reading Passage 3 has seven paragraphs, A-G.

Which paragraph contains the following information?

Write the correct number, A-G, in boxes 27-31 on your answer sheet.

- 27 Reference of an artist’s dreams who has versatile talents
- 28 The dream actually happens to many animals
- 29 Dreams are related to benefit and happiness
- 30 advanced scientific technology applied in the investigation of the REM stage.
- 31 questioning concern raised about the usefulness of investigation on dreams

Questions 35-40

Look at the following people and the list of statements below.

Match each statement with the correct person, A-G.

Write the correct letter, A-G, in boxes 35-40 on your answer sheet.

List of people

- A Sigmund Freud
- B Allan Hobson (Harvard)
- C Robert McCarley
- D Eric Nofzinger
- E Jerry Siegel
- F Clara Hill
- G Rosalind Cartwright

- 35 Dreams sometimes come along with REM as no more than a trivial attachment

- 36 Exploring patients' dreams would be beneficial for treatment as it reveals the unconscious thinking
- 37 Dreams help people cope with the difficulties they meet in the daytime
- 38 Decoding dreams would provide a reminder to human desire in the early days
- 39 Dreams are a body function to control strong emotion
- 40 Dreams seem to be as randomly occurring and have limited research significance.

答案

27. E
28. F
29. G
30. D
31. C
35. E
36. F
37. D
38. A
39. G
40. B

Test 29

READING PASSAGE 1

You should spend about 20 minutes on Questions 1-14 which are based on Reading Passage 1 below.

The culture of Chimpanzee!

A

The similarities between chimpanzees and humans have been studied for years, but in the past decade, researchers have determined that these resemblances run much deeper than anyone first thought. For instance, the nut-cracking observed in the Taï Forest is far from a simple chimpanzee behavior; rather it is a singular adaptation found only in that particular part of Africa and a trait that biologists consider being an expression of chimpanzee culture. Scientists frequently use the term “culture” to describe elementary animal behaviors – such as the regional dialects of different populations of songbirds – but as it turns out, the rich and varied cultural traditions found among chimpanzees are second in complexity only to human traditions.

B

During the past two years, an unprecedented scientific collaboration, involving every major research group studying chimpanzees, has documented a multitude of distinct cultural patterns extending across Africa, in actions ranging from the animals’ use of tools to their forms of communication and social customs. This emerging picture of chimpanzees not only affects how we think of these amazing creatures but also alters human beings’ conception of our own uniqueness and hints at ancient foundations for extraordinary capacity for culture.

C

Homo sapiens and Pan troglodytes have coexisted for hundreds of millennia and share more than 98 percent of their genetic material, yet only 40 years ago we still knew next to nothing about chimpanzee behavior in the wild. That began to change in the 1960s when Toshisada Nishida of Kyoto University in Japan and Jane Goodall began their studies of wild chimpanzees at two field sites in Tanzania. (Goodall’s research station at Gombe – the first of its kind – is more famous, but Nishida’s site at Mahale is the second oldest chimpanzee research site in the world.)

D

In these initial studies, as the chimpanzees became accustomed to close observation, the remarkable discoveries began. Researchers witnessed a range of unexpected behaviors, including fashioning and using tools, hunting, meat-eating, food sharing and lethal fights between members of neighboring communities. In the years that followed, other primatologists set up camp elsewhere, and, despite all the financial, political and logistical problems that can beset African fieldwork, several of these outposts became truly long-term projects. As a result, we live in an unprecedented

time, when an intimate and comprehensive scientific record of chimpanzees' lives, at last, exists not just for one but for several communities spread across Africa.

E

As early as 1973, Goodall recorded 13 forms of tool use as well as eight social activities that appeared to differ between the Gombe chimpanzees and chimpanzee populations elsewhere. She ventured that some variations had what she termed a cultural origin. But what exactly did Goodall mean by "culture"? According to the Oxford Encyclopedic English Dictionary, culture is defined as "the customs ... and achievements of a particular time or people." The diversity of human cultures extends from technological variations to marriage rituals, from culinary habits to myths and legends. Animals do not have myths and legends, of course. But they do have the capacity to pass on behavioral traits from generation to generation, not through their genes but by learning. For biologists, this is the fundamental criterion for a cultural trait: it must be something that can be learned by observing the established skills of others and thus passed on to future generations.

F

What of the implications for chimpanzees themselves? We must highlight the tragic loss of chimpanzees, whose populations are being decimated just when we are at last coming to appreciate these astonishing animals more completely. Populations have plummeted in the past century and continue to fall as a result of illegal trapping, logging and, most recently, the bushmeat trade. The latter is particularly alarming: logging has driven roadways into the forests that are now used to ship wild-animal meat-including chimpanzee meat to consumers as far afield as Europe. Such destruction threatens not only the animals themselves but also a host of fascinatingly different ape cultures.

G

Perhaps the cultural richness of the ape may yet help in its salvation, however. Some conservation efforts have already altered the attitudes of some local people. A few organizations have begun to show videotapes illustrating the cognitive prowess of chimpanzees. One Zairian viewer was heard to exclaim, "Ah, this ape is so like me, I can no longer eat him."

H

How an international team of chimpanzee experts conducted the most comprehensive survey of the animals ever attempted. Scientists have been investigating chimpanzee culture for several decades, but too often their studies contained a crucial flaw. Most attempts to document cultural diversity among chimpanzees have relied solely on officially published accounts of the behaviors recorded at each research site. But this approach probably overlooks a good deal of cultural variation for three reasons.

I

First, scientists typically don't publish an extensive list of all the activities they do not see at a particular location. Yet this is exactly what we need to know-which behaviors were and were not observed at each site. Second, many reports describe chimpanzee behaviors without saying how common they are; without this information, we can't determine whether a particular action was a once-in-a-lifetime aberration or

a routine event that should be considered part of the animals' culture. Finally, researchers' descriptions of potentially significant chimpanzee behaviors frequently lack sufficient detail, making it difficult for scientists working at other spots to record the presence or absence of the activities.

J

To remedy these problems, the two of us decided to take a new approach. We asked field researchers at each site for a list of all the behaviors they suspected were local traditions. With this information in hand, we pulled together a comprehensive list of 65 candidates for cultural behaviors.

K

Then we distributed our list to the team leaders at each site. In consultation with their colleagues, they classified each behavior in terms of its occurrence or absence in the chimpanzee community studied. The key categories were customary behavior (occurs in most or all of the able-bodied members of at least one age or sex class, such as all adult males), habitual (less common than customary but occurs repeatedly in several individuals), present (seen at the site but not habitual), absent (never seen), and unknown.

Questions 1-5

The Reading Passage has seven paragraphs 1-5.

Which paragraph contains the following information?

Write the correct letter, A-K, in boxes 1-5 on your answer sheet.

- 1 A problem of researchers on chimpanzee culture which are only based on official sources.
- 2 Design a new system by two scientists aims to solve the problem.
- 3 Reasons why previous research on ape culture is problematic.
- 4 Classification of data observed or collected.
- 5 An example that showing the tragic outcome of animals leading to an indication of the change in local people's attitude in preservation.

答案

1. H
2. J
3. I
4. K
5. G

READING PASSAGE 3

You should spend about 20 minutes on Questions 28-40 which are based on Reading Passage 3 below.

Company Innovation

A

In a scruffy office in midtown Manhattan, a team of 30 artificial-intelligence programmers is trying to simulate the brains of an eminent sexologist, a well-known dietician, a celebrity fitness trainer and several other experts. Umagic Systems is a young firm, setting up websites that will allow clients to consult the virtual versions of these personalities. Subscribers will feed in details about themselves and their goals; Umagic's software will come up with the advice that the star expert would give. Although few people have lost money betting on the neuroses of the American consumer, Umagic's prospects are hard to gauge (in ten years' time, consulting a computer about your sex life might seem natural, or it might seem absurd). But the company and others like it are beginning to spook large American firms because they see such half-barmy "innovative" ideas as the key to their own future success.

B

Innovation has become the buzz-word of American management. Firms have found that most of the things that can be outsourced or re-engineered have been (worryingly, by their competitors as well). The stars of American business tend today to be innovators such as Dell, Amazon and Wal-Mart, which have produced ideas or products that have changed their industries.

C

A new book by two consultants from Arthur D. Little records that, over the past 15 years, the top 20% of firms in an annual innovation poll by Fortune magazine have achieved double the shareholder returns of their peers. Much of today's merger boom is driven by a desperate search for new ideas. So is the fortune now spent on licensing and buying others' intellectual property. According to the Pasadena-based Patent & Licence Exchange, trading in intangible assets in the United States has risen from \$15 billion in 1990 to \$100 billion in 1998, with an increasing proportion of the rewards going to small firms and individuals.

D

And therein lies the terror for big companies: that innovation seems to work best outside them. Several big established "ideas factories", including 3M, Procter & Gamble and Rubbermaid, have had dry spells recently. Gillette spent ten years and \$1 billion developing its new Mach 3 razor; it took a British supermarket only a year or so to produce a reasonable imitation. "In the management of creativity, size is your enemy," argues Peter Chernin, who runs the Fox TV and film empire for News Corporation. One person managing 20 movies is never going to be as involved as one doing five movies. He has thus tried to break down the studio into smaller units – even at the risk of incurring higher costs.

E

It is easier for ideas to thrive outside big firms these days. In the past, if a clever scientist had an idea he wanted to commercialise, he would take it first to a big company. Now, with plenty of cheap venture capital, he is more likely to set up on his own. Umagic has already raised \$5m and is about to raise \$25m more. Even in capital-intensive businesses such as pharmaceuticals, entrepreneurs can conduct early-stage research, selling out to the big firms when they reach expensive, risky clinical trials. Around a third of drug firms' total revenue now comes from licensed-in technology.

F

Some giants, including General Electric and Cisco, have been remarkably successful at snapping up and integrating scores of small companies. But many others worry about the prices they have to pay and the difficulty in hanging on to the talent that dreamt up the idea. Everybody would like to develop more ideas in-house. Procter & Gamble is now shifting its entire business focus from countries to products; one aim is to get innovations accepted across the company. Elsewhere, the search for innovation has led to a craze for “intrapreneurship” – devolving power and setting up internal ideas-factories and tracking stocks so that talented staff will not leave.

G

Some people think that such restructuring is not enough. In a new book, Clayton Christensen argues that many things which established firms do well, such as looking after their current customers, can hinder the sort of innovative behaviour needed to deal with disruptive technologies. Hence the fashion for cannibalization – setting up businesses that will actually fight your existing ones. Bank One, for instance, has established Wingspan, an Internet bank that competes with its real branches (see article). Jack Welch's Internet initiative at General Electric is called “Destroyyourbusiness.com”.

H

Nobody could doubt that innovation matters. But need large firms to be quite so pessimistic? A recent survey of the top 50 innovations in America, by Industry Week, a journal, suggested that ideas are as likely to come from big firms as from small ones. Another sceptical note is sounded by Amar Bhidé, a colleague of Mr Christensen's at the Harvard Business School and the author of another book on entrepreneurship. Rather than having to reinvent themselves, big companies, he believes, should concentrate on projects with high costs and low uncertainty, leaving those with low costs and high uncertainty to small entrepreneurs. As ideas mature and the risks and rewards become more quantifiable, big companies can adopt them.

I

At Kimberly-Clark, Mr Sanders had to discredit the view that jobs working on new products were for “those who couldn't hack it in the real business.” He has tried to change the culture not just by preaching fuzzy concepts but also by introducing hard incentives, such as increasing the rewards for those who come up with successful new ideas and, particularly, not punishing those whose experiments fail. The genesis of one of the firm's current hits, Depend, a more dignified incontinence garment, lay in a previous miss, Kotex Personals, a form of disposable underwear for menstruating

women.

J

Will all this creative destruction, cannibalization and culture tweaking make big firms more creative? David Post, the founder of Umagic, is skeptical: “The only successful intrapreneurs are ones who leave and become entrepreneurs.” He also recalls with glee the looks of total incomprehension when he tried to hawk his “virtual experts” idea three years ago to the idea labs of firms such as IBM – though, as he cheerfully adds, “of course, they could have been right.” Innovation – unlike, apparently, sex, parenting and fitness – is one area where a computer cannot tell you what to do.

Questions 28-33

The Reading Passage has ten paragraphs A-J.

Which paragraph contains the following information?

Write the correct letter A-J, in boxes 28-33 on your answer sheet.

NB You may use any letter more than once.

- 28 Approach to retain the best employees
- 29 Safeguarding expenses on an innovative idea
- 30 Integrating outside firms might produce a certain counter effect
- 31 Example of three famous American companies’ innovation
- 32 Example of one company changing its focus
- 33 Example of a company resolving financial difficulties itself

答案

- 28. F
- 29. C
- 30. G
- 31. B
- 32. F
- 33. E

Test 30

READING PASSAGE 2

You should spend about 20 minutes on Questions 15-27 which are based on Reading Passage 2 below.

Aqua product: New Zealand's Algae Biodiesel

A

The world's first wild algae biodiesel, produced in New Zealand by Aquaflow Bionomic Corporation, was successfully test-driven in Wellington by the Minister for Energy and Climate Change Issues, David Parker. In front of a crowd of invited guests, media and members of the public, the Minister filled up a diesel-powered Land Rover with Aquaflow B5 blend bio-diesel and then drove the car around the forecourt of Parliament Buildings in Central Wellington. Green Party co-leader, Jeanette Fitzsimons was also on board. Marlborough-based Aquaflow announced on May 2006 that it had produced the world's first bio-diesel derived from wild microalgae sourced from local sewage ponds.

B

"We believe we are the first company in the world to test drive a car powered by wild algae-based biodiesel. This will come as a surprise to some international bio-diesel industry people who believe that this break-through is still years away," explains Aquaflow spokesperson Barrie Leay. "A bunch of inventive Kiwis and an Aussie have developed this fuel in just over a year", he comments. "This is a huge opportunity for New Zealand and a great credit to the team of people who saw the potential in this technology from day one."

C

Bio-diesel based on algae could eventually become a sustainable, low cost, cleaner-burning fuel alternative for New Zealand, powering family cars, trucks, buses and boats. It can also be used for other purposes such as heating or distributed electricity generation. There is now a global demand for billions of litres of biodiesel per year. Algae are also readily available and produced in huge volumes in nutrient-rich waste streams such as at the settling ponds of Effluent Management Systems (EMS). It is a renewable indigenous resource ideally suited to the production of fuel and other useful by-products. The breakthrough comes after technology start-up, Aquaflow, agreed to undertake a pilot with Marlborough District Council late last year to extract algae from the settling ponds of its EMS based in Blenheim. By removing the main contaminant to use as a fuel feedstock, Aquaflow is also helping clean up the council's water discharge – a process known as bio-remediation. Dairy farmers, and many food processors too, can benefit in similar ways by applying the harvesting technology to their nutrient-rich waste streams.

D

Blended with conventional mineral diesel, bio-diesel can run vehicles without the need for vehicle modifications. Fuel derived from algae can also help meet the Government B5 (5% blended) target, with the prospect of this increase over time as bio-fuel production increases. “Our next step is to increase capacity to produce one million litres of bio-diesel from the Marlborough sewerage ponds over the next year,” says Leay. Aquaflow will launch a prospectus pre-Christmas as the company has already attracted considerable interest from potential investors. The test drive bio-diesel was used successfully in a static engine test at Massey University’s Wellington campus on Monday, December 11.

E

Today Algae are used by humans in many ways; for example, as fertilizers, soil conditioners and livestock feed. Aquatic and microscopic species are cultured in clear tanks or ponds and are either harvested or used to treat effluents pumped through the ponds. Algalculture on a large scale is an important type of aquaculture in some places. Naturally growing seaweeds are an important source of food, especially in Asia. They provide many vitamins including A, B, B2, B6, niacin and C, and are rich in iodine, potassium, iron, magnesium and calcium. In addition, commercially cultivated microalgae, including both Algae and Cyan-bacteria, are marketed as nutritional supplements, such as Spirulina, Chlorella and the Vitamin-C supplement, Dunaliella, high in beta-carotene. Algae are national foods of many nations: China consumes more than 70 species, including fat choy, a cyanobacterium considered a vegetable; Japan, over 20 species. The natural pigments produced by algae can be used as an alternative to chemical dyes and coloring agents.

F

Algae are the simplest plant organisms that convert sunlight and carbon dioxide in the air around us into stored energy through the well-understood process of photosynthesis. Algae are rich in lipids and other combustible elements and Aquaflow is developing technology that will allow these elements to be extracted in a cost-effective way. The proposed process is the subject of a provisional patent. Although algae are good at taking most of the nutrients out of sewage, too many algae can taint the water and make it smell. So, councils have to find a way of cleaning up the excess algae in their sewerage outflows and then either dispose of it or find alternative uses for it. And that’s where Aquaflow comes in.

G

Unlike some bio-fuels which require crops to be specially grown and thereby compete for land use with food production, and use other scarce resources of fuel, chemicals and fertiliser, the source for algae-based biodiesel already exists extensively and the process produces a sustainable net energy gain by capturing free solar energy from the sun.

Questions 15-19

Reading Passage 2 contains seven paragraphs A-G.

Which paragraph states the following information?

Write the appropriate letter A-G, in boxes 15-19 on your answer sheet.

- NB You may use any letter more than once.
- 15 It is unnecessary to modify vehicles driven by bio-diesel.
 - 16 Some algae are considered edible plants.
 - 17 Algae could be part of a sustainable and recycled source.
 - 18 Algae biodiesel is superior to other bio-fuels in a lot of ways.
 - 19 overgrown algae also can be a potential threat to the environment

答案

- 15. D
- 16. E
- 17. C
- 18. G
- 19. F

Test 31

READING PASSAGE 1

You should spend about 20 minutes on Questions 1-13 which are based on Reading Passage 1 below.

Magnetic Therapy

A

Magnetic therapy, which is a \$5-billion market worldwide, is a form of alternative medicine which claims that magnetic fields have healing powers. Magnetic devices that are claimed to be therapeutic include magnetic bracelets, insoles, wrist and knee bands, back and neck braces, and even pillows and mattresses. Their annual sales are estimated at \$300 million in the United States and more than a billion dollars globally. They have been advertised to cure a vast array of ills, particularly pain.

B

The therapy works on the principle of balancing electrical energy in the body by pulsating magnetic waves through different parts of the body. The electrical currents generated by magnets increase the blood flow and oxygen which helps to heal many of the ailments. The natural effects of the Earth's magnetic field are considered to play an essential role in the health of humans and animals. It is generally accepted that our body draws some benefit from the Earth's magnetic field. To restore the balance within our body allows us to function at our optimum level. For example, when the first astronauts returned to earth sick, NASA concluded that their illness resulted from the lack of a planetary magnetic field in outer space. To resolve the problem, NASA placed magnets in the astronauts' space suits and space travel vehicles, and astronauts have returned to Earth healthy ever since.

C

Historically it is reported that magnets have been around for an extremely long time. The therapeutic power of magnets was known to physicians in ancient Greece, Egypt and China over 4000 years ago, who used naturally magnetic rock – lodestone – to treat a variety of physical and psychological ailments. Cleopatra the beautiful Egyptian queen was probably the first celebrity to use magnets. It is documented that in order to prevent from aging, she slept on a Lodestone to keep her skin youthful. Ancient Romans also used magnet therapy to treat eye disease.

D

The popularity of magnet therapy in the United States began to rise during the 1800s and soared in the post-Civil War era. Sears-Roebuck advertised magnetic jewelry in its catalog for the healing of virtually any ailment. An Austrian psychoanalyst by the name of Wilhelm Reich immigrated to the United States in 1939 and researched the effects of electromagnetism on humans. Today, Germany, Japan, Israel, Russia and at least 45 other countries consider magnetic therapy to be an official medical procedure

for the treatment of numerous ailments, including various inflammatory and neurological problems.

E

For those who practice magnetic therapy, strongly believe that certain ailments can be treated if the patient is exposed to magnetic fields while at the same time there is a strong resentment from the medical establishment and critics claim that most magnets don't have the strength to affect the various organs and tissues within the body and it is a product of Pseudoscience and is not based on proper research and analysis. There are few reported complications of magnetic therapy and the World Health Organization says a low level of magnetic energy is not harmful. Documented side effects are not life-threatening and include pain, nausea and dizziness that disappeared when the magnets were removed. If considering magnet therapy, as with any medical treatment, it is always advisable to consult one's regular physician first. Magnet therapy is gaining popularity; however, scientific evidence to support the success of this therapy is lacking. More scientifically sound studies are needed in order to fully understand the effects that magnets can have on the body and the possible benefits or dangers that could result from their use.

F

Researchers at Baylor University Medical Center recently conducted a double-blind study on the use of concentric-circle magnets to relieve chronic pain in 50 post-polio patients. A static magnetic device or a placebo device was applied to the patient's skin for 45 minutes. The patients were asked to rate how much pain they experienced when a "trigger point was touched." The researchers reported that the 29 patients exposed to the magnetic device achieved lower pain scores than did the 21 who were exposed to the placebo device. However, this study had significant flaws in their design. Although the groups were said to be selected randomly, the ratio of women to men in the experimental group was twice that of the control group; the age of the placebo group was four years higher than that of the control group; there were just one brief exposure and no systematic follow-up of patients.

G

Magnet therapy is gaining popularity; however, scientific evidence to support the success of this therapy is lacking. More scientifically sound studies are needed in order to fully understand the effects that magnets can have on the body and the possible benefits or dangers that could result from their use.

Questions 1-6

Reading Passage 1 has seven paragraphs, A-G.

Choose the correct heading for paragraphs A-G from the list of headings below.

Write the correct number, i-x, in boxes 1-6 on your answer sheet.

List of headings

- i Earth itself as the biggest magnet
- ii The commercial magnetic products
- iii Utilize the power from the natural magnetic field
- iv Early application of the magnet

- v Brief introduction of how the magnetic therapy works
- vi pain-reducing effect
- vii Arguments for and against the therapy
- viii An experiment on post-polio patients
- ix Conditions of magnet use today
- 1 Paragraph A
- 2 Paragraph B
- 3 Paragraph C
- 4 Paragraph D
- 5 Paragraph E
- 6 Paragraph F

Questions 11-13

Complete each sentence with the correct ending, A-F, below.

Write the correct letters, A-F, in boxes 11-13 on your answer sheet.

- 11 The first NASA astronauts' sickness
 - 12 According to the WHO, under the physician' s instruction, a small amount of magnetic energy
 - 13 The author holds that in order to fully understand the magnetic effects, we
- A has no negative side effect.
 - B resulted from physical ailment.
 - C should have more sophisticated studies
 - D is exposed to the placebo device.
 - E must select the subjects randomly.
 - F came from the absence of a magnetic field.

答案

- 1. ii
- 2. v
- 3. iv
- 4. ix
- 5. vii
- 6. viii
- 11. F
- 12. A
- 13. C

READING PASSAGE 2

You should spend about 20 minutes on Questions 14-27 which are based on Reading Passage 2 below.

When the Tulip Bubble Burst

Tulips are spring-blooming perennials that grow from bulbs. Depending on the species, tulip plants can grow as short as 4 inches (10 cm) or as high as 28 inches (71 cm). The tulip's large flowers usually bloom on scapes or sub-scapes stems that lack bracts. Most tulips produce only one flower per stem, but a few species bear multiple flowers on their scapes (e.g. *Tulipa turkestanica*). The showy, generally cup or star-shaped tulip flower has three petals and three sepals, which are often termed tepals because they are nearly identical. These six tepals are often marked on the interior surface near the bases with darker colorings. Tulip flowers come in a wide variety of colors, except pure blue (several tulips with "blue" in the name have a faint violet hue)

A

Long before anyone ever heard of Qualcomm, CMGI, Cisco Systems, or the other high-tech stocks that have soared during the current bull market, there was *Semper Augustus*. Both more prosaic and more subline than any stock or bond, it was a tulip of extraordinary beauty, its midnight-blue petals topped by a band of pure white and accented with crimson flares. To denizens of 17th century Holland, little was as desirable.

B

Around 1624, the Amsterdam man who owned the only dozen specimens was offered 3,000 guilders for one bulb. While there's no accurate way to render that in today's greenbacks, the sum was roughly equal to the annual income of a wealthy merchant. (A few years later, Rembrandt received about half that amount for painting *The Night Watch*.) Yet the bulb's owner, whose name is now lost to history, nixed the offer.

C

Who was crazier, the tulip lover who refused to sell for a small fortune or the one who was willing to splurge. That's a question that springs to mind after reading *Tulipmania: The Story of the World's Most Coveted Flower and the Extraordinary Passions It Aroused* by British journalist Mike Dash. In recent years, as investors have intentionally forgotten everything they learned in *Investing 101* in order to load up on unproven, unprofitable dot-com issues, tulip mania has been invoked frequently. In this concise, artfully written account, Dash tells the real history behind the buzzword and in doing so, offers a cautionary tale for our times.

D

The Dutch were not the first to go gaga over the tulip. Long before the first tulip bloomed in Europe – in Bavaria, it turns out, in 1559 – the flower had enchanted the Persians and bewitched the rulers of the Ottoman Empire. It was in Holland, however, that the passion for tulips found its most fertile ground, for reasons that had little to do with horticulture.

E

Holland in the early 17th century was embarking on its Golden Age. Resources that had just a few years earlier gone toward fighting for independence from Spain now flowed into commerce. Amsterdam merchants were at the center of the lucrative East Indies trade, where a single voyage could yield profits of 400%. They displayed their success by erecting grand estates surrounded by flower gardens. The Dutch population seemed torn by two contradictory impulses: a horror of living beyond one's means and the love of a long shot.

F

Enter the tulip. “It is impossible to comprehend the tulip mania without understanding just how different tulips were from every other flower known to horticulturists in the 17th century,” says Dash. “The colors they exhibited were more intense and more concentrated than those of ordinary plants.” Despite the outlandish prices commanded by rare bulbs, ordinary tulips were sold by the pound. Around 1630, however, a new type of tulip fancier appeared, lured by tales of fat profits. These “florists,” or professional tulip traders, sought out flower lovers and speculators alike. But if the supply of tulip buyers grew quickly, the supply of bulbs did not. The tulip was a conspirator in the supply squeeze: It takes seven years to grow one from seed. And while bulbs can produce two or three clones, or “offsets,” annually, the mother bulb only lasts a few years.

G

Bulb prices rose steadily throughout the 1630s, as ever more speculators wedged into the market. Weavers and farmers mortgaged whatever they could to raise cash to begin trading. In 1633, a farmhouse in Hoorn changed hands for three rare bulbs. By 1636 any tulip—even bulbs recently considered garbage — could be sold off, often for hundreds of guilders. A futures market for bulbs existed, and tulip traders could be found conducting their business in hundreds of Dutch taverns. Tulipmania reached its peak during the winter of 1636-37 when some bulbs were changing hands ten times in a day. The zenith came early that winter, at an auction to benefit seven orphans whose only asset was 70 fine tulips left by their father. One, a rare Violetten Admiraal van Enkhuizen bulb that was about to split in two, sold for 5,200 guilders, the all-time record. All told, the flowers brought in nearly 53,000 guilders.

H

Soon after, the tulip market crashed utterly, spectacularly. It began in Haarlem, at a routine bulb auction when, for the first time, the greater fool refused to show up and pay. Within days, the panic had spread across the country. Despite the efforts of traders to prop up demand, the market for tulips evaporated. Flowers that had commanded 5,000 guilders a few weeks before now fetched one-hundredth that amount. Tulipmania is not without flaws. Dash dwells too long on the tulip's migration from Asia to Holland. But he does a service with this illuminating, accessible account of incredible financial folly.

I

Tulipmania differed in one crucial aspect from the dot-com craze that grips our attention today: even at its height, the Amsterdam Stock Exchange, well-established

in 1630, wouldn't touch tulips. "The speculation in tulip bulbs always existed at the margins of Dutch economic life," Dash writes. After the market crashed, a compromise was brokered that let most traders settle their debts for a fraction of their liability. The overall fallout on the Dutch economy was negligible. Will we say the same when Wall Street's current obsession finally runs its course?

Questions 14-18

The Reading Passage has nine paragraphs A-I.

Which paragraph contains the following information?

Write the correct letter A-I, in boxes 14-18 on your answer sheet.

- 14 Difference between bubble burst impacts by tulip and by high-tech shares.
- 15 Spread of tulip before 17th century
- 16 Indication of money offered for the rare bulb in the 17th century
- 17 Tulip was treated as money in Holland
- 18 The comparison made between a tulip and other plants

答案

- 14. I
- 15. D
- 16. B
- 17. G
- 18. F

READING PASSAGE 3

You should spend about 20 minutes on Questions 28-40 which are based on Reading Passage 3 below.

Does An IQ Test Prove Creativity?

Everyone has creativity, some a lot more than others. The development of humans, and possibly the universe, depends on it. Yet creativity is an elusive creature. What do we mean by it? What is going on in our brains when ideas form? Does it feel the same for artists and scientists? We asked writers and neuroscientists, pop stars and AI gurus to try to deconstruct the creative process – and learn how we can all ignite the spark within.

A

In the early 1970s, creativity was still seen as a type of intelligence. But when more subtle tests of IQ and creative skills were developed in the 1970s, particularly by the father of creativity testing, Paul Torrance, it became clear that the link was not so simple. Creative people are intelligent, in terms of IQ tests at least, but only averagely or just above. While it depends on the discipline, in general beyond a certain level IQ does not help boost creativity; it is necessary, but not sufficient to make someone creative.

B

Because of the difficulty of studying the actual process, most early attempts to study creativity concentrated on personality. According to creativity specialist Mark Runco of California State University, Fullerton, the “creative personality” tends to place a high value on aesthetic qualities and to have broad interests, providing lots of resources to draw on and knowledge to recombine into novel solutions. “Creatives” have an attraction to the complexity and an ability to handle conflict. They are also usually highly self-motivated, perhaps even a little obsessive. Less creative people, on the other hand, tend to become irritated if they cannot immediately fit all the pieces together. They are less tolerant of confusion. Creativity comes to those who wait, but only to those who are happy to do so in a bit of a fog.

C

But there may be a price to pay for having a creative personality. For centuries, a link has been made between creativity and mental illness. Psychiatrist Jamison of Johns Hopkins University in Baltimore, Maryland, found that established artists are significantly more likely to have mood disorders. But she also suggests that a change of mood state might be the key to triggering a creative event, rather than the negative mood itself. Intelligence can help channel this thought style into great creativity, but when combined with emotional problems, lateral, divergent or open thinking can lead to mental illness instead.

D

Jordan Peterson, a psychologist at the University of Toronto, Canada, believes he has identified a mechanism that could help explain this. He says that the brains of creative people seem more open to incoming stimuli than less creative types. Our senses are

continuously feeding a mass of information into our brains, which have to block or ignore most of it to save us from being snowed under. Peterson calls this process latent inhibition, and argues that people who have less of it, and who have a reasonably high IQ with a good working memory can juggle more of the data, and so may be open to more possibilities and ideas. The downside of extremely low latent inhibition may be a confusing thought style that predisposes people to mental illness. So for Peterson, mental illness is not a prerequisite for creativity, but it shares some cognitive traits.

E

But what of the creative act itself? One of the first studies of the creative brain at work was by Colin Martindale, a psychologist from the University of Maine in Orono. Back in 1978, he used a network of scalp electrodes to record an electroencephalogram, a record of the pattern of brain waves, as people made up stories. Creativity has two stages: inspiration and elaboration, each characterised by very different states of mind. While people were dreaming up their stories, he found their brains were surprisingly quiet. The dominant activity was alpha waves, indicating a very low level of cortical arousal: a relaxed state, as though the conscious mind was quiet while the brain was making connections behind the scenes. It's the same sort of brain activity as in some stages of sleep, dreaming or rest, which could explain why sleep and relaxation can help people be creative. However, when these quiet-minded people were asked to work on their stories, the alpha wave activity dropped off and the brain became busier, revealing increased cortical arousal, more corraling of activity and more organised thinking. Strikingly, it was the people who showed the biggest difference in brain activity between the inspiration and development stages who produced the most creative storylines. Nothing in their background brain activity marked them as creative or uncreative. "It's as if the less creative person can't shift gear," says Guy Claxton, a psychologist at the University of Bristol, UK. "Creativity requires different kinds of thinking. Very creative people move between these states intuitively." Creativity, it seems, is about mental flexibility: perhaps not a two-step process, but a toggling between two states. In a later study, Martindale found that communication between the sides of the brain is also important.

F

Paul Howard-Jones, who works with Claxton at Bristol, believes he has found another aspect of creativity. He asked people to make up a story based on three words and scanned their brains using functional magnetic resonance imaging. In one trial, people were asked not to try too hard and just report the most obvious story suggested by the words. In another, they were asked to be inventive. He also varied the words so it was easier or harder to link them. As people tried harder and came up with more creative tales, there was a lot more activity in a particular prefrontal brain region on the right-hand side. These regions are probably important in monitoring for conflict, helping us to filter out many of the unhelpful ways of combining the words and allowing us to pull out just the desirable connections, Howard-Jones suggests. It shows that there is another side to creativity, he says. The story-making task,

particularly when we are stretched, produces many options which we have to assess. So part of creativity is a conscious process of evaluating and analysing ideas. The test also shows that the more we try and are stretched, the more creative our minds can be.

G

And creativity need not always be a solitary, tortured affair, according to Teresa Amabile of Harvard Business School. Though there is a slight association between solitary writing or painting and negative moods or emotional disturbances, scientific creativity and workplace creativity seem much more likely to occur when people are positive and buoyant. In a decade-long study of real businesses, to be published soon, Amabile found that positive moods relate positively to creativity in organisations and that the relationship is a simple linear one. Creative thought also improves people's moods, her team found, so the process is circular. Time pressures, financial pressures and hard-earned bonus schemes, on the other hand, do not boost workplace creativity" internal motivation, not coercion, produces the best work.

H

Another often forgotten aspect of creativity is social. Vera John-Steiner of the University of New Mexico says that to be really creative you need strong social networks and trusting relationships, not just active neural networks. One vital characteristic of a highly creative person, she says, is that they have at least one other person in their life who doesn't think they are completely nuts.

Questions 32-36

Use the information in the passage to match the people (listed A-F) with opinions or deeds below. Write the appropriate letters A-F in boxes 32-36 on your answer sheet.

- A Jamison
- B Jordan Peterson
- C Guy Claxton
- D Howard-Jone
- E Teresa Amabile
- F Vera John-Steiner

32 Instead of producing a negative mood, a shift of mood state might be the one important factor of inducing creative thinking.

33 Where the more positive moods individuals achieve, there is higher creativity in organizations.

34 Good interpersonal relationship and trust contribute to a person with more

35 Creativity demands different kinds of thinking that can be easily changed back and forth.

36 A certain creative mind can be upgraded if we are put into more practice in assessing and processing ideas.

答案

32. A

33. E

34. F

35. C 36. D